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V6501 ABC
Ammonia Process Flare KOD
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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\Ammonia Flare KOD - V6501 ABC -7%.pvdb

Date of Analysis : Sep 16,2026 2:48pm

PV Elite 28 U1, March 2026

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Vessel Design Summary: Step: 43 2:48pm Sep 16,2026



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Vessel Design Summary:**ASME Code, Section VIII Division 1, 2025**

Diameter Spec : 6000.000 mm. ID
 Vessel Design Length, Tangent to Tangent 13500.00 mm.
 Specified Datum Line Distance 0.00 mm.
 Internal Design Temperature 270 °C
 Internal Design Pressure 3.500 bars
 External Design Temperature 0 °C
 Maximum Allowable Working Pressure 4 bars
 Shop Test Pressure 6.486 bars
 Field Test Pressure 5.040 bars
 Required Minimum Design Metal Temperature -45.0 °C
 Warmest Computed Minimum Design Metal Temperature -45.0 °C
 Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+IT
Head	SA-516 70	...	...	K02700	+IT
Nozzle	SA-350 LF2	1	...	K03011	+IT
Nozzle	SA-516 70	...	...	K02700	+IT
Nozzle	SA-333 3	...	...	K31918	+IT
Nozzle Flg	SA-350 LF2	1	...	K03011	+IT
Nozzle Flg	SA-105				
Hrz Bolting	SA-36	...	...	K02600	
Saddles	SA-516 70	...	...	K02700	+IT

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Left Dish	3.530	0.00	4.000	3.0000	No	No
S1	3.530	0.00	5.000	3.0000	N/A	No
S2	3.530	0.00	5.000	3.0000	N/A	No
S3	3.530	0.00	5.000	3.0000	N/A	No
S4	3.530	0.00	5.000	3.0000	N/A	No
S5	3.530	0.00	5.000	3.0000	N/A	No
Right Dish	3.530	0.00	4.000	3.0000	No	No

Liquid Level: 450.00 mm. Dens.: 0.001 kg./cm3 Sp. Gr.: 0.682

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
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Vessel Design Summary:

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Ellipse	50.0	50.0	16.0	12.0	11.1	...	1.00	1.00
Cylinder	3050.0	3000.0	14.0	14.0	10.7	...	1.00	1.00
Cylinder	6050.0	3000.0	14.0	14.0	10.7	...	1.00	1.00
Cylinder	7450.0	1400.0	14.0	14.0	10.7	...	1.00	1.00
Cylinder	10450.0	3000.0	14.0	14.0	10.7	...	1.00	1.00
Cylinder	13450.0	3000.0	14.0	14.0	10.7	...	1.00	1.00
Ellipse	13500.0	50.0	16.0	12.0	11.1	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
Trunnion	WRC-107/537	1.013	n/a	Failed
Trunnion	WRC-107/537	1.033	n/a	Failed
N1A-34in	WRC-297	0.933	Shell	Passed
N2-42in	WRC-297	1.045	Shell	Failed
N1B-34in	WRC-297	0.933	Shell	Passed
N3-4in	WRC-297	0.773	Shell	Passed

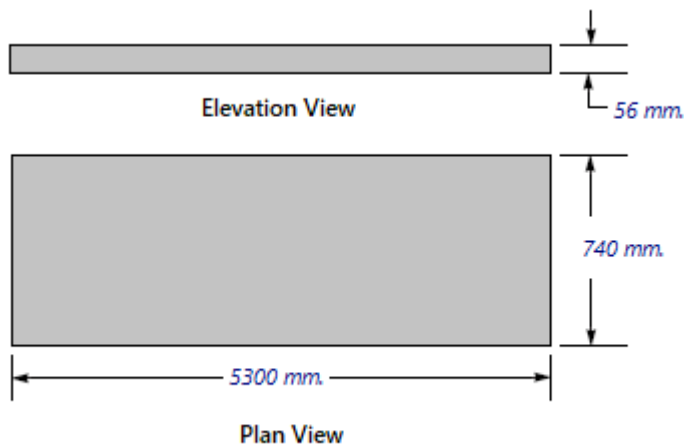
Foundation/Support Summary

Saddle Parameters:

Saddle Width	350.000 mm.
Saddle Bearing Angle	132.000 deg.
Centerline Dimension	3300.000 mm.
Wear Pad Width	900.000 mm.
Wear Pad Thickness	14.000 mm.
Wear Pad Bearing Angle	144.000 deg.
Distance from Saddle to Tangent	2700.000 mm.

Baseplate Length	5300.000 mm.
Baseplate Thickness	56.000 mm.
Baseplate Width	740.000 mm.
Number of Ribs (including outside ribs)	6
Rib Thickness	28.000 mm.
Web Thickness	28.000 mm.
Height of Center Web	216.000 mm.
Number of Bolts in Baseplate	4

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	49741.40 Kg
Maximum Transverse Saddle Shear Load	2948.12 Kg

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Maximum Longitudinal Saddle Shear Load	5896.23	Kgf
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Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	259911.72	Kgf
------------------------------	-----------	-----

Maximum Transverse Saddle Shear Load	1364.92	Kgf
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Maximum Longitudinal Saddle Shear Load	2024.73	Kgf
--	---------	-----

Weights:

Fabricated - Bare w/o removable internals	60152.7	kg.
---	---------	-----

Shop Test - Fabricated + water (full)	498215.8	kg.
---	----------	-----

Shipping - Fab. + rem. intls.+ shipping app.	60152.7	kg.
--	---------	-----

Erected - Fab. + rem. intls.+ insul. (etc)	75663.1	kg.
--	---------	-----

Empty - Fab. + intls. + details + wghts.	75663.1	kg.
--	---------	-----

Operating - Empty + operating liquid (no ca)	85112.6	kg.
--	---------	-----

Field Test - Empty weight + test medium (full)	513726.2	kg.
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**Nozzle Calculation Summary (bars & mm.):**

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
K1A-2in	4	...	...	OK	7.80	OK	No Calc[*]
K1B-2in	4	...	...	OK	7.80	OK	No Calc[*]
K2A-2in	4	...	...	OK	7.80	OK	No Calc[*]
K2B-2in	4	...	...	OK	7.80	OK	No Calc[*]
K3A-2in	4	...	...	OK	7.80	OK	No Calc[*]
K3B-2in	4	...	...	OK	7.80	OK	No Calc[*]
K4A-2in	4	...	...	OK	7.80	OK	No Calc[*]
K4B-2in	4	...	...	OK	7.80	OK	No Calc[*]
K5A-2in	4	...	...	OK	7.80	OK	No Calc[*]
K5B-2in	4	...	...	OK	7.80	OK	No Calc[*]
N1A-34in	5	...	...	OK	10.67	OK	Passed
N2-42in	5	...	...	OK	10.67	OK	Passed
N1B-34in	5	...	...	OK	10.67	OK	Passed
N3-4in	5	...	...	OK	8.26	OK	Passed
M1_24in	4	...	...	...		OK	Passed
K6-2in	4	...	...	OK	7.80	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 4 Nozzle : M124in
 Minimum MAWP Shells/Flanges : 4 Element : Right Dish
 Minimum MAPnc Shells/Flanges : 6 Element : Right Dish

Computed Vessel M.A.W.P. : 4 bars

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm.	Layout Angle deg	Dia. Limit mm.
10	K1A-2in	0.000	0.000	113.600
10	K1B-2in	0.000	15.000	113.600
10	K2A-2in	0.000	30.000	113.600
10	K2B-2in	0.000	45.000	113.600
10	K3A-2in	0.000	60.000	113.600
10	K3B-2in	0.000	75.000	113.600
10	K4A-2in	0.000	85.000	113.600
10	K4B-2in	0.000	95.000	113.600
10	K5A-2in	0.000	110.000	113.600
10	K5B-2in	0.000	130.000	113.600
20	N1A-34in	1300.000	0.000	1643.200
40	N2-42in	6750.000	0.000	2041.600
60	N1B-34in	12199.999	0.000	1643.200
60	N3-4in	12999.999	180.000	193.394
70	M1_24in	0.000	0.000	1122.920
70	K6-2in	0.000	0.000	113.600

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

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Nozzle Summary:

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If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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**Nozzle Schedule:**

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
K1A-2in	2.000 in	Actual	LWN	3.323	16.800	...	...	226.70	300
K1B-2in	2.000 in	Actual	LWN	3.323	16.800	...	...	226.70	300
K2A-2in	2.000 in	Actual	LWN	3.323	16.800	...	...	226.70	300
K2B-2in	2.000 in	Actual	LWN	3.323	16.800	...	...	226.70	300
K3A-2in	2.000 in	Actual	LWN	3.323	16.800	...	...	226.70	300
K3B-2in	2.000 in	Actual	LWN	3.323	16.800	...	...	226.70	300
K4A-2in	2.000 in	Actual	LWN	3.323	16.800	...	...	226.70	300
K4B-2in	2.000 in	Actual	LWN	3.323	16.800	...	...	226.70	300
K5A-2in	2.000 in	Actual	LWN	3.323	16.800	...	...	226.70	300
K5B-2in	2.000 in	Actual	LWN	3.323	16.800	...	...	226.70	300
K6-2in	2.000 in	Actual	LWN	3.323	16.800	...	...	226.70	300
N3-4in	4.000 in	160	WNF	4.500	13.487	...	...	178.69	300
M1_24in	24.000 in	80	WNF	24.000	30.937	...	...	376.29	150
N1A-34in	34.000 in	Actual	WNF	34.000	24.000	...	...	412.93	150
N1B-34in	34.000 in	Actual	WNF	34.000	24.000	...	...	412.93	150
N2-42in	42.000 in	Actual	WNF	42.000	26.000	...	...	407.40	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K1A-2in	SA-350 LF2	12.000	10.000	...	...	...
K1B-2in	SA-350 LF2	12.000	10.000	...	...	...
K2A-2in	SA-350 LF2	12.000	10.000	...	...	...
K2B-2in	SA-350 LF2	12.000	10.000	...	...	...
K3A-2in	SA-350 LF2	12.000	10.000	...	...	...
K3B-2in	SA-350 LF2	12.000	10.000	...	...	...
K4A-2in	SA-350 LF2	12.000	10.000	...	...	...
K4B-2in	SA-350 LF2	12.000	10.000	...	...	...
K5A-2in	SA-350 LF2	12.000	10.000	...	...	...
K5B-2in	SA-350 LF2	12.000	10.000	...	...	...
K6-2in	SA-350 LF2	12.000	10.000	...	...	...
N3-4in	SA-333 3	12.000	10.000	...	...	...
M1_24in	SA-333 3	12.000	10.000	...	...	...
N1A-34in	SA-516 70	12.000	10.000	...	...	...
N1B-34in	SA-516 70	12.000	10.000	...	...	...
N2-42in	SA-516 70	12.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
K1A-2in	...	0.0	206.38	0.00	Left Dish
K1B-2in	...	15.0	206.38	0.00	Left Dish

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K2A-2in	...	30.0	206.38	0.00	Left Dish
K2B-2in	...	45.0	206.38	0.00	Left Dish
K3A-2in	...	60.0	206.38	0.00	Left Dish
K3B-2in	...	75.0	206.38	0.00	Left Dish
K4A-2in	...	85.0	206.38	0.00	Left Dish
K4B-2in	...	95.0	206.38	0.00	Left Dish
K5A-2in	...	110.0	206.38	0.00	Left Dish
K5B-2in	...	130.0	206.38	0.00	Left Dish
K6-2in	...	0.0	206.38	0.00	Right Dish
N3-4in	12999.999	180.0	164.15	0.00	S5
M1_24in	...	0.0	353.70	0.00	Right Dish
N1A-34in	1300.000	0.0	353.70	0.00	S1
N1B-34in	12199.999	0.0	353.70	0.00	S5
N2-42in	6750.000	0.0	331.60	0.00	S3

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Bill of Material:

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 16.0mm. THK X 6000.0mm. ID X 50.0mm.	SA-516 70
4	CYLINDER: 14.0mm. THK X 6000.0mm. ID X 3000.0mm.	SA-516 70
1	CYLINDER: 14.0mm. THK X 6000.0mm. ID X 1400.0mm.	SA-516 70
2	SADDLE: 350.0mm X 132 DEG	SA-516 70
1	PLATFORM: 1000mm X 6028mm WIDE	
1	LADDER: 7000.0mm	...
4	TRUNNIONS:	SA-516 70
4	TRUNNION PADS: 1400.0mm OD X 650.0mm ID X 14.0mm THK	
1	CLASS 150 GR 1.1, 24.0" BLIND FLANGE(S)	SA-350 LF2
1	NAMEPLATE	...

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Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
Left Dish	[10]	!	-45	-48		0.938	12.000	1.00	Yes
Left Dish	[7]	!	-45	-64		0.651	16.000	1.00	Yes
S1	[8]	!	-45	-58		0.770	14.000	1.00	Yes
S2	[8]	!	-45	-58		0.770	14.000	1.00	Yes
S3	[8]	!	-45	-58		0.770	14.000	1.00	Yes
S4	[8]	!	-45	-58		0.770	14.000	1.00	Yes
S5	[8]	!	-45	-58		0.770	14.000	1.00	Yes
Right Dish	[10]	!	-45	-48		0.938	12.000	1.00	Yes
Right Dish	[7]	!	-45	-64		0.651	16.000	1.00	Yes
K1A-2in	[1]	!	-46	-46		0.761	12.000	1.00	Yes
Nozzle Flg	[5]	!	-46	-46					
K1B-2in	[1]	!	-46	-46		0.761	12.000	1.00	Yes
Nozzle Flg	[5]	!	-46	-46					
K2A-2in	[1]	!	-46	-46		0.761	12.000	1.00	Yes
Nozzle Flg	[5]	!	-46	-46					
K2B-2in	[1]	!	-46	-46		0.761	12.000	1.00	Yes
Nozzle Flg	[5]	!	-46	-46					
K3A-2in	[1]	!	-46	-46		0.761	12.000	1.00	Yes
Nozzle Flg	[5]	!	-46	-46					
K3B-2in	[1]	!	-46	-46		0.761	12.000	1.00	Yes
Nozzle Flg	[5]	!	-46	-46					
K4A-2in	[1]	!	-46	-46		0.761	12.000	1.00	Yes
Nozzle Flg	[5]	!	-46	-46					
K4B-2in	[1]	!	-46	-46		0.761	12.000	1.00	Yes
Nozzle Flg	[5]	!	-46	-46					
K5A-2in	[1]	!	-46	-46		0.761	12.000	1.00	Yes
Nozzle Flg	[5]	!	-46	-46					
K5B-2in	[1]	!	-46	-46		0.761	12.000	1.00	Yes
Nozzle Flg	[5]	!	-46	-46					
N1A-34in	[1]	!	-45	-104		0.052	24.000	1.00	Yes
Nozzle Flg	[5]	!	-46	-104					
N2-42in	[1]	!	-45	-104		0.059	26.000	1.00	Yes
Nozzle Flg	[5]	!	-46	-104					
N1B-34in	[1]	!	-45	-104		0.052	24.000	1.00	Yes
Nozzle Flg	[5]	!	-46	-104					
N3-4in	[1]	!	-101	-101		0.699	14.000	1.00	Yes
Nozzle Flg	[5]	!	-46	-104					
M124in	[1]	!	-101	-101		0.623	14.000	1.00	Yes
Nozzle Flg	[5]	!	-46	-104					
K6-2in	[1]	!	-46	-46		0.761	12.000	1.00	Yes
Nozzle Flg	[5]	!	-45	-45					
Nozl. Bolting[22]			-48			1.000	64.000	1.00	Yes

Warmest MDMT: -45 -45

Required Minimum Design Metal Temperature -45.0 °C

Warmest Computed Minimum Design Metal Temperature -45.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3

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MDMT Summary:

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- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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V6501 ABC

Ammonia Process Flare KOD

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Units used in this Analysis (barsUniCode):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm2
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm2
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm3
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm2
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m2
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm3
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

V6501 ABC

Ammonia Process Flare KOD

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Design Internal Pressure (for Hydrotest)	3.5	bars
Design Internal Temperature	270.0	°C
Projection of Nozzle from Vessel Top	500	mm.
Projection of Nozzle from Vessel Bottom	500	mm.
Minimum Design Metal Temperature	-45.0	°C
Type of Construction	Welded	
Special Service	Lethal	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS



Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW
Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	0 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.
Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	1.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000
Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No



Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	Left Dish
Distance "FROM" to "TO"	50 mm.
Inside Diameter	6000 mm.
Element Thickness	12 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	16 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	3.5 bars
Design Temperature Internal Pressure	270 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	137.2 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm.
Yield Stress, Operating	211.2 N./mm ²
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	10
Detail Type	Liquid
Detail ID	Liquid 10
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	450 mm.
Liquid Density	0.0006809 kg./cm ³

Element From Node	10
Detail Type	Nozzle
Detail ID	K1A-2in
Dist. from "FROM" Node / Offset dist	1000 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	10.562 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT

Element From Node	10
Detail Type	Nozzle
Detail ID	K1B-2in
Dist. from "FROM" Node / Offset dist	1000 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300



Layout Angle	15.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	10.562	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT

Element From Node	10	
Detail Type	Nozzle	
Detail ID	K2A-2in	
Dist. from "FROM" Node / Offset dist	1000	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	30.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	10.562	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT

Element From Node	10	
Detail Type	Nozzle	
Detail ID	K2B-2in	
Dist. from "FROM" Node / Offset dist	1000	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	45.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	10.562	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT

Element From Node	10	
Detail Type	Nozzle	
Detail ID	K3A-2in	
Dist. from "FROM" Node / Offset dist	1000	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	60.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	10.562	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT

Element From Node	10	
Detail Type	Nozzle	
Detail ID	K3B-2in	
Dist. from "FROM" Node / Offset dist	1000	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	75.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	10.562	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT

Element From Node	10	
Detail Type	Nozzle	
Detail ID	K4A-2in	
Dist. from "FROM" Node / Offset dist	1000	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	85.0	
Blind Flange (Y/N)	N	



Weight of Nozzle (Used if > 0)	10.562	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT

Element From Node	10	
Detail Type	Nozzle	
Detail ID	K4B-2in	
Dist. from "FROM" Node / Offset dist	1000	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	95.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	10.562	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT

Element From Node	10	
Detail Type	Nozzle	
Detail ID	K5A-2in	
Dist. from "FROM" Node / Offset dist	1000	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	110.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	10.562	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT

Element From Node	10	
Detail Type	Nozzle	
Detail ID	K5B-2in	
Dist. from "FROM" Node / Offset dist	1000	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	130.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	10.562	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT

Element From Node	10	
Detail Type	Weight	
Detail ID	Wear plate 1	
Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	50	Kgf
Offset from Element Centerline	1500	mm.

Element#	2/7	
Element From Node	20	
Element To Node	30	
Element Type	Cylinder	
Description	S1	
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	6000	mm.
Element Thickness	14	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	14	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	3.5	bars
Design Temperature Internal Pressure	270	°C
Design External Pressure	0	bars
Design Temperature External Pressure	0	°C



Effective Diameter Multiplier	1.2	
Material Name	SA-516 70 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	20	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	2650	mm.
Width of Saddle	350	mm.
Height of Saddle at Bottom	3300	mm.
Saddle Contact Angle	132.0	
Height of Composite Ring Stiffener	0	mm.
Width of Wear Plate	900	mm.
Thickness of Wear Plate	14	mm.
Contact Angle, Wear Plate (degrees)	144.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm.
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm.
Element From Node	20	
Detail Type	Platform	
Detail ID	PLAT:[1 OF 1]	
Dist. from "FROM" Node / Offset dist	0	mm.
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	135000	cm2
Platform Weight	14100	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm2
Platform Width	6028	mm.
Platform Height	1000	mm.
Platform Clearance or End Offset	105	mm.
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm.
Ladder End Elevation	7000	mm.
Unit Weight of Ladder	0.037	Kgf/mm.
Platform Length (top head platform)	13500	mm.
Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid 20	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	450	mm.
Liquid Density	0.0006809	kg./cm3
Element From Node	20	
Detail Type	Nozzle	
Detail ID	N1A-34in	
Dist. from "FROM" Node / Offset dist	1250	mm.
Nozzle Diameter	34	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	357.39	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70 +IT	
Element From Node	20	
Detail Type	Weight	
Detail ID	N1A-Elbow	

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Dist. from "FROM" Node / Offset dist	1000	mm.
Miscellaneous Weight	560	Kgf
Offset from Element Centerline	1000	mm.

Element#	3/7	
Element From Node	30	
Element To Node	40	
Element Type	Cylinder	
Description	S2	
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	6000	mm.
Element Thickness	14	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	14	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	3.5	bars
Design Temperature Internal Pressure	270	°C
Design External Pressure	0	bars
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid 30	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	450	mm.
Liquid Density	0.0006809	kg./cm ³

Element#	4/7	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	S3	
Distance "FROM" to "TO"	1400	mm.
Inside Diameter	6000	mm.
Element Thickness	14	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	14	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	3.5	bars
Design Temperature Internal Pressure	270	°C
Design External Pressure	0	bars
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid 40	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	450	mm.
Liquid Density	0.0006809	kg./cm ³

Element From Node	40
Detail Type	Nozzle
Detail ID	N2-42in

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Dist. from "FROM" Node / Offset dist	700	mm.
Nozzle Diameter	42	in.
Nozzle Schedule	STD	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	893.9	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	+IT

Element From Node	40	
Detail Type	Weight	
Detail ID	N1AB-Insert Pla	
Dist. from "FROM" Node / Offset dist	700	mm.
Miscellaneous Weight	515	Kgf
Offset from Element Centerline	0	mm.

Element From Node	40	
Detail Type	Weight	
Detail ID	N2-insert plate	
Dist. from "FROM" Node / Offset dist	0.0005822	mm.
Miscellaneous Weight	395	Kgf
Offset from Element Centerline	0	mm.

Element From Node	40	
Detail Type	Weight	
Detail ID	N1AB Internal E	
Dist. from "FROM" Node / Offset dist	0.0005822	mm.
Miscellaneous Weight	1270	Kgf
Offset from Element Centerline	0	mm.

Element#	5/7	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	S4	
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	6000	mm.
Element Thickness	14	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	14	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	3.5	bars
Design Temperature Internal Pressure	270	°C
Design External Pressure	0	bars
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid 50	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	450	mm.
Liquid Density	0.0006809	kg./cm3

Element From Node	50	
Detail Type	Weight	
Detail ID	Clip Wt	
Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	130	Kgf
Offset from Element Centerline	0	mm.



Element#	6/7
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	S5
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	6000 mm.
Element Thickness	14 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	14 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	3.5 bars
Design Temperature Internal Pressure	270 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	60
Detail Type	Saddle
Detail ID	New Saddle
Dist. from "FROM" Node / Offset dist	350 mm.
Width of Saddle	350 mm.
Height of Saddle at Bottom	3300 mm.
Saddle Contact Angle	132.0
Height of Composite Ring Stiffener	0 mm.
Width of Wear Plate	900 mm.
Thickness of Wear Plate	14 mm.
Contact Angle, Wear Plate (degrees)	144.0
Friction coefficient	0.30000001
Moment Factor	3.0
Dimension E at base (optional)	0 mm.
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm.

Element From Node	60
Detail Type	Liquid
Detail ID	Liquid 60
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	450 mm.
Liquid Density	0.0006809 kg./cm3

Element From Node	60
Detail Type	Nozzle
Detail ID	N1B-34in
Dist. from "FROM" Node / Offset dist	1750 mm.
Nozzle Diameter	34 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	357.39 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70 +IT

Element From Node	60
Detail Type	Nozzle
Detail ID	N3-4in
Dist. from "FROM" Node / Offset dist	2550 mm.
Nozzle Diameter	4 in.

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Nozzle Schedule	160
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	31.743 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-333 3 +IT

Element From Node	60
Detail Type	Weight
Detail ID	N1B-Elbow
Dist. from "FROM" Node / Offset dist	2000 mm.
Miscellaneous Weight	560 Kg
Offset from Element Centerline	1500 mm.

Element#	7/7
Element From Node	70
Element To Node	80
Element Type	Elliptical
Description	Right Dish
Distance "FROM" to "TO"	50 mm.
Inside Diameter	6000 mm.
Element Thickness	12 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	16 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	3.5 bars
Design Temperature Internal Pressure	270 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	70
Detail Type	Liquid
Detail ID	Liquid 70
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	450 mm.
Liquid Density	0.0006809 kg./cm ³

Element From Node	70
Detail Type	Nozzle
Detail ID	M124in
Dist. from "FROM" Node / Offset dist	0 mm.
Nozzle Diameter	24 in.
Nozzle Schedule	80
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	503.35 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-333 3 +IT

Element From Node	70
Detail Type	Nozzle
Detail ID	K6-2in
Dist. from "FROM" Node / Offset dist	1000 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0

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Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	10.562 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT

Element From Node	70
Detail Type	Weight
Detail ID	M1Davit24"
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	50 Kgf
Offset from Element Centerline	1500 mm.

Element From Node	70
Detail Type	Weight
Detail ID	Wear Plate 2
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	50 Kgf
Offset from Element Centerline	300 mm.

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V6501 ABC

Ammonia Process Flare KOD

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FileName : Ammonia Flare KOD - V6501 ABC -7%

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XY Coordinate Calculations: Step: 2 2:48pm Sep 16,2026



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XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
Left Dish		50	...	50	...
S1		3050	...	3000	...
S2		6050	...	3000	...
S3		7450	...	1400	...
S4		10450	...	3000	...
S5		13450	...	3000	...
Right Dish		13500	...	50	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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V6501 ABC

Ammonia Process Flare KOD

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FileName : Ammonia Flare KOD - V6501 ABC -7%

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Internal Pressure Calculations: Step: 3 2:48pm Sep 16,2026



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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
Left Dish		3.5303	16	3	6000	137.2	1
	S1	3.5303	14	3	6000	137.2	2
	S2	3.5303	14	3	6000	137.2	3
	S3	3.5303	14	3	6000	137.2	4
	S4	3.5303	14	3	6000	137.2	5
	S5	3.5303	14	3	6000	137.2	6
Right Dish		3.5303	16	3	6000	137.2	7

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
Left Dish		3.5	4.08555	5.51735	12	11.1213	1
	S1	3.5	4.98397	6.42151	14	10.7395	2
	S2	3.5	4.98397	6.42151	14	10.7395	3
	S3	3.5	4.98397	6.42151	14	10.7395	4
	S4	3.5	4.98397	6.42151	14	10.7395	5
	S5	3.5	4.98397	6.42151	14	10.7395	6
Right Dish		3.5	4.08555	5.51735	12	11.1213	7

Minimum

4

6

MAWP: 4 bars, limited by: Nozzle Reinforcement.

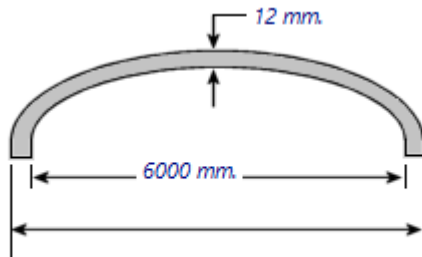
Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Left Dish

Elliptical Head Design Information:					
Design Pressure	P	3.530 bars	Int. Design Temperature		270 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	137.2 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	6000.000 mm.	Actual Stress		117.7 N./mm^2
Spec. Min. Thk t _e , or s _a		12.0000 mm.	Spec. Nominal Thk.		16.000 mm.
Surface Area		403883.4 cm^2	SG of Contents		0.682
Element Volume		29693370.0 cm3	Weight of Contents		324.4 Kg
Empty Weight		5488.8 Kg	Operating Weight		5813.3 Kg
			Comp. All. Ope. Stress		41.4 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 75.000 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 37.500 mm.

Material treatment options: +IT

Appendix 1-4(f) Calculations ($t_s/L = 0.00167$)

Note: Please check the temperature limit given in Table 1-4.3 of the code.

If the max. design temp. exceeds the temp. limit, see U-2(g).

Note: Calculation of L and r is from ASME VIII, Division 2, 4.3.7.1

$$r = D(0.5/k - 0.08) \text{ and } L = D(0.44k + 0.02) \quad k = D/2h$$

$$L = 5403.000 : r = 1023.0001 : r/D = 0.1703 : C1 = 0.7229 : C2 = 1.0171$$

Required Thickness Calculation:

Final iteration:



Elastic Buckling Stress (Se):

$$= C1 \cdot Et \cdot (ts/r)$$

$$= (0.723 \cdot 187242.906 \cdot 0.008)$$

$$= 1074.523 \text{ N./mm}^2$$

$$a = 0.5 \cdot D - r = 1980.000 \text{ mm.}$$

$$b = L - r = 4380.000 \text{ mm.}$$

$$\text{Beta} = \text{COS}(A/B) = 1.102 \text{ rad.}$$

$$\text{Ph1} = \text{SQRT}(L \cdot ts) / r = 0.205 \text{ rad.}$$

$$c = a / \text{COS}(\text{Beta}-\text{Phi}) = 3173.151 \text{ mm.}$$

$$\text{Re} = c + r = 4196.151 \text{ mm.}$$

Buckling Internal Pressure (Pe):

$$= (Se \cdot ts) / (C2 \cdot \text{Re} \cdot ((0.5 \cdot \text{Re} / r) - 1))$$

$$= (1074.5 \cdot 8.121) / (1.017 \cdot 4196.151 \cdot ((0.5 \cdot 4196.151 / 1023.0) - 1))$$

$$= 19.455 \text{ bars}$$

Yield Internal Pressure (Py):

$$= (Sy \cdot ts) / (C2 \cdot \text{Re} \cdot ((0.5 \cdot \text{Re} / r) - 1))$$

$$= (211.2 \cdot 8.121) / (1.017 \cdot 4196.151 \cdot ((0.5 \cdot 4196.151 / 1023.0) - 1))$$

$$= 3.824 \text{ bars}$$

Knuckle Failure Internal Pressure (Pck):

$$= 0.408 \cdot Py + 0.192 \cdot Pe$$

$$= 0.408 \cdot 3.824 + 0.192 \cdot 19.455$$

$$= 5.295 \text{ bars}$$

Allowable Pressure (Pa):

$$= Pck / 1.5$$

$$= 5.295 / 1.5$$

$$= 3.530 \text{ bars}$$

App 1-4(f) Calculated Required Thick. (TR) : 8.1213 mm.

Note: Calculation of L and r is from ASME VIII, Division 2, 4.3.7.1

$$r = D(0.5/k - 0.08) \text{ and } L = D(0.44k + 0.02) \quad k = D/2h$$

MAWP Calculation (ts/L = 0.00167)

Elastic Buckling Stress (Se):

$$= C1 \cdot Et \cdot (ts/r)$$

$$= (0.723 \cdot 187242.906 \cdot 0.009)$$

$$= 1190.779 \text{ N./mm}^2$$

$$a = 0.5 \cdot D - r = 1980.000 \text{ mm.}$$

$$b = L - r = 4380.000 \text{ mm.}$$

$$\text{Beta} = \text{COS}(A/B) = 1.102 \text{ rad.}$$

$$\text{Ph1} = \text{SQRT}(L \cdot ts) / r = 0.216 \text{ rad.}$$

$$c = a / \text{COS}(\text{Beta}-\text{Phi}) = 3131.016 \text{ mm.}$$

$$\text{Re} = c + r = 4154.016 \text{ mm.}$$

Buckling Internal Pressure (Pe):

$$= (Se \cdot ts) / (C2 \cdot \text{Re} \cdot ((0.5 \cdot \text{Re} / r) - 1))$$

$$= (1190.8 \cdot 9.0) / (1.017 \cdot 4154.016 \cdot ((0.5 \cdot 4154.016 / 1023.0) - 1))$$

$$= 24.617 \text{ bars}$$

Yield Internal Pressure (Py):

$$= (Sy \cdot ts) / (C2 \cdot \text{Re} \cdot ((0.5 \cdot \text{Re} / r) - 1))$$

$$= (211.2 \cdot 9.0) / (1.017 \cdot 4154.016 \cdot ((0.5 \cdot 4154.016 / 1023.0) - 1))$$

$$= 4.366 \text{ bars}$$

Knuckle Failure Internal Pressure (Pck):

$$= 0.408 \cdot Py + 0.192 \cdot Pe$$

$$= 0.408 \cdot 4.366 + 0.192 \cdot 24.617$$

V6501 ABC

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Element: Left Dish

Step: 4 2:48pm Sep 16,2026



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= 6.508 bars

Maximum Allowable Working Pressure (MAWP):

Less Operating Hydrostatic Head Pressure of 0.03 bars

= Pck / 1.5

= 6.508/1.5

= 4.339 - 0.030 = 4.308 bars

Straight Flange Required Thickness:

= 10.740 mm.

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 1.167 %

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 75.000 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 37.500 mm.

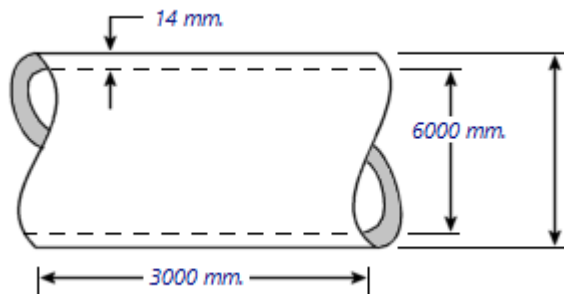
Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: S1

Cylindrical Shell Design Information:					
Design Pressure	P	3.530 bars	Int. Design Temperature		270 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	137.2 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	6000.000 mm.	Actual Stress		96.6 N./mm^2
Spec. Min. Thk t _e , or s _a		14.0000 mm.	Spec. Nominal Thk.		14.000 mm.
Surface Area		568125.6 cm^2	SG of Contents		0.682
Element Volume		84838200.0 cm3	Weight of Contents		1970.3 Kg
Empty Weight		6150.2 Kg	Operating Weight		8120.5 Kg
			Comp. All. Ope. Stress		50.5 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (3.53 \cdot 3003.0) / (137.2 \cdot 1.0 - 0.6 \cdot 3.53)$$

$$= 7.7395 + 3.0000 = 10.7395 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.030 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.2 \cdot 1.0 \cdot 11.0) / (3003.0 + 0.6 \cdot 11.0)$$

$$= 5 - 0 = 5 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 14.0) / (3000.0 + 0.6 \cdot 14.0)$$

$$= 6 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (3.53 \cdot (3003.0 + 0.6 \cdot 11.0)) / (1.0 \cdot 11.0)$$

$$= 96.594 \text{ N./mm}^2$$

V6501 ABC

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FileName : Ammonia Flare KOD - V6501 ABC -7%

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Element: S1

Step: 5 2:48pm Sep 16,2026

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% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.233 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

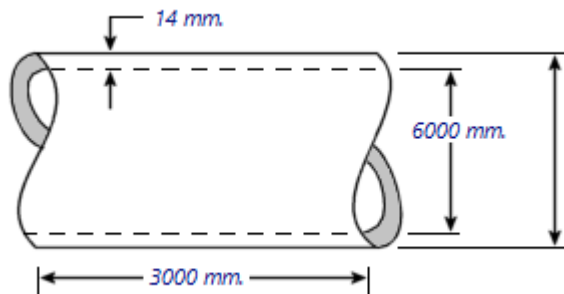
Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: S2

Cylindrical Shell Design Information:					
Design Pressure	P	3.530 bars	Int. Design Temperature		270 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	137.2 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	6000.000 mm.	Actual Stress		96.6 N./mm^2
Spec. Min. Thk t _e , or s _a		14.0000 mm.	Spec. Nominal Thk.		14.000 mm.
Surface Area		568125.6 cm^2	SG of Contents		0.682
Element Volume		84838200.0 cm3	Weight of Contents		1970.3 Kg
Empty Weight		6150.2 Kg	Operating Weight		8120.5 Kg
			Comp. All. Ope. Stress		50.5 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (3.53 \cdot 3003.0) / (137.2 \cdot 1.0 - 0.6 \cdot 3.53)$$

$$= 7.7395 + 3.0000 = 10.7395 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.030 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.2 \cdot 1.0 \cdot 11.0) / (3003.0 + 0.6 \cdot 11.0)$$

$$= 5 - 0 = 5 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 14.0) / (3000.0 + 0.6 \cdot 14.0)$$

$$= 6 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (3.53 \cdot (3003.0 + 0.6 \cdot 11.0)) / (1.0 \cdot 11.0)$$

$$= 96.594 \text{ N./mm}^2$$

V6501 ABC

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Element: S2

Step: 6 2:48pm Sep 16,2026

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% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.233 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

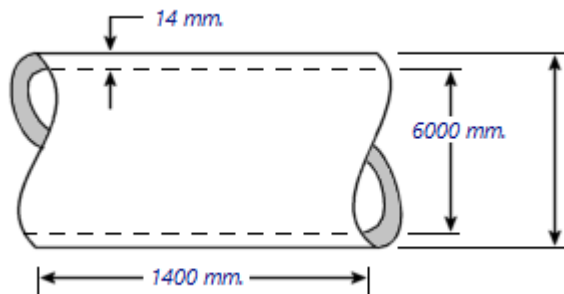
Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: S3

Cylindrical Shell Design Information:					
Design Pressure	P	3.530 bars	Int. Design Temperature		270 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		1400.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	137.2 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	6000.000 mm.	Actual Stress		96.6 N./mm^2
Spec. Min. Thk t _e , or s _a		14.0000 mm.	Spec. Nominal Thk.		14.000 mm.
Surface Area		265125.3 cm^2	SG of Contents		0.682
Element Volume		39591164.0 cm3	Weight of Contents		919.5 Kg
Empty Weight		2870.1 Kg	Operating Weight		3789.5 Kg
			Comp. All. Ope. Stress		50.5 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (3.53 \cdot 3003.0) / (137.2 \cdot 1.0 - 0.6 \cdot 3.53)$$

$$= 7.7395 + 3.0000 = 10.7395 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.030 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.2 \cdot 1.0 \cdot 11.0) / (3003.0 + 0.6 \cdot 11.0)$$

$$= 5 - 0 = 5 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 14.0) / (3000.0 + 0.6 \cdot 14.0)$$

$$= 6 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (3.53 \cdot (3003.0 + 0.6 \cdot 11.0)) / (1.0 \cdot 11.0)$$

$$= 96.594 \text{ N./mm}^2$$

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Element: S3

Step: 7 2:48pm Sep 16,2026

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% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.233 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

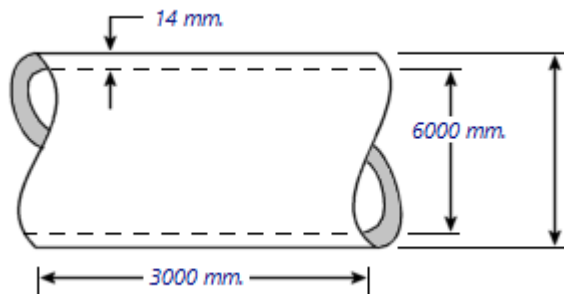
Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: S4

Cylindrical Shell Design Information:					
Design Pressure	P	3.530 bars	Int. Design Temperature		270 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	137.2 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	6000.000 mm.	Actual Stress		96.6 N./mm^2
Spec. Min. Thk t _e , or s _a		14.0000 mm.	Spec. Nominal Thk.		14.000 mm.
Surface Area		568125.6 cm^2	SG of Contents		0.682
Element Volume		84838200.0 cm3	Weight of Contents		1970.3 Kg
Empty Weight		6150.2 Kg	Operating Weight		8120.5 Kg
			Comp. All. Ope. Stress		50.5 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (3.53 \cdot 3003.0) / (137.2 \cdot 1.0 - 0.6 \cdot 3.53)$$

$$= 7.7395 + 3.0000 = 10.7395 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.030 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.2 \cdot 1.0 \cdot 11.0) / (3003.0 + 0.6 \cdot 11.0)$$

$$= 5 - 0 = 5 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 14.0) / (3000.0 + 0.6 \cdot 14.0)$$

$$= 6 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (3.53 \cdot (3003.0 + 0.6 \cdot 11.0)) / (1.0 \cdot 11.0)$$

$$= 96.594 \text{ N./mm}^2$$

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Element: S4

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% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.233 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

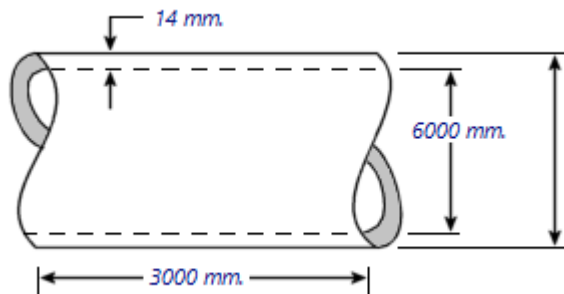
Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: S5

Cylindrical Shell Design Information:					
Design Pressure	P	3.530 bars	Int. Design Temperature		270 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	137.2 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	6000.000 mm.	Actual Stress		96.6 N./mm^2
Spec. Min. Thk t _e , or s _a		14.0000 mm.	Spec. Nominal Thk.		14.000 mm.
Surface Area		568125.6 cm^2	SG of Contents		0.682
Element Volume		84838200.0 cm3	Weight of Contents		1970.3 Kg
Empty Weight		6150.2 Kg	Operating Weight		8120.5 Kg
			Comp. All. Ope. Stress		50.5 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (3.53 \cdot 3003.0) / (137.2 \cdot 1.0 - 0.6 \cdot 3.53)$$

$$= 7.7395 + 3.0000 = 10.7395 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.030 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.2 \cdot 1.0 \cdot 11.0) / (3003.0 + 0.6 \cdot 11.0)$$

$$= 5 - 0 = 5 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 14.0) / (3000.0 + 0.6 \cdot 14.0)$$

$$= 6 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (3.53 \cdot (3003.0 + 0.6 \cdot 11.0)) / (1.0 \cdot 11.0)$$

$$= 96.594 \text{ N./mm}^2$$

V6501 ABC

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% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.233 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

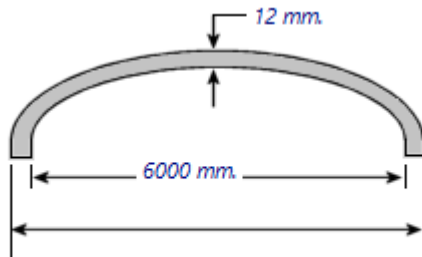
Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Right Dish

Elliptical Head Design Information:					
Design Pressure	P	3.530 bars	Int. Design Temperature		270 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	137.2 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	6000.000 mm.	Actual Stress		117.7 N./mm^2
Spec. Min. Thk t _e , or s _a		12.0000 mm.	Spec. Nominal Thk.		16.000 mm.
Surface Area		403883.4 cm^2	SG of Contents		0.682
Element Volume		29693370.0 cm3	Weight of Contents		324.4 Kg
Empty Weight		5488.8 Kg	Operating Weight		5813.3 Kg
			Comp. All. Ope. Stress		50.5 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 75.000 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 37.500 mm.

Material treatment options: +IT

Appendix 1-4(f) Calculations ($t_s/L = 0.00167$)

Note: Please check the temperature limit given in Table 1-4.3 of the code.

If the max. design temp. exceeds the temp. limit, see U-2(g).

Note: Calculation of L and r is from ASME VIII, Division 2, 4.3.7.1

$$r = D(0.5/k - 0.08) \text{ and } L = D(0.44k + 0.02) \quad k = D/2h$$

$$L = 5403.000 : r = 1023.0001 : r/D = 0.1703 : C1 = 0.7229 : C2 = 1.0171$$

Required Thickness Calculation:

Final iteration:



Elastic Buckling Stress (Se):

$$= C1 \cdot Et \cdot (ts/r)$$

$$= (0.723 \cdot 187242.906 \cdot 0.008)$$

$$= 1074.523 \text{ N./mm}^2$$

$$a = 0.5 \cdot D - r = 1980.000 \text{ mm.}$$

$$b = L - r = 4380.000 \text{ mm.}$$

$$\text{Beta} = \text{COS}(A/B) = 1.102 \text{ rad.}$$

$$\text{Ph1} = \text{SQRT}(L \cdot ts) / r = 0.205 \text{ rad.}$$

$$c = a / \text{COS}(\text{Beta}-\text{Phi}) = 3173.151 \text{ mm.}$$

$$\text{Re} = c + r = 4196.151 \text{ mm.}$$

Buckling Internal Pressure (Pe):

$$= (Se \cdot ts) / (C2 \cdot \text{Re} \cdot ((0.5 \cdot \text{Re} / r) - 1))$$

$$= (1074.5 \cdot 8.121) / (1.017 \cdot 4196.151 \cdot ((0.5 \cdot 4196.151 / 1023.0) - 1))$$

$$= 19.455 \text{ bars}$$

Yield Internal Pressure (Py):

$$= (Sy \cdot ts) / (C2 \cdot \text{Re} \cdot ((0.5 \cdot \text{Re} / r) - 1))$$

$$= (211.2 \cdot 8.121) / (1.017 \cdot 4196.151 \cdot ((0.5 \cdot 4196.151 / 1023.0) - 1))$$

$$= 3.824 \text{ bars}$$

Knuckle Failure Internal Pressure (Pck):

$$= 0.408 \cdot Py + 0.192 \cdot Pe$$

$$= 0.408 \cdot 3.824 + 0.192 \cdot 19.455$$

$$= 5.295 \text{ bars}$$

Allowable Pressure (Pa):

$$= Pck / 1.5$$

$$= 5.295 / 1.5$$

$$= 3.530 \text{ bars}$$

App 1-4(f) Calculated Required Thick. (TR) : 8.1213 mm.

Note: Calculation of L and r is from ASME VIII, Division 2, 4.3.7.1

$$r = D(0.5/k - 0.08) \text{ and } L = D(0.44k + 0.02) \quad k = D/2h$$

MAWP Calculation (ts/L = 0.00167)

Elastic Buckling Stress (Se):

$$= C1 \cdot Et \cdot (ts/r)$$

$$= (0.723 \cdot 187242.906 \cdot 0.009)$$

$$= 1190.779 \text{ N./mm}^2$$

$$a = 0.5 \cdot D - r = 1980.000 \text{ mm.}$$

$$b = L - r = 4380.000 \text{ mm.}$$

$$\text{Beta} = \text{COS}(A/B) = 1.102 \text{ rad.}$$

$$\text{Ph1} = \text{SQRT}(L \cdot ts) / r = 0.216 \text{ rad.}$$

$$c = a / \text{COS}(\text{Beta}-\text{Phi}) = 3131.016 \text{ mm.}$$

$$\text{Re} = c + r = 4154.016 \text{ mm.}$$

Buckling Internal Pressure (Pe):

$$= (Se \cdot ts) / (C2 \cdot \text{Re} \cdot ((0.5 \cdot \text{Re} / r) - 1))$$

$$= (1190.8 \cdot 9.0) / (1.017 \cdot 4154.016 \cdot ((0.5 \cdot 4154.016 / 1023.0) - 1))$$

$$= 24.617 \text{ bars}$$

Yield Internal Pressure (Py):

$$= (Sy \cdot ts) / (C2 \cdot \text{Re} \cdot ((0.5 \cdot \text{Re} / r) - 1))$$

$$= (211.2 \cdot 9.0) / (1.017 \cdot 4154.016 \cdot ((0.5 \cdot 4154.016 / 1023.0) - 1))$$

$$= 4.366 \text{ bars}$$

Knuckle Failure Internal Pressure (Pck):

$$= 0.408 \cdot Py + 0.192 \cdot Pe$$

$$= 0.408 \cdot 4.366 + 0.192 \cdot 24.617$$

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Element: Right Dish

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= 6.508 bars

Maximum Allowable Working Pressure (MAWP):

Less Operating Hydrostatic Head Pressure of 0.03 bars

= Pck / 1.5

= 6.508/1.5

= 4.339 - 0.030 = 4.308 bars

Straight Flange Required Thickness:

= 10.740 mm.

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 1.167 %

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 75.000 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 37.500 mm.

Material treatment options: +IT

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**Hydrostatic Test Pressure Results:**

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	5.040 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	4.576 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6.486 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	4.265 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	5.005 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	5.005 bars
App 27-4	= M.A.W.P.	3.854 bars

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Left Dish	176.9	234.0	0.756	7.07
S1	152.0	234.0	0.650	7.07
S2	152.0	234.0	0.650	7.07
S3	152.0	234.0	0.650	7.07
S4	152.0	234.0	0.650	7.07
S5	152.0	234.0	0.650	7.07
Right Dish	176.9	234.0	0.756	7.07

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
K1A-2in	Nozzle	138.00	133.20	1.036
K1B-2in	Nozzle	138.00	133.20	1.036
K2A-2in	Nozzle	138.00	133.20	1.036
K2B-2in	Nozzle	138.00	133.20	1.036
K3A-2in	Nozzle	138.00	133.20	1.036
K3B-2in	Nozzle	138.00	133.20	1.036
K4A-2in	Nozzle	138.00	133.20	1.036
K4B-2in	Nozzle	138.00	133.20	1.036
K5A-2in	Nozzle	138.00	133.20	1.036
K5B-2in	Nozzle	138.00	133.20	1.036
N1A-34in	Nozzle	138.00	137.20	1.006
N2-42in	Nozzle	138.00	137.20	1.006
N1B-34in	Nozzle	138.00	137.20	1.006
N3-4in	Nozzle	128.00	126.40	1.013
M1_24in	Nozzle	128.00	126.40	1.013
K6-2in	Nozzle	138.00	133.20	1.036

Minimum

1.006

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Left Dish	138.00	137.20	1.006
S1	138.00	137.20	1.006
S2	138.00	137.20	1.006
S3	138.00	137.20	1.006
S4	138.00	137.20	1.006
S5	138.00	137.20	1.006
Right Dish	138.00	137.20	1.006

Minimum

1.006

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Shop Pressure Test Results:

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Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
K1A-2in	1.49	225.00	0.007
K1B-2in	1.49	225.00	0.007
K2A-2in	1.49	225.00	0.007
K2B-2in	1.49	225.00	0.007
K3A-2in	1.49	225.00	0.007
K3B-2in	1.49	225.00	0.007
K4A-2in	1.49	225.00	0.007
K4B-2in	1.49	225.00	0.007
K5A-2in	1.49	225.00	0.007
K5B-2in	1.49	225.00	0.007
N1A-34in	12.45	234.00	0.053
N2-42in	14.23	234.00	0.061
N1B-34in	12.45	234.00	0.053
N3-4in	3.14	216.00	0.015
M1 24in	7.68	216.00	0.036
K6-2in	1.49	225.00	0.007

Elements Suitable for Test Pressure.

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**Hydrostatic Test Pressure Results:**

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	5.040 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	4.576 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6.486 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	4.265 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	5.005 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	5.005 bars
App 27-4	= M.A.W.P.	3.854 bars

UG-99(b), Test Pressure Calculation:

= Test Factor • MAWP • Stress Ratio

= 1.3 • 3.854 • 1.006

= 5.040 bars

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Left Dish	189.2	234.0	0.809	5.68
S1	155.3	234.0	0.664	5.68
S2	155.3	234.0	0.664	5.68
S3	155.3	234.0	0.664	5.68
S4	155.3	234.0	0.664	5.68
S5	155.3	234.0	0.664	5.68
Right Dish	189.2	234.0	0.809	5.68

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
K1A-2in	Nozzle	138.00	133.20	1.036
K1B-2in	Nozzle	138.00	133.20	1.036
K2A-2in	Nozzle	138.00	133.20	1.036
K2B-2in	Nozzle	138.00	133.20	1.036
K3A-2in	Nozzle	138.00	133.20	1.036
K3B-2in	Nozzle	138.00	133.20	1.036
K4A-2in	Nozzle	138.00	133.20	1.036
K4B-2in	Nozzle	138.00	133.20	1.036
K5A-2in	Nozzle	138.00	133.20	1.036
K5B-2in	Nozzle	138.00	133.20	1.036
N1A-34in	Nozzle	138.00	137.20	1.006
N2-42in	Nozzle	138.00	137.20	1.006
N1B-34in	Nozzle	138.00	137.20	1.006
N3-4in	Nozzle	128.00	126.40	1.013
M1_24in	Nozzle	128.00	126.40	1.013
K6-2in	Nozzle	138.00	133.20	1.036

Minimum

1.006

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Left Dish	138.00	137.20	1.006
S1	138.00	137.20	1.006
S2	138.00	137.20	1.006
S3	138.00	137.20	1.006
S4	138.00	137.20	1.006

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Field Pressure Test Results:

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S5	138.00	137.20	1.006
Right Dish	138.00	137.20	1.006

Minimum

1.006

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
K1A-2in	1.20	225.00	0.005
K1B-2in	1.20	225.00	0.005
K2A-2in	1.20	225.00	0.005
K2B-2in	1.20	225.00	0.005
K3A-2in	1.20	225.00	0.005
K3B-2in	1.20	225.00	0.005
K4A-2in	1.20	225.00	0.005
K4B-2in	1.20	225.00	0.005
K5A-2in	1.20	225.00	0.005
K5B-2in	1.20	225.00	0.005
N1A-34in	9.99	234.00	0.043
N2-42in	11.42	234.00	0.049
N1B-34in	9.99	234.00	0.043
N3-4in	2.52	216.00	0.012
M1_24in	6.17	216.00	0.029
K6-2in	1.20	225.00	0.005

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	6024	9	0.0002075	20.7457
20	30	14500	6028	11	0.41978E-04	4.19684
30	40	14500	6028	11	0.41978E-04	4.19684
40	50	14500	6028	11	0.41978E-04	4.19684
50	60	14500	6028	11	0.41978E-04	4.19684
60	70	14500	6028	11	0.41978E-04	4.19684
70	80	No Calc	6024	9	0.0002075	20.7457

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	12	No Calc	...	0.34436
20	30	14	No Calc	...	0.10211
30	40	14	No Calc	...	0.10211
40	50	14	No Calc	...	0.10211
50	60	14	No Calc	...	0.10211
60	70	14	No Calc	...	0.10211
70	80	12	No Calc	...	0.34436

Minimum

0

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	14500	No Calc	No Calc	No Calc
30	40	14500	No Calc	No Calc	No Calc
40	50	14500	No Calc	No Calc	No Calc
50	60	14500	No Calc	No Calc	No Calc
60	70	14500	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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V6501 ABC

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Element and Detail Weights:

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm3	Corroded Metal Wgt. kg.	Corroded ID Volume cm3	Extra due Misc % kg.
10	20	5488.82	29693370	4459.67	29781124	548.882
20	30	6150.17	84838200	4834.62	85007968	615.017
30	40	6150.17	84838200	4834.62	85007968	615.017
40	50	2870.06	39591164	2256.17	39670384	287.006
50	60	6150.17	84838200	4834.62	85007968	615.017
60	70	6150.17	84838200	4834.62	85007968	615.017
70	80	5488.82	29693370	4459.67	29781124	548.882
Total		38448	0.438E+09	30513	0.439E+09	3844

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
10	Liqd	324.449	-500	2775	...	Liquid 10
10	Nozl	11.6183	-1414.21	1000	...	K1A-2in
10	Nozl	11.6183	-1414.21	965.926	258.819	K1B-2in
10	Nozl	11.6183	-1414.21	866.025	500	K2A-2in
10	Nozl	11.6183	-1414.21	707.107	707.107	K2B-2in
10	Nozl	11.6183	-1414.21	500	866.025	K3A-2in
10	Nozl	11.6183	-1414.21	258.819	965.926	K3B-2in
10	Nozl	11.6183	-1414.21	87.1557	996.195	K4A-2in
10	Nozl	11.6183	-1414.21	-87.1557	996.195	K4B-2in
10	Nozl	11.6183	-1414.21	-342.02	939.693	K5A-2in
10	Nozl	11.6183	-1414.21	-642.788	766.044	K5B-2in
10	Wght	50	-1414.21	1500	...	Wear plate 1
20	Sadl	4952.79	2650	3136	...	Left Saddle
20	Plat	15510.4	6750	3090.2	...	PLAT:[1 OF 1]
20	Liqd	1970.29	1500	2775	...	Liquid 20
20	Nozl	393.128	1250	3431.8	...	N1A-34in
20	Wght	560	1000	1000	...	N1A-Elbow
30	Liqd	1970.29	1500	2775	...	Liquid 30
40	Liqd	919.457	700	2775	...	Liquid 40
40	Nozl	983.29	700	3533.4	...	N2-42in
40	Wght	515	700	...	...	N1AB-Insert Pla
40	Wght	395	0.00058217	...	...	N2-insert plate
40	Wght	1270	0.00058217	...	...	N1AB Internal E
50	Liqd	1970.29	1500	2775	...	Liquid 50
50	Wght	130	...	...	...	Clip Wt
60	Sadl	4952.79	350	3136	...	New Saddle
60	Liqd	1970.29	1500	2775	...	Liquid 60
60	Nozl	393.128	1750	3431.8	...	N1B-34in
60	Nozl	34.9173	2550	-3057.15	...	N3-4in
60	Wght	560	2000	1500	...	N1B-Elbow
70	Liqd	324.449	550	2775	...	Liquid 70
70	Nozl	553.689	1550	...	...	M1_24in
70	Nozl	11.6183	1464.21	1000	...	K6-2in
70	Wght	50	1464.21	1500	...	M1_Davit_24"
70	Wght	50	1464.21	...	300	Wear Plate 2
20	Trun	943.96	950	...	...	Trunnions
60	Trun	943.96	2050	...	...	Trunnions

Total Weight of Each Detail Type:

Saddles 9905.6
 Platforms 15510.4
 Liquid 9449.5

V6501 ABC

Ammonia Process Flare KOD

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Element and Detail Weights: Step: 14 2:48pm Sep 16,2026



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Nozzles	2486.0
Weights	3580.0
Trunnions	1887.9

Sum of the Detail Weights	42819.4 kg.
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Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	42293.2	42293.2	42293.2	42293.2	42293.2	42293.2
Saddles	9905.6	9905.6	9905.6	9905.6	9905.6	9905.6
Nozzles	2486.0	2486.0	2486.0	2486.0	2486.0	2486.0
Wld Weights	3580.0	3580.0	3580.0	3580.0	3580.0	3580.0
Trunnions	1887.9	1887.9	1887.9	1887.9	1887.9	1887.9
Platforms	...	...	...	15510.4	15510.4	15510.4
Ope. Liquid	...	...	...	...	...	9449.5
Test Liquid	...	438063.2	...	...	...	...
Totals	60152.7	498215.8	60152.7	75663.1	75663.1	85112.6

Field Installation Options:

Platforms installed before lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	60152.7 kg.
Shop Test Wt.	- Fabricated weight + water (full)	498215.8 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	60152.7 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	75663.1 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	75663.1 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	85112.6 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	76384.8 kg.
Field Test Wt.	- Empty weight + test medium (full)	513726.2 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm^2	Inside Surface Area cm^2
10	20	403883	399660
20	30	568126	565487
30	40	568126	565487
40	50	265125	263894
50	60	568126	565487
60	70	568126	565487
70	80	403883	399660

Total	3345394.5	3325160.0 cm2
Total	3601.0	3579.2 ft²

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Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Op.	Ambient					UG-44	50%	DNV
K1A-2in	41.16	51.00	270	300	GR 1.1	...	...	...	...
K1B-2in	41.16	51.00	270	300	GR 1.1	...	...	...	...
K2A-2in	41.16	51.00	270	300	GR 1.1	...	...	...	...
K2B-2in	41.16	51.00	270	300	GR 1.1	...	...	...	...
K3A-2in	41.16	51.00	270	300	GR 1.1	...	...	...	...
K3B-2in	41.16	51.00	270	300	GR 1.1	...	...	...	...
K4A-2in	41.16	51.00	270	300	GR 1.1	...	...	...	...
K4B-2in	41.16	51.00	270	300	GR 1.1	...	...	...	...
K5A-2in	41.16	51.00	270	300	GR 1.1	...	...	...	...
K5B-2in	41.16	51.00	270	300	GR 1.1	...	...	...	...
N1A-34in	11.34	19.60	270	150	GR 1.1	...	...	...	...
N2-42in	11.34	19.60	270	150	GR 1.1	...	...	...	...
N1B-34in	11.34	19.60	270	150	GR 1.1	...	...	...	...
N3-4in	41.16	51.00	270	300	GR 1.1	38.210	23.531	41.160	23.531
M1 24in	11.34	19.60	270	150	GR 1.1	...	...	...	...
K6-2in	41.16	51.00	270	300	GR 1.1	...	...	...	...
Min Rating	11.340	19.600, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm.	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kgf	MAWP PD bars	Maximum Pressure Pmax bars	Pass Fail
N3-4in	138.176	0.50	202	0	3.854	23.531	Pass

Evaluating Flange on Nozzle [N3-4in]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(202) + 4 \cdot \text{abs}(0) \cdot 138.176 \leq \pi \cdot 138.176^3 \cdot ((41.160 - 3.854) + 0.500 \cdot 41.160)$$

$$= 0.32294E+04 \leq 0.48925E+04 \quad \text{Check Passes}$$

Maximum Pressure for Flange on Nozzle N3-4in [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(41.160, 0.500 \cdot 41.160 + 41.160 - 3229.4 / (\pi \cdot 138.176^3))$$

$$= \min(41.160, 23.531)$$

$$= 23.531 \text{ bars}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	0 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.228 \cdot 0.957) / (1 + 1.7 \cdot 3.4 \cdot 0.228)))$
 $= \min(0.85, 0.903)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.526
Structure Height to Diameter ratio	2.546

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$; $Zg = 274320.0$ mm.

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 3300.0 + 0.0 = 3300.0$ mm.
 $= 10.827$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (10.827 ft.) < 15 ft.
 $= 2.01(15 / Zg)^{2/\alpha}$
 $= 2.01(15 / 274320.0)^{2/9.5}$
 $= 0.849$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1$, per Table 26.6-1



As there is No Hill Present: [Kzt]:

$K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:

$= (1 + K1 \cdot K2 \cdot K3)^2$

$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$

$= 1.0$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$

$= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$

$= 16.0 \text{ psf } [78.12] \text{ Kgs/m}^2$

Force on the first element [F]:

$= qz \cdot G \cdot Cf \cdot \text{WindArea}$

$= 16.0 \cdot 0.85 \cdot 0.526 \cdot 80.892$

$= 578.4 \text{ lbs. } [262.4] \text{ Kg}$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Left Dish	3300.0	0.000	0.000	0.000	0.849	1.000	78.120
S1	3300.0	0.000	0.000	0.000	0.849	1.000	78.120
S2	3300.0	0.000	0.000	0.000	0.849	1.000	78.120
S3	3300.0	0.000	0.000	0.000	0.849	1.000	78.120
S4	3300.0	0.000	0.000	0.000	0.849	1.000	78.120
S5	3300.0	0.000	0.000	0.000	0.849	1.000	78.120
Right Dish	3300.0	0.000	0.000	0.000	0.849	1.000	78.120

Platform Load Calculations

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf	Cf
PLAT:[1 OF 1]	135000.02	3300.00	78.12	1054.62	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf
Wear plate 1	0.00	4800.00	78.12	0.00
N1A-Elbow	0.00	4300.00	78.12	0.00
N1AB-Insert P	0.00	3300.00	78.12	0.00
N2-insert pla	0.00	3300.00	78.12	0.00
N1AB Internal	0.00	3300.00	78.12	0.00
Clip Wt	0.00	3300.00	78.12	0.00
N1B-Elbow	0.00	4800.00	78.12	0.00
M1Davit24"	0.00	4800.00	78.12	0.00
Wear Plate 2	0.00	3300.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	3300	7228.8	75150.7	78.12	288.599
20	30	3300	7233.6	217008	78.12	1993.45
30	40	3300	7233.6	217008	78.12	833.369
40	50	3300	7233.6	101270	78.12	388.905
50	60	3300	7233.6	217008	78.12	833.369
60	70	3300	7233.6	217008	78.12	833.369
70	80	3300	7228.8	75150.7	78.12	288.599

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Wind Load Calculation:

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The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	1.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}
 SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\
 SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\
 SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\
 SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104
 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot hn^x$ where $Ct = 0.020$, $x = 0.75$ and hn = Structural Height (ft.)
 $= 0.020(20.66930.75)$
 $= 0.194$ seconds

The Coefficient Cu from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/33.0)$
 $= 0.030$

Check the Value of T which is the smaller of $Cu \cdot Ta$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.194, 0.03) \text{ per 12.8.2}$
 $= 0.030$

Loads on Building/Structure Supported Equipment:

$$\begin{aligned}
 &\text{Seismic Force per equation 13.3-1, [Fp]:} \\
 &= 0.4 \cdot Ap \cdot Sds \cdot W(1 + 2 \cdot (z/h)) / (R / Ie) \\
 &= 0.4 \cdot 1.0 \cdot 0.157 \cdot 75207(1 + 2 \cdot 1.0) / (3.0/1.25) \\
 &= 5896.232 \text{ Kgf}
 \end{aligned}$$

Maximum value of Fp per equation 13.3-2:
 $= 1.6 \cdot Sds \cdot I \cdot W$
 $= 1.6 \cdot 0.157 \cdot 1.25 \cdot 75207 = 23584.93 \text{ Kgf}$

Minimum value of Fp per equation 13.3-3:
 $= 0.3 \cdot 0.16 \cdot 1.25 \cdot 75207$
 $= 4422.175 \text{ Kgf}$

Total Base Shear $V = Fp$, [V]:
 $= 5896.232 \text{ Kgf}$

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.157 \cdot 75207$

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Earthquake Load Calculation:

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= 2358.5 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	3000	8356.34	655.137
20	30	3000	8356.34	655.137
20	30	3000	8356.34	655.137
30	40	3000	8356.34	655.137
40	50	3000	8356.34	655.137
50	60	3000	8356.34	655.137
60	0	3000	8356.34	655.137
60	70	3000	8356.34	655.137
70	80	3000	8356.34	655.137

The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Platform(s) installed in the Field before being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	6750.000 mm.
Center of Gravity of the Platforms	6800.000 mm.
Center of Gravity of the Liquid	6750.000 mm.
Center of Gravity of the Nozzles	8331.880 mm.
Center of Gravity of the Added Weights (Operating)	6563.884 mm.
Center of Gravity of the Added Weights (Empty)	6563.884 mm.
Center of Gravity of the Trunnions	6750.000 mm.
Center of Gravity of Bare Shell New and Cold	6749.999 mm.
Center of Gravity of Bare Shell Corroded	6750.000 mm.
Vessel CG in the Operating Condition	6802.913 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	6804.297 mm.
Vessel CG in the Test Condition	6756.555 mm.

Rigging Analysis Results:

- Lifting weight based on the user selected calculated erected weight

Total Effective Length of Vessel for this analysis	13500.00 mm.
Total vessel weight (No Liquid)	Twt 75663.08 Kgf
Impact weight multiplication factor	Imp 1.50
Design lifting weight, DWT = Imp * Twt	113494.62 Kgf
Distance to the Lug on the Left End	1000.00 mm.
Distance to the Lug on the Right End	12500.00 mm.
Design Reaction force at the Left End Trunnion	56211.44 Kgf
Design Reaction force at the Right End Trunnion	57283.17 Kgf
CG distance during field lifting operation	6804.30 mm.
CG distance from the left end lug/trunnion	5804.30 mm.
CG distance from the nearer lifting lug/trunnion	5695.70 mm.

Compressive Allowable Increase Factor 1.00

Critical Values: (N./mm², mm.)

	Max Stress	Elevation	Allowables
Bending	1.80	6890.00	58.05 (UG-23)
Shear	1.48	12250.00	96.60 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm.	Bending Moment Kg-m.	Bending Stress N./mm ²	Shear Force Kgf	Shear Stress N./mm ²
0.00	0.0	0.0	21858.3	0.9
650.00	19992.3	0.5	28583.9	1.1
4250.00	50406.0	1.2	7898.9	0.3
6890.00	72672.0	1.8	7233.7	0.3
9850.00	45543.1	1.1	22366.3	0.8
13450.00	7717.8	0.2	1681.4	0.1

Unity Check (Actual Stress / Allowable Stress):

Maximum Unity Check is 0.0309 at elevation 6890.0 mm. - Must be <=1

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Center of Gravity Calculation: Step: 18 2:48pm Sep 16,2026



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Note: The rigging analysis is performed using a uniformly distributed load.

--- Plot data successfully generated ...----

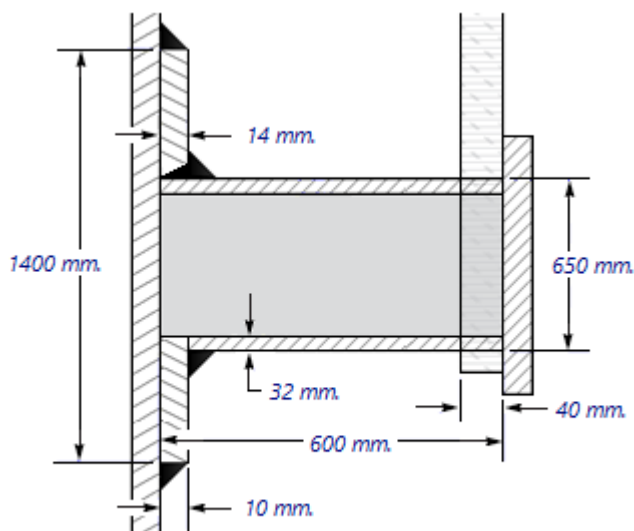
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Trunnion Analysis: Left End of Vessel

Input Data:

Lifting Lug Material	SA-516 70
Lifting Lug Yield Stress	Yield 262.00 N./mm ²
Trunnion Outside Diameter	Tod 650.000 mm.
Trunnion Thickness	Tthk 32.0000 mm.
Trunnion Length	Tl 600.0000 mm.
Reinforcing Pad Used	Yes
Pad Outside Diameter	Dp 1400.000 mm.
Pad Thickness	Thkp 14.0000 mm.
Pad Edge Fillet Weld Thk	Tfp 10.0000 mm.
Bail/Sling Width	Tt 40.0000 mm.
Sling Angle from Horizontal	90.0 degrees
Occasional Load Factor	Occfac 1.0
Impact Factor	Impfac 1.50
Number of Lugs in Group	2



Trunnion may be inserted or abutting

Trunnion Calculation Results:

Trunnion Lifting Load [Tlf]:
 = Resultant Load • Impact Factor
 = 18737.1 • 1.5
 = 28105.7 Kg

Trunnion Section Modulus [Z]:
 = $\pi (Tod^4 - Tid^4) / (32 \cdot Tod)$
 = $\pi (650.0^4 - 586.0^4) / (32 \cdot 650.0)$
 = 9150711.0 mm.³

Trunnion Bending Moment [Tmom]:
 = $(Tl - 0.5 \cdot Tt) / 9.807 \cdot Tlf$
 = $(600.0 - 0.5 \cdot 40.0) / 9.807 \cdot 28105.72$
 = 16301.6 Kg-m.

Trunnion Calculated Bending Stress [Sbc]:
 = $Tmom \cdot 9.807 / Z$
 = $16301.64 \cdot 9.807 / 9150711$
 = 17.47 N./mm [Passed] Sbc < min(0.66 • SY • Occfac, 0.75 • Sy) (172.92)

Trunnion Cross Section Area [Txsa]:



$$= \pi/4 (T_{od}^2 - T_{id}^2)$$

$$= \pi/4 (650.0^2 - 586.0^2)$$

$$= 621.3 \text{ cm}^2$$

Trunnion Calculated Shear Stress [Ssc]:

$$= T_{lf}/T_{xsa}$$

$$= 28105.72/621.28$$

$$= 4.44 \text{ N./mm [Passed] } Ssc < 0.4 \cdot S_y \cdot Occfac (104.80)$$

Trunnion Unity Check [Uchk]:

$$= S_{bc}/S_{ba} + S_{sc}/S_{sa}$$

$$= 17.47/172.92 + 4.44/104.8$$

$$= 0.143 \text{ [Passed]}$$

Weld Section Modulus at Edge of Trunnion [Sw]:

$$= \pi (T_{od}/2)^2$$

$$= \pi (650.0/2)^2$$

$$= 331830.5 \text{ mm.}^2$$

Shear Stress in Weld due to Bending Moment [fs]:

$$= T_{mom}/S_w$$

$$= 16301.6/331830.5$$

$$= 49.125 \text{ Kgf/mm.}$$

Required Fillet Weld Leg Size at Edge of Trunnion [Wsr]:

$$= \max(10 \text{ mm}, fs/(0.4 \cdot S_y))$$

$$= \max(10 \text{ mm}, 49.125/(0.4 \cdot 262.0))$$

$$= 10.000 \text{ mm.}$$

Trunnion Bearing Contact Area [Tbca]:

$$= T_{od} \cdot T_t$$

$$= 650.0 \cdot 40.0$$

$$= 260.0 \text{ cm}^2$$

Trunnion Bearing Stress [Sbec]:

$$= T_{lf}/T_{bca}$$

$$= 28105.72/260.0$$

$$= 10.60 \text{ N./mm [Passed] } S_{bec} < \min(0.75 \cdot S_Y \cdot Occfac, 0.9 \cdot S_y) (196.50)$$

Summary of Results:

Trunnion Unity Check Value: 0.1434 Must be less than or equal to 1

Comparison of Stress Values versus Allowables: (N./mm²)

Type of Stress	Actual	Allowable	Pass/Fail
Bending Stress	17.47	172.92	Pass
Shear Stress	4.44	104.80	Pass
Bearing Stress	10.60	196.50	Pass

Results of local shell stress analysis due to loads on the trunnion:

This following Stress Analysis assumes that the Trunnion passes through the pad and is welded to the shell via a groove weld. Additionally the re-pad is welded to the shell at the outside edge via a fillet weld.

In the local stress results below, the 'Nozzle' is the 'Trunnion'.

WRC 107/537 Stress Analysis for the Lifting Lug to Shell Junction in the new and Cold Condition (no corrosion applied).

Input Echo, WRC107/537 Item 1, Description: Trunnion

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Cylindrical

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Trunnion Calcs: Left End

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Internal Corrosion Allowance	Cas	0.0000	mm.
Vessel Diameter	Dv	6000.000	mm.
Vessel Thickness	Tv	14.000	mm.
Design Temperature	T1	37.8	C
Attachment Type	Type	Round	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	0.0000	mm.
Nozzle Diameter	Dn	650.000	mm.
Nozzle Thickness	Tn	32.000	mm.
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm2
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm2
Thickness of Reinforcing Pad	Tpad	14.000	mm.
Diameter of Reinforcing Pad	Dpad	1420.000	mm.
Design Internal Pressure	Dp	0.000	bars
Include Pressure Thrust		No	

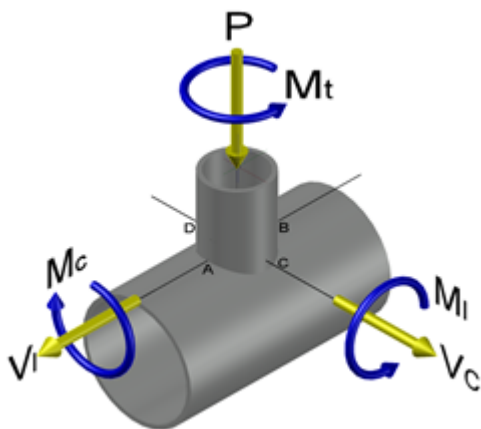
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	0.0	Kgf
Longitudinal Shear (SUS)	Vl	0.0	Kgf
Circumferential Shear (SUS)	Vc	28105.7	Kgf
Circumferential Moment (SUS)	Mc	16301.6	Kg-m.
Longitudinal Moment (SUS)	Ml	0.0	Kg-m.
Torsional Moment (SUS)	Mt	0.0	Kg-m.

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))} \\
 &= 650.0 + 2 \cdot 1.65 \cdot \sqrt{(3007.0 (14.0 - 0))} \\
 &= 1327.087 \text{ mm.}
 \end{aligned}$$



WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	28105.7	Kgf
Longitudinal Shear	VL	0.0	Kgf
Circumferential Moment	MC	16301.6	Kg-m.
Longitudinal Moment	ML	0.0	Kg-m.
Torsional Moment	MT	0.0	Kg-m.

Dimensionless Parameters used : Gamma = 107.64

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.094	4C	16.459	(A,B)
N(PHI) / (P/Rm)	0.094	3C	12.784	(C,D)
M(PHI) / (P)	0.094	2C1	0.055	(A,B)
M(PHI) / (P)	0.094	1C	0.089	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.094	3A	4.021	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.094	1A	0.083	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.094	3B	12.039	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.094	1B	0.033	(A,B,C,D)
N(x) / (P/Rm)	0.094	3C	12.784	(A,B)
N(x) / (P/Rm)	0.094	4C	16.459	(C,D)
M(x) / (P)	0.094	1C1	0.092	(A,B)
M(x) / (P)	0.094	2C	0.056	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.094	4A	6.684	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.094	2A	0.044	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.094	4B	4.072	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.094	2B	0.048	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-26.8	-26.8	26.8	26.8
Circ. Bend. MC		0.0	0.0	0.0	0.0	-359.0	359.0	359.0	-359.0
Circ. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Circ. Str.		0.0	0.0	0.0	0.0	-385.8	332.2	385.8	-332.2
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-44.5	-44.5	44.5	44.5
Long. Bend. MC		0.0	0.0	0.0	0.0	-187.2	187.2	187.2	-187.2
Long. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Long. Str.		0.0	0.0	0.0	0.0	-231.7	142.7	231.7	-142.7
Shear VC		9.6	9.6	-9.6	-9.6	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Shear MT		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear		9.6	9.6	-9.6	-9.6	0.0	0.0	0.0	0.0
Str. Int.		19.3	19.3	19.3	19.3	385.8	332.2	385.8	332.2

V6501 ABC

Ammonia Process Flare KOD

QUEP

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Trunnion Calcs: Left End

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Dimensionless Parameters used : Gamma = 214.79

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.207	4C	17.769	(A,B)
N(PHI) / (P/Rm)	0.207	3C	5.478	(C,D)
M(PHI) / (P)	0.207	2C1	0.009	(A,B)
M(PHI) / (P)	0.207	1C !	0.068	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.207	3A	5.133	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.207	1A	0.049	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.207	3B	10.760	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.207	1B	0.006	(A,B,C,D)
N(x) / (P/Rm)	0.207	3C	5.478	(A,B)
N(x) / (P/Rm)	0.207	4C	17.769	(C,D)
M(x) / (P)	0.207	1C1	0.025	(A,B)
M(x) / (P)	0.207	2C !	0.037	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.207	4A	17.561	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.207	2A	0.019	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.207	4B	4.885	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.207	2B	0.007	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-31.4	-31.4	31.4	31.4
Circ. Bend. MC		0.0	0.0	0.0	0.0	-388.1	388.1	388.1	-388.1
Circ. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Circ. Str.		0.0	0.0	0.0	0.0	-419.5	356.8	419.5	-356.8
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-107.3	-107.3	107.3	107.3
Long. Bend. MC		0.0	0.0	0.0	0.0	-149.2	149.2	149.2	-149.2
Long. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Long. Str.		0.0	0.0	0.0	0.0	-256.6	41.9	256.6	-41.9
Shear VC		8.8	8.8	-8.8	-8.8	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Shear MT		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear		8.8	8.8	-8.8	-8.8	0.0	0.0	0.0	0.0
Str. Int.		17.7	17.7	17.7	17.7	419.5	356.8	419.5	356.8

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N./mm2)**

V6501 ABC

Ammonia Process Flare KOD

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Trunnion Calcs: Left End

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Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		0.0	0.0	0.0	0.0	-26.8	-26.8	26.8	26.8
Circ. Q (SUS)		0.0	0.0	0.0	0.0	-359.0	359.0	359.0	-359.0
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		0.0	0.0	0.0	0.0	-44.5	-44.5	44.5	44.5
Long. Q (SUS)		0.0	0.0	0.0	0.0	-187.2	187.2	187.2	-187.2
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		9.6	9.6	-9.6	-9.6	0.0	0.0	0.0	0.0
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		19.3	19.3	19.3	19.3	44.5	44.5	44.5	44.5
Pm+Pl+Q (Total)		19.3	19.3	19.3	19.3	385.8	332.2	385.8	332.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	44.52	207.00	Passed
Pm+Pl+Q (TOTAL)	385.78	413.99	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		0.0	0.0	0.0	0.0	-31.4	-31.4	31.4	31.4
Circ. Q (SUS)		0.0	0.0	0.0	0.0	-388.1	388.1	388.1	-388.1
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		0.0	0.0	0.0	0.0	-107.3	-107.3	107.3	107.3
Long. Q (SUS)		0.0	0.0	0.0	0.0	-149.2	149.2	149.2	-149.2
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		8.8	8.8	-8.8	-8.8	0.0	0.0	0.0	0.0
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		17.7	17.7	17.7	17.7	107.3	107.3	107.3	107.3
Pm+Pl+Q (Total)		17.7	17.7	17.7	17.7	419.5	356.8	419.5	356.8

Vessel Stress Summation Comparison (N./mm²):

V6501 ABC

Ammonia Process Flare KOD

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Trunnion Calcs: Left End

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Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	107.34	207.00	Passed
Pm+Pl+Q (TOTAL)	419.52	413.99	Failed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

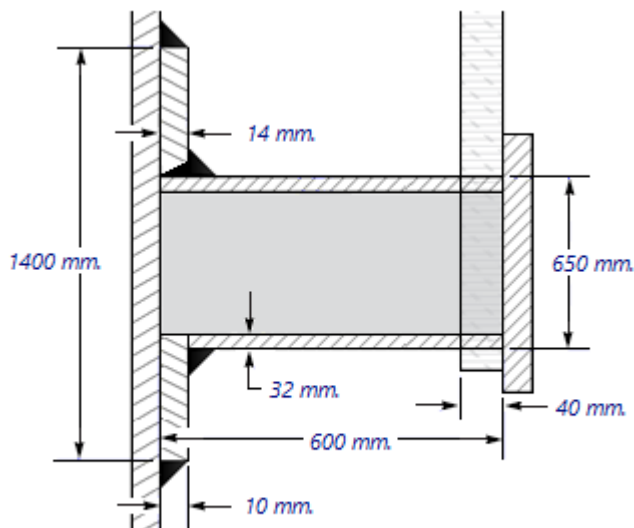
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Trunnion Analysis: Right End of Vessel

Input Data:

Lifting Lug Material	SA-516 70
Lifting Lug Yield Stress	Yield 262.00 N./mm ²
Trunnion Outside Diameter	Tod 650.000 mm.
Trunnion Thickness	Tthk 32.0000 mm.
Trunnion Length	Tl 600.0000 mm.
Reinforcing Pad Used	Yes
Pad Outside Diameter	Dp 1400.000 mm.
Pad Thickness	Thkp 14.0000 mm.
Pad Edge Fillet Weld Thk	Tfp 10.0000 mm.
Bail/Sling Width	Tt 40.0000 mm.
Occasional Load Factor	Occfac 1.0
Impact Factor	Impfac 1.50
Number of Lugs in Group	2



Trunnion may be inserted or abutting

Trunnion Calculation Results:

Sling Angle from Horizontal: 0.0 deg

Trunnion Lifting Load [Tlf]:
 = Resultant Load • Impact Factor
 = 19094.4 • 1.5
 = 28641.6 Kgf

Trunnion Section Modulus [Z]:
 = $\pi (Tod^4 - Tid^4) / (32 \cdot Tod)$
 = $\pi (650.0^4 - 586.0^4) / (32 \cdot 650.0)$
 = 9150711.0 mm.³

Trunnion Bending Moment [Tmom]:
 = $(Tl - 0.5 \cdot Tt) / 9.807 \cdot Tlf$
 = $(600.0 - 0.5 \cdot 40.0) / 9.807 \cdot 28641.59$
 = 16612.4 Kg-m.

Trunnion Calculated Bending Stress [Sbc]:
 = $Tmom \cdot 9.807 / Z$
 = $16612.45 \cdot 9.807 / 9150711$
 = 17.80 N/mm [Passed] $Sbc < \min(0.66 \cdot SY \cdot Occfac, 0.75 \cdot Sy) (172.92)$



Trunnion Cross Section Area [Txsa]:

$$= \pi/4 (T_{od}^2 - T_{id}^2)$$

$$= \pi/4 (650.0^2 - 586.0^2)$$

$$= 621.3 \text{ cm}^2$$

Trunnion Calculated Shear Stress [Ssc]:

$$= T_{lf}/T_{xsa}$$

$$= 28641.59/621.28$$

$$= 4.52 \text{ N./mm [Passed] } Ssc < 0.4 \cdot S_y \cdot Occfac (104.80)$$

Trunnion Unity Check [Uchk]:

$$= S_{bc}/S_{ba} + S_{sc}/S_{sa}$$

$$= 17.8/172.92 + 4.52/104.8$$

$$= 0.146 \text{ [Passed]}$$

Weld Section Modulus at Edge of Trunnion [Sw]:

$$= \pi (T_{od}/2)^2$$

$$= \pi (650.0/2)^2$$

$$= 331830.5 \text{ mm.}^2$$

Shear Stress in Weld due to Bending Moment [fs]:

$$= T_{mom}/S_w$$

$$= 16612.4/331830.5$$

$$= 50.062 \text{ Kg/mm.}$$

Required Fillet Weld Leg Size at Edge of Trunnion [Wsr]:

$$= \max(10 \text{ mm}, f_s/(0.4 \cdot S_y))$$

$$= \max(10 \text{ mm}, 50.062/(0.4 \cdot 262.0))$$

$$= 10.000 \text{ mm.}$$

Trunnion Bearing Contact Area [Tbca]:

$$= T_{od} \cdot T_t$$

$$= 650.0 \cdot 40.0$$

$$= 260.0 \text{ cm}^2$$

Trunnion Bearing Stress [Sbec]:

$$= T_{lf}/T_{bca}$$

$$= 28641.59/260.0$$

$$= 10.80 \text{ N./mm [Passed] } S_{bec} < \min(0.75 \cdot S_Y \cdot Occfac, 0.9 \cdot S_y) (196.50)$$

Summary of Results:

Trunnion Unity Check Value: 0.1461 Must be less than or equal to 1

Comparison of Stress Values versus Allowables: (N./mm²)

Type of Stress	Actual	Allowable	Pass/Fail
Bending Stress	17.80	172.92	Pass
Shear Stress	4.52	104.80	Pass
Bearing Stress	10.80	196.50	Pass

Results of local shell stress analysis due to loads on the trunnion:

This following Stress Analysis assumes that the Trunnion passes through the pad and is welded to the shell via a groove weld. Additionally the re-pad is welded to the shell at the outside edge via a fillet weld.

In the local stress results below, the 'Nozzle' is the 'Trunnion'.

WRC 107/537 Stress Analysis for the Lifting Lug to Shell Junction in the new and Cold Condition (no corrosion applied).

Input Echo, WRC107/537 Item 1, Description: Trunnion

Diameter Basis for Vessel

Vbasis

ID



Cylindrical or Spherical Vessel	Cylsph	Cylindrical
Internal Corrosion Allowance	Cas	0.0000 mm.
Vessel Diameter	Dv	6000.000 mm.
Vessel Thickness	Tv	14.000 mm.
Design Temperature	T1	37.8 C
Attachment Type	Type	Round
Diameter Basis for Nozzle	Nbasis	OD
Corrosion Allowance for Nozzle	Can	0.0000 mm.
Nozzle Diameter	Dn	650.000 mm.
Nozzle Thickness	Tn	32.000 mm.
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm2
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm2
Thickness of Reinforcing Pad	Tpad	14.000 mm.
Diameter of Reinforcing Pad	Dpad	1420.000 mm.
Design Internal Pressure	Dp	0.000 bars
Include Pressure Thrust	No	

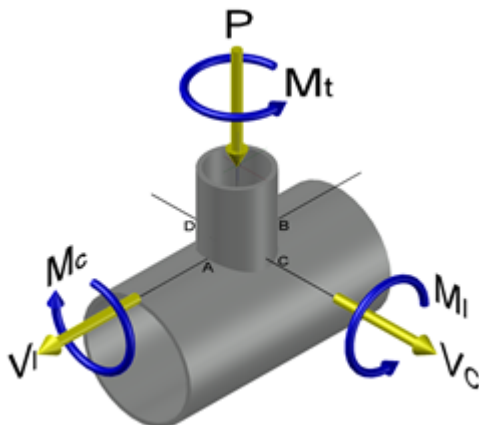
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	0.0	Kgf
Longitudinal Shear (SUS)	Vl	0.0	Kgf
Circumferential Shear (SUS)	Vc	28641.6	Kgf
Circumferential Moment (SUS)	Mc	16612.4	Kg-m.
Longitudinal Moment (SUS)	Ml	0.0	Kg-m.
Torsional Moment (SUS)	Mt	0.0	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))} \\
 &= 650.0 + 2 \cdot 1.65 \cdot \sqrt{(3007.0 (14.0 - 0))} \\
 &= 1327.087 \text{ mm.}
 \end{aligned}$$



WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	28641.6	Kgf
Longitudinal Shear	VL	0.0	Kgf
Circumferential Moment	MC	16612.4	Kg-m.
Longitudinal Moment	ML	0.0	Kg-m.
Torsional Moment	MT	0.0	Kg-m.

Dimensionless Parameters used : Gamma = 107.64

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.094	4C	16.459	(A,B)
N(PHI) / (P/Rm)	0.094	3C	12.784	(C,D)
M(PHI) / (P)	0.094	2C1	0.055	(A,B)
M(PHI) / (P)	0.094	1C	0.089	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.094	3A	4.021	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.094	1A	0.083	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.094	3B	12.039	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.094	1B	0.033	(A,B,C,D)
N(x) / (P/Rm)	0.094	3C	12.784	(A,B)
N(x) / (P/Rm)	0.094	4C	16.459	(C,D)
M(x) / (P)	0.094	1C1	0.092	(A,B)
M(x) / (P)	0.094	2C	0.056	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.094	4A	6.684	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.094	2A	0.044	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.094	4B	4.072	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.094	2B	0.048	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-27.3	-27.3	27.3	27.3
Circ. Bend. MC		0.0	0.0	0.0	0.0	-365.8	365.8	365.8	-365.8
Circ. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Circ. Str.		0.0	0.0	0.0	0.0	-393.1	338.5	393.1	-338.5
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-45.4	-45.4	45.4	45.4
Long. Bend. MC		0.0	0.0	0.0	0.0	-190.8	190.8	190.8	-190.8
Long. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Long. Str.		0.0	0.0	0.0	0.0	-236.1	145.4	236.1	-145.4
Shear VC		9.8	9.8	-9.8	-9.8	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Shear MT		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear		9.8	9.8	-9.8	-9.8	0.0	0.0	0.0	0.0
Str. Int.		19.7	19.7	19.7	19.7	393.1	338.5	393.1	338.5



Dimensionless Parameters used : Gamma = 214.79

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.207	4C	17.769	(A,B)
N(PHI) / (P/Rm)	0.207	3C	5.478	(C,D)
M(PHI) / (P)	0.207	2C1	0.009	(A,B)
M(PHI) / (P)	0.207	1C !	0.068	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.207	3A	5.133	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.207	1A	0.049	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.207	3B	10.760	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.207	1B	0.006	(A,B,C,D)
N(x) / (P/Rm)	0.207	3C	5.478	(A,B)
N(x) / (P/Rm)	0.207	4C	17.769	(C,D)
M(x) / (P)	0.207	1C1	0.025	(A,B)
M(x) / (P)	0.207	2C !	0.037	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.207	4A	17.561	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.207	2A	0.019	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.207	4B	4.885	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.207	2B	0.007	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-32.0	-32.0	32.0	32.0
Circ. Bend. MC		0.0	0.0	0.0	0.0	-395.5	395.5	395.5	-395.5
Circ. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Circ. Str.		0.0	0.0	0.0	0.0	-427.5	363.6	427.5	-363.6
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-109.4	-109.4	109.4	109.4
Long. Bend. MC		0.0	0.0	0.0	0.0	-152.1	152.1	152.1	-152.1
Long. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Long. Str.		0.0	0.0	0.0	0.0	-261.4	42.7	261.4	-42.7
Shear VC		9.0	9.0	-9.0	-9.0	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Shear MT		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear		9.0	9.0	-9.0	-9.0	0.0	0.0	0.0	0.0
Str. Int.		18.0	18.0	18.0	18.0	427.5	363.6	427.5	363.6

WRC 107/537 Stress Summations:

V6501 ABC

Ammonia Process Flare KOD

QUEP

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Trunnion Calcs: Right End

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Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		0.0	0.0	0.0	0.0	-27.3	-27.3	27.3	27.3
Circ. Q (SUS)		0.0	0.0	0.0	0.0	-365.8	365.8	365.8	-365.8
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		0.0	0.0	0.0	0.0	-45.4	-45.4	45.4	45.4
Long. Q (SUS)		0.0	0.0	0.0	0.0	-190.8	190.8	190.8	-190.8
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		9.8	9.8	-9.8	-9.8	0.0	0.0	0.0	0.0
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		19.7	19.7	19.7	19.7	45.4	45.4	45.4	45.4
Pm+Pl+Q (Total)		19.7	19.7	19.7	19.7	393.1	338.5	393.1	338.5

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	45.37	207.00	Passed
Pm+Pl+Q (TOTAL)	393.13	413.99	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		0.0	0.0	0.0	0.0	-32.0	-32.0	32.0	32.0
Circ. Q (SUS)		0.0	0.0	0.0	0.0	-395.5	395.5	395.5	-395.5
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		0.0	0.0	0.0	0.0	-109.4	-109.4	109.4	109.4
Long. Q (SUS)		0.0	0.0	0.0	0.0	-152.1	152.1	152.1	-152.1
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		9.0	9.0	-9.0	-9.0	0.0	0.0	0.0	0.0
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		18.0	18.0	18.0	18.0	109.4	109.4	109.4	109.4
Pm+Pl+Q (Total)		18.0	18.0	18.0	18.0	427.5	363.6	427.5	363.6

Vessel Stress Summation Comparison (N./mm²):

V6501 ABC

Ammonia Process Flare KOD

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Trunnion Calcs: Right End

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Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	109.39	207.00	Passed
Pm+Pl+Q (TOTAL)	427.52	413.99	Failed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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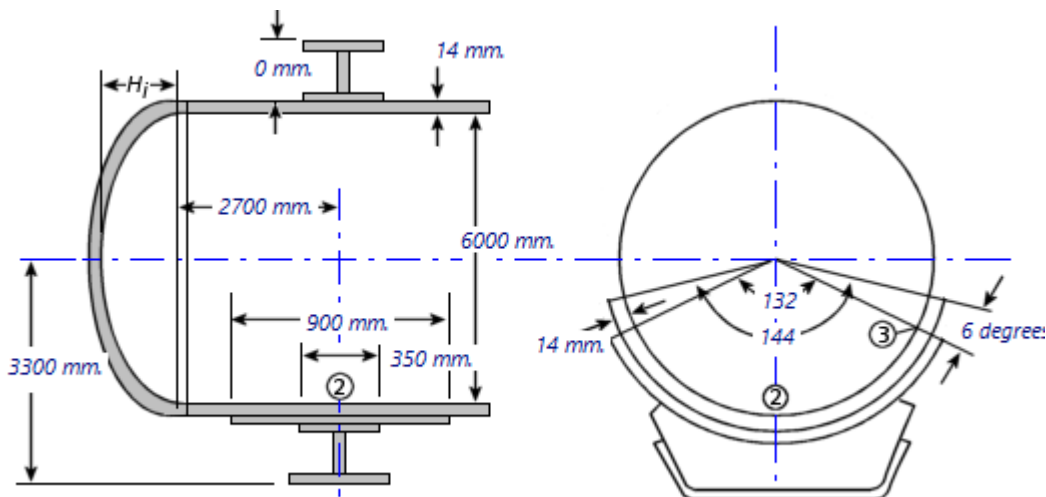


ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	3008.50	mm.
Shell Thickness used in this Case	t	14.000	mm.
Stiffened Vessel Length per 4.15.6	L	13500.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	2700.00	mm.
Saddle Width	b1 or b	350.00	mm.
Saddle Bearing Angle	delta or theta	132.00	degrees
Wear Plate Width	b2 or b1	900.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	144.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm.
Wear Plate Allowable Stress	fw or Sr	137.20	N./mm ²
Inside or Mean Depth of Head	Hi or hm	1507.50	mm.
Shell Allowable Stress used in Calculation		137.20	N./mm ²
Head Allowable Stress used in Calculation		137.20	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	3300.00	mm.
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	43790.64	Kgf
Pressure used in Analysis	P	3.500 bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	47.24	137.20
Long. Stress at Bottom of Midspan	48.49	137.20
Long. Stress at Top of Saddles	54.39	137.20
Long. Stress at Bottom of Saddles	44.20	137.20
Tangential Shear in Shell	6.75	109.76
Circ. Stress at Horn of Saddle	74.33	171.50
Circ. Compressive Stress in Shell	1.95	137.20



Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (5459.7/2 + 0) \cdot 3300.0/5300.0$$

$$= 5099.1 \text{ Kg}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(4049.46, 0) \cdot 3300.0/8100.0$$

$$= 1649.8 \text{ Kg}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(5896.23, 0) \cdot 3300.0/8100.0$$

$$= 2402.2 \text{ Kg}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (5896/2 + 0) \cdot 3300.0/5300.0$$

$$= 5506.9 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 38284 + \max(1650, 5099, 2402, 5507)$$

$$= 43790.6 \text{ Kg}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08 (6.028 + \pi/4 (6.028 \cdot 1.2)^2)$$

$$= 39711.066 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	48743.42	Kgf
Transverse Shear Load	2948.12	Kgf
Longitudinal Shear Load	5896.23	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 5300.000 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1266	K2 = 0.9959	K3 = 0.6937	K4 = 0.3546
K5 = 0.7196	K6 = 0.0434	K7 = 0.0368	K8 = 0.3272
K9 = 0.2516	K10 = 0.0493	K1* = 0.2252	K6p = 0.0353
K7p = 0.0299			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -43791 \cdot 2700.0 [1 - (1 - 2700.0/13500.0 + (3008.5^2 - 1507.5^2)/(2 \cdot 2700.0 \cdot 13500.0))/(1 + (4 \cdot 1507.5)/(3 \cdot 13500.0))]$$

$$= -26336.3 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 43791 \cdot 13500/4 ((1 + 2(3009^2 - 1507^2)/(13500^2))/(1 + (4 \cdot 1507)/(3 \cdot 13500)) - 4 \cdot 2700/13500)$$

$$= 19975.2 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= 3.5 \cdot 3008.5 / (2 \cdot 11.0) - 19975.2 / (\pi \cdot 3008.5^2 \cdot 11.0)$$

$$= 47.24 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 3.5 \cdot 3008.5 / (2 \cdot 11.0) + 19975.2 / (\pi \cdot 3008.5^2 \cdot 11.0)$$

$$= 48.49 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 3.5 \cdot 3008.5 / (2 \cdot 11.0) - 26336.3 / (0.1266 \cdot \pi \cdot 3008.5^2 \cdot 11.0)$$

$$= 54.39 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 3.5 \cdot 3008.5 / (2 \cdot 11.0) + 26336.3 / (0.2252 \cdot \pi \cdot 3008.5^2 \cdot 11.0)$$

$$= 44.20 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 43791(13500.0 - 2 \cdot 2700.0) / (13500.0 + (4 \cdot 1507.5 / 3))$$

$$= 22869.4 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.9959 \cdot 22869.38 / (3008.5 \cdot 11.0)$$

$$= 6.75 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(3008.5 \cdot 11.0)}$$

$$= 141.895 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(350.0 + 1.56 \cdot \sqrt{(3008.5 \cdot 11.0)}, 2 \cdot 2700.0)$$

$$= 633.79 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(S_r / S, 1)$$

$$= \min(137.198 / 137.198, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot t_r))$$

$$= -0.7196 \cdot 43791 \cdot 0.1 / (633.789(11.0 + 1.0 \cdot 14.0))$$

$$= -1.95 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r_{*}]:

$$= -Q / (4(t + \eta \cdot t_r) b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot t_r)^2)$$

$$= -43791 / (4(11.0 + 1.0 \cdot 14.0) 633.789) -$$

$$12 \cdot 0.037 \cdot 43791 \cdot 3008.5 / (13500.0(11.0 + 1.0 \cdot 14.0)^2)$$

$$= -74.33 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:

$$= 8100.0 \text{ mm.}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000013 \cdot 8100.0(270.0 - 21.1)$$

$$= 26.488 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	5300.0000	mm.
Baseplate Thickness	Bpthk	56.0000	mm.

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Saddle Calcs: Operating Case: Step: 21 2:48pm Sep 16,2026



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Baseplate Width	Bpwid	740.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	28.0000	mm.
Web Thickness	Webtk	28.0000	mm.
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	246.0	N./mm2
Height of Web at Center	Hw,c	216.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(3003.0 \cdot 11.0) + 900.0}) 11.0^3) / 12$$

$$= 13.127 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 13.127 + 130.188 (167.565 - 5.5)^2$$

$$= 34207.160 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	1183.5	11.0	5.5	130.2	71603.6	0.342E+05
Wearplate	900.0	14.0	18.0	126.0	226800.0	0.282E+05
Web	28.0	216.0	133.0	60.5	804384.4	0.307E+04
BasePlate	740.0	56.0	269.0	414.4	11147363.0	0.437E+05
Totals	...	...	...	731.1	12250151.0	0.109E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 12250151 / 731.068$$

$$= 167.565 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 132.0 / 2$$

$$= 114.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(114.0) - 0.5 \cdot \sin(114.0)^2) / (\pi - 1.99 + \sin(114.0) \cos(114.0))$$

$$= 0.2255$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.226 \cdot 43790.637$$

$$= 9875.5098 \text{ Kgf}$$

Tension Stress, St = (Fh/As)	=	1.6118	N./mm2
Allowed Stress, Sa = 0.6 * Yield Str	=	147.6000	N./mm2

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 3300.0 - 3003.0 \cdot \sin(132.0/2) / 1.1519$$

$$= 918.426 \text{ mm.}$$

Bending Moment, M = Fh * d	=	9070.0996	Kg-m.
----------------------------	---	-----------	-------

Bending Stress, Sb = (M * C1 / I)	=	13.6477	N./mm2
Allowed Stress, Sa = 2/3 * Yield Str	=	164.0000	N./mm2

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(43791 + 4953) 740.0 / (2 \cdot 5300.0 \cdot 164.0)}$$



= 24.707 mm.

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

= $2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$

= $2 \cdot \cos(90 - 132.0/2) (3000.0 + 14.0 + 14.0)$

= 5532.431 mm.

Distance between Ribs [e]:

= Web Length / (Nribs - 1)

= $5532.4307 / (6 - 1)$

= 1106.486 mm.

Baseplate Pressure Area [Ap]:

= $e \cdot \text{Bpwid} / 2$

= $1106.4861 \cdot 740.0 / 2$

= 4093.999 cm²

Bearing Pressure [Bp]:

= $(Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$

= $48743.422 / (5300.0 \cdot 740.0)$

= 1.243 Kg/cm²

Axial Load [P]:

= $A_p \cdot B_p$

= $4094.0 \cdot 1.24$

= 5088.106 Kg

Area of the Rib and Web [Ar]:

= Rib Area + Web Area

= $90.16 + 154.908$

= 245.068 cm²

Compressive Stress [Sc]:

= P / A_r

= $5088.1 / 245.0681$

= 2.036 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	28.0	322.0	189.0	90.2	1704024.0	0.188E+05
Web	553.2	28.0	14.0	154.9	216871.3	0.652E+04
Totals	...	...	...	245.1	1920895.2	0.253E+05

Rib dimension [D]:

= Saddle Width - Web Thickness

= $350.0 - 28.0$

= 322.000 mm.

Distance to Centroid from Datum [ytot]:

= AY / A

= $1920895 / 245.068$

= 78.382 mm.

Distance to Centroid [C1]:

= $\max(y_{tot}, \text{Saddle Width} - y_{tot})$

= $\max(78.382, 350.0 - 78.382)$

= 271.618 mm.

Radius of Gyration [r]:

= $\sqrt{(\text{Total Inertia} / \text{Total Area})}$

= $\sqrt{(25344.6 / 245.068)}$



$$= 101.695 \text{ mm.}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 3300.0 - \cos(132.0/2)(3000.0 + 14.0 + 14.0) - 56.0$$

$$= 2012.402 \text{ mm.}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(132.0/2)(3000.0 + 14.0 + 14.0)$$

$$= 5532.431 \text{ mm.}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 5300.0 - 50.8$$

$$= 5249.200 \text{ mm.}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (5532.431 - 5249.2) / 2$$

$$= 141.615 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \sqrt{Lv^2 + Dd^2}$$

$$= \sqrt{(2012.402^2 + 141.615^2)}$$

$$= 2017.378 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 246.0)}$$

$$= 126.667$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 2017.378 / 101.695$$

$$= 19.838$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 5896.2 / (2 \cdot 5300.0) \cdot 1106.486 \cdot 2017.38 / 2$$

$$= 620.841 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($19.8375 < 126.6669$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (19.84)^2 / (2 \cdot 126.67^2)) 246 / (5/3 + 3 \cdot (19.84) / (8 \cdot 126.67) - (19.84^3) / (8 \cdot 126.67^3))$$

$$= 140.9 \text{ N/mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 2.04 / 140.87 + (620.84 \cdot 271.618 / 25345\text{E}+09) / 164.0$$

$$= 0.054$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	28.0	322.0	189.0	90.2	1704024.0	0.244E+05
Web	1106.5	28.0	14.0	309.8	433742.6	0.502E+04
Totals	...	...	...	400.0	2137766.5	0.294E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 2137767 / 399.976$$

$$= 53.447 \text{ mm.}$$



Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(53.447, \text{abs}(350.0 - 53.447))$$

$$= 296.553 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + W_{pdthk})^2 - (Pitch/2)^2 } - B_{pthk}$$

$$= 3300.0 - \sqrt{ (3028.0 + 14.0)^2 - (1106.486/2)^2 } - 56.0$$

$$= 266.971 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 29380.0/399.976 }$$

$$= 85.706 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 266.971/85.706$$

$$= 3.115$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 5896.232 / (2 \cdot 5300.0)$$

$$= 0.556 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.556 \cdot 1106.486 \cdot 266.971$$

$$= 164.319 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 164.319/990719$$

$$= 1.627 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($3.115 < 126.6669$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (3.11)^2 / (2 \cdot 126.67^2)) 246 / (5/3 + 3 \cdot (3.11) / (8 \cdot 126.67) - (3.11^3) / (8 \cdot 126.67^3))$$

$$= 146.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 1.99/146.74 + (164.32 \cdot 296.553/29380.008) / 164.0$$

$$= 0.023$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	138.00 N./mm2
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	71.0 mm.
Nominal Bolt Diameter	Bnd	64.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm2
Area Available in a Single Bolt	BlArea	22.1008 cm2
Saddle Load Q0 (Weight)	Q0	43236.6 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	2402.2 Kg
Maximum Transverse Force	Ft	2948.1 Kg
Maximum Longitudinal Force	Fl	5896.2 Kg
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

$$= 13270.4 \text{ Kg}$$



Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= \max(F_f, F_l) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 13270.4 / (22.101 \cdot 4)$$

$$= 14.7 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 2948.1 / (22.101 \cdot 4)$$

$$= 3.3 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$$

$$= 5896.23 / (82.8 \cdot 4.0)$$

$$= 1.7459 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 3300.0 \cdot 2948.12 + 0.0$$

$$= 9728.98 \text{ Kg-m.}$$

Eccentricity (e):

$$= R_{mom} / QO$$

$$= 9728.98 / 43236.56$$

$$= 225.01 \text{ mm.} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 1.7459, 0.0]$$

$$= 1.7459 \text{ cm}^2$$

Note: Saddle and wear plate material are both specified as impact tested.

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	3008.50	mm.
Shell Thickness used in this Case	t	14.000	mm.
Stiffened Vessel Length per 4.15.6	L	13500.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	2700.00	mm.
Saddle Width	b1 or b	350.00	mm.
Saddle Bearing Angle	delta or theta	132.00	degrees
Wear Plate Width	b2 or b1	900.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	144.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm.
Wear Plate Allowable Stress	fw or Sr	137.20	N./mm2
Inside or Mean Depth of Head	Hi or hm	1507.50	mm.
Shell Allowable Stress used in Calculation		137.20	N./mm2
Head Allowable Stress used in Calculation		137.20	N./mm2
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	

V6501 ABC

Ammonia Process Flare KOD

QUEP

PV Elite 28 U1 Licensee: SL User

FileName : Ammonia Flare KOD - V6501 ABC -7%

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Saddle Calcs: Operating Case: Step: 21 2:48pm Sep 16,2026



HEXAGON

Distance from Saddle Base to Centerline B 3300.00 mm.
Coefficient of Friction mu 0.30

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case 44788.61 Kg
Pressure used in Analysis P 3.500 bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	47.22	137.20
Long. Stress at Bottom of Midspan	48.51	137.20
Long. Stress at Top of Saddles	54.53	137.20
Long. Stress at Bottom of Saddles	44.12	137.20
Tangential Shear in Shell	6.90	109.76
Circ. Stress at Horn of Saddle	76.02	171.50
Circ. Compressive Stress in Shell	1.99	137.20

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
= $F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
= $3.0 (5459.7/2 + 0) \cdot 3300.0/5300.0$
= 5099.1 Kg

Longitudinal Saddle Reaction Force [Fwl]:
= $\max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
= $\max(4049.46, 0) \cdot 3300.0/8100.0$
= 1649.8 Kg

Saddle Reaction Force due to Earthquake Fl [Fsl]:
= $\max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$
= $\max(5896.23, 0) \cdot 3300.0/8100.0$
= 2402.2 Kg

Saddle Reaction Force due to Earthquake Ft [Fst]:
= $F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
= $3.0 (5896/2 + 0) \cdot 3300.0/5300.0$
= 5506.9 Kg

Load Combination Results for Q + Wind or Seismic [Q]:
= Saddle Load + $\max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
= $39282 + \max(1650, 5099, 2402, 5507)$
= 44788.6 Kg

Longitudinal Wind Force [Fl]:
= $\text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
= $1.1 \cdot 766.08 (6.028 + \pi/4 (6.028 \cdot 1.2)^2)$
= 39711.066 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 49741.40 Kg
Transverse Shear Load 2948.12 Kg
Longitudinal Shear Load 5896.23 Kg

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
= Baseplate Length
= 5300.000 mm.

The Computed K values from Table 4.15.1:

K1 = 0.1266 K2 = 0.9959 K3 = 0.6937 K4 = 0.3546



K5 = 0.7196 K6 = 0.0434 K7 = 0.0368 K8 = 0.3272
 K9 = 0.2516 K10 = 0.0493 K1* = 0.2252 K6p = 0.0353
 K7p = 0.0299

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)] \\
 &= -44789 \cdot 2700.0 [1 - (1 - 2700.0/13500.0 + (3008.5^2 - 1507.5^2)/(2 \cdot 2700.0 \cdot 13500.0))/(1 + (4 \cdot 1507.5)/(3 \cdot 13500.0))] \\
 &= -26936.5 \text{ Kg-m.}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L) \\
 &= 44789 \cdot 13500/4 ((1 + 2(3009^2 - 1507^2)/(13500^2))/(1 + (4 \cdot 1507)/(3 \cdot 13500)) - 4 \cdot 2700/13500) \\
 &= 20430.4 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t) \\
 &= 3.5 \cdot 3008.5/(2 \cdot 11.0) - 20430.4/(\pi \cdot 3008.5^2 \cdot 11.0) \\
 &= 47.22 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t) \\
 &= 3.5 \cdot 3008.5/(2 \cdot 11.0) + 20430.4/(\pi \cdot 3008.5^2 \cdot 11.0) \\
 &= 48.51 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t) \\
 &= 3.5 \cdot 3008.5/(2 \cdot 11.0) - 26936.5/(0.1266 \cdot \pi \cdot 3008.5^2 \cdot 11.0) \\
 &= 54.53 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t) \\
 &= 3.5 \cdot 3008.5/(2 \cdot 11.0) + 26936.5/(0.2252 \cdot \pi \cdot 3008.5^2 \cdot 11.0) \\
 &= 44.12 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a)/(L + (4 \cdot h_m/3)) \\
 &= 44789(13500.0 - 2 \cdot 2700.0)/(13500.0 + (4 \cdot 1507.5/3)) \\
 &= 23390.6 \text{ Kg}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (R_m \cdot t) \\
 &= 0.9959 \cdot 23390.57 / (3008.5 \cdot 11.0) \\
 &= 6.90 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(R_m \cdot t)} \\
 &= 0.78 \cdot \sqrt{(3008.5 \cdot 11.0)} \\
 &= 141.895 \text{ mm.}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a) \\
 &= \min(350.0 + 1.56 \cdot \sqrt{(3008.5 \cdot 11.0)}, 2 \cdot 2700.0) \\
 &= 633.79 \text{ mm.}
 \end{aligned}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$\begin{aligned}
 &= \min(S_r/S, 1) \\
 &= \min(137.198/137.198, 1) \\
 &= 1.0000
 \end{aligned}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [$\sigma_{6,r}$]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot t_r))$$



$$= -0.7196 \cdot 44789 \cdot 0.1 / (633.789(11.0 + 1.0 \cdot 14.0))$$

$$= -1.99 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) $[\sigma_7, r^*]$:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -44789 / (4(11.0 + 1.0 \cdot 14.0)633.789) -$$

$$12 \cdot 0.037 \cdot 44789 \cdot 3008.5 / (13500.0(11.0 + 1.0 \cdot 14.0)^2)$$

$$= -76.02 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	5300.0000	mm.
Baseplate Thickness	Bpthk	56.0000	mm.
Baseplate Width	Bpwid	740.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	28.0000	mm.
Web Thickness	Webtk	28.0000	mm.
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	246.0	N./mm ²
Height of Web at Center	Hw,c	216.0	mm.
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(3003.0 \cdot 11.0)} + 900.0)^3) / 12$$

$$= 13.127 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 13.127 + 130.188(167.565 - 5.5)^2$$

$$= 34207.160 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	1183.5	11.0	5.5	130.2	71603.6	0.342E+05
Wearplate	900.0	14.0	18.0	126.0	226800.0	0.282E+05
Web	28.0	216.0	133.0	60.5	804384.4	0.307E+04
BasePlate	740.0	56.0	269.0	414.4	11147363.0	0.437E+05
Totals	...	...	...	731.1	12250151.0	0.109E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 12250151 / 731.068$$

$$= 167.565 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 132.0 / 2$$

$$= 114.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(114.0) - 0.5 \cdot \sin(114.0)^2) / (\pi - 1.99 + \sin(114.0) \cos(114.0))$$

$$= 0.2255$$

Saddle Splitting Force [Fh]:

$$= K_1 \cdot Q$$

$$= 0.226 \cdot 44788.613$$

$$= 10100.5703 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.6485 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 147.6000 \text{ N./mm}^2$$

**Saddle Splitting Dimension [d]:**

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 3300.0 - 3003.0 \cdot \sin(132.0/2) / 1.1519$$

$$= 918.426 \text{ mm.}$$

$$\text{Bending Moment, } M = F_h \cdot d = 9276.8057 \text{ Kg-m.}$$

$$\text{Bending Stress, } S_b = (M \cdot C_1 / I) = 13.9587 \text{ N./mm}^2$$

$$\text{Allowed Stress, } S_a = 2/3 \cdot \text{Yield Str} = 164.0000 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(44789 + 4953)740.0 / (2 \cdot 5300.0 \cdot 164.0)}$$

$$= 24.959 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:**Web Length Dimension [Web Length]:**

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 132.0/2) (3000.0 + 14.0 + 14.0)$$

$$= 5532.431 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 5532.4307 / (6 - 1)$$

$$= 1106.486 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 1106.4861 \cdot 740.0 / 2$$

$$= 4093.999 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 49741.398 / (5300.0 \cdot 740.0)$$

$$= 1.268 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 4094.0 \cdot 1.27$$

$$= 5192.280 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 90.16 + 154.908$$

$$= 245.068 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 5192.3 / 245.0681$$

$$= 2.078 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	28.0	322.0	189.0	90.2	1704024.0	0.188E+05
Web	553.2	28.0	14.0	154.9	216871.3	0.652E+04
Totals	...	...	...	245.1	1920895.2	0.253E+05

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 350.0 - 28.0$$

$$= 322.000 \text{ mm.}$$



Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 1920895/245.068$$

$$= 78.382 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{Saddle Width} - ytot)$$

$$= \max(78.382, 350.0 - 78.382)$$

$$= 271.618 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(25344.6/245.068)}$$

$$= 101.695 \text{ mm.}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 3300.0 - \cos(132.0/2) (3000.0 + 14.0 + 14.0) - 56.0$$

$$= 2012.402 \text{ mm.}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(132.0/2) (3000.0 + 14.0 + 14.0)$$

$$= 5532.431 \text{ mm.}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 5300.0 - 50.8$$

$$= 5249.200 \text{ mm.}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (5532.431 - 5249.2) / 2$$

$$= 141.615 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \sqrt{Lv^2 + Dd^2}$$

$$= \sqrt{2012.402^2 + 141.615^2}$$

$$= 2017.378 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 246.0)}$$

$$= 126.667$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 2017.378 / 101.695$$

$$= 19.838$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 5896.2 / (2 \cdot 5300.0) \cdot 1106.486 \cdot 2017.38 / 2$$

$$= 620.841 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($19.8375 < 126.6669$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (19.84)^2 / (2 \cdot 126.67^2)) 246 / (5/3 + 3 \cdot (19.84) / (8 \cdot 126.67) - (19.84^3) / (8 \cdot 126.67^3))$$

$$= 140.9 \text{ N/mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 2.08/140.87 + (620.84 \cdot 271.618 / .25345\text{E}+09) / 164.0$$

$$= 0.055$$

Check of Inside Ribs:



Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	28.0	322.0	189.0	90.2	1704024.0	0.244E+05
Web	1106.5	28.0	14.0	309.8	433742.6	0.502E+04
Totals	...	...	...	400.0	2137766.5	0.294E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 2137767/399.976$$

$$= 53.447 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs(Saddle Width - ytot)})$$

$$= \max(53.447, \text{abs(350.0 - 53.447)})$$

$$= 296.553 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2} - Bpthk$$

$$= 3300.0 - \sqrt{(3028.0 + 14.0)^2 - (1106.486/2)^2} - 56.0$$

$$= 266.971 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(29380.0/399.976)}$$

$$= 85.706 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 266.971/85.706$$

$$= 3.115$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 5896.232/(2 \cdot 5300.0)$$

$$= 0.556 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.556 \cdot 1106.486 \cdot 266.971$$

$$= 164.319 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 164.319/990719$$

$$= 1.627 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (3.115 < 126.6669) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (3.11)^2 / (2 \cdot 126.67^2)) 246 / (5/3 + 3 \cdot (3.11) / (8 \cdot 126.67) - (3.11^3) / (8 \cdot 126.67^3))$$

$$= 146.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 2.03/146.74 + (164.32 \cdot 296.553/29380.008) / 164.0$$

$$= 0.024$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	138.00 N./mm2
Bolt Corrosion Allowance	Bca	1.5 mm.



Distance from Bolts to Edge	Edgedis	71.0 mm.
Nominal Bolt Diameter	Bnd	64.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm2
Area Available in a Single Bolt	BltArea	22.1008 cm2
Saddle Load QO (Weight)	QO	44234.5 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	2402.2 Kgf
Maximum Transverse Force	Ft	2948.1 Kgf
Maximum Longitudinal Force	Fl	5896.2 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 5896.2 / (22.101 \cdot 4)$$

$$= 6.5 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 2948.1 / (22.101 \cdot 4)$$

$$= 3.3 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbols})$$

$$= 5896.23 / (82.8 \cdot 4.0)$$

$$= 1.7459 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 3300.0 \cdot 2948.12 + 0.0$$

$$= 9728.98 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 9728.98 / 44234.54$$

$$= 219.94 \text{ mm.} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 1.7459, 0.0]$$

$$= 1.7459 \text{ cm}^2$$

Note: Saddle and wear plate material are both specified as impact tested.

Review notes about nozzle loadings and supports in the Warnings report.

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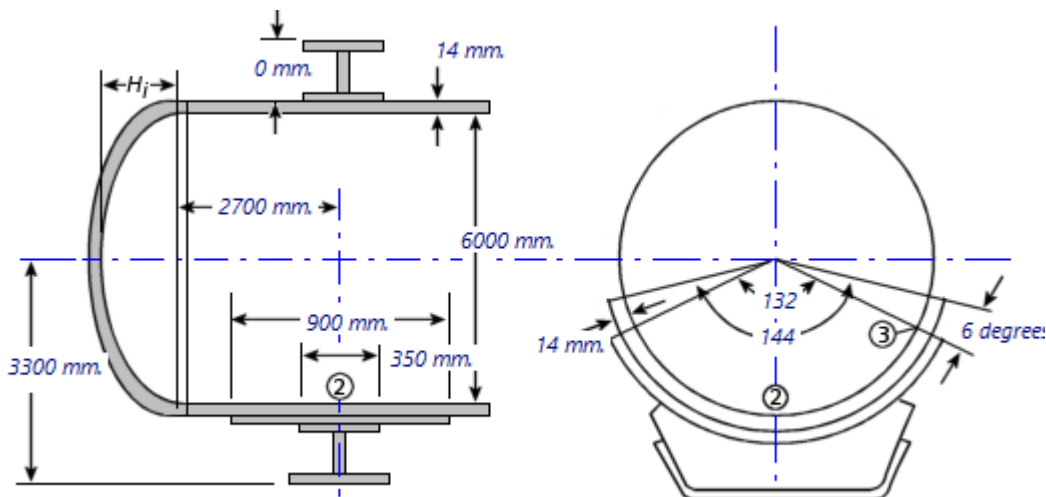


ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	3007.00	mm.
Shell Thickness used in this Case	t	14.000	mm.
Stiffened Vessel Length per 4.15.6	L	13500.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	2700.00	mm.
Saddle Width	b1 or b	350.00	mm.
Saddle Bearing Angle	delta or theta	132.00	degrees
Wear Plate Width	b2 or b1	900.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	144.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm.
Wear Plate Allowable Stress	fw or Sr	137.20	N./mm ²
Inside or Mean Depth of Head	Hi or hm	1506.00	mm.
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	3300.00	mm.
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	253960.91	Kgf
Pressure used in Analysis	P 5.335	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	54.44	234.00
Long. Stress at Bottom of Midspan	60.15	234.00
Long. Stress at Top of Saddles	87.03	234.00
Long. Stress at Bottom of Saddles	40.58	234.00
Tangential Shear in Shell	30.78	138.58
Circ. Stress at Horn of Saddle	345.58	350.99
Circ. Compressive Stress in Shell	9.55	234.00



Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$$

$$= 3.0(2729.8/2 + 0) \cdot 3300.0/5300.0$$

$$= 2549.6\ Kg$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(2024.73, 0) \cdot 3300.0/8100.0$$

$$= 824.9\ Kg$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 251411 + \max(825, 2550, 0, 0)$$

$$= 253960.9\ Kg$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08(6.028 + \pi/4(6.028 \cdot 1.2)^2)$$

$$= 39711.066\ N$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	258913.69	Kgf
Transverse Shear Load	1364.92	Kgf
Longitudinal Shear Load	2024.73	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 5300.000\ mm$$

The Computed K values from Table 4.15.1:

K1 = 0.1266	K2 = 0.9959	K3 = 0.6937	K4 = 0.3546
K5 = 0.7196	K6 = 0.0434	K7 = 0.0368	K8 = 0.3272
K9 = 0.2516	K10 = 0.0493	K1* = 0.2252	K6p = 0.0353
K7p = 0.0299			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -253961 \cdot 2700.0 [1 - (1 - 2700.0/13500.0 + (3007.0^2 - 1506.0^2) / (2 \cdot 2700.0 \cdot 13500.0)) / (1 + (4 \cdot 1506.0) / (3 \cdot 13500.0))]$$

$$= -152703.7\ Kg \cdot m$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 [(1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m)/(3L)) - 4a/L]$$

$$= 253961 \cdot 13500/4 [(1 + 2(3007^2 - 1506^2)/(13500^2)) / (1 + (4 \cdot 1506)/(3 \cdot 13500)) - 4 \cdot 2700/13500]$$

$$= 115911.0\ Kg \cdot m$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 5.335 \cdot 3007.0 / (2 \cdot 14.0) - 115911.0 / (\pi \cdot 3007.0^2 \cdot 14.0)$$

$$= 54.44\ N/mm^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 5.335 \cdot 3007.0 / (2 \cdot 14.0) + 115911.0 / (\pi \cdot 3007.0^2 \cdot 14.0)$$

$$= 60.15\ N/mm^2$$



Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 5.335 \cdot 3007.0 / (2 \cdot 14.0) - 152703.7 / (0.1266 \cdot \pi \cdot 3007.0^2 \cdot 14.0)$$

$$= 87.03 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 5.335 \cdot 3007.0 / (2 \cdot 14.0) + 152703.7 / (0.2252 \cdot \pi \cdot 3007.0^2 \cdot 14.0)$$

$$= 40.58 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 253961(13500.0 - 2 \cdot 2700.0) / (13500.0 + (4 \cdot 1506.0 / 3))$$

$$= 132646.6 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.9959 \cdot 132646.59 / (3007.0 \cdot 14.0)$$

$$= 30.78 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(3007.0 \cdot 14.0)}$$

$$= 160.039 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(350.0 + 1.56 \cdot \sqrt{(3007.0 \cdot 14.0)}, 2 \cdot 2700.0)$$

$$= 670.08 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7196 \cdot 253961 \cdot 0.1 / (670.078(14.0 + 1.0 \cdot 14.0))$$

$$= -9.55 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -253961 / (4(14.0 + 1.0 \cdot 14.0)670.078) -$$

$$12 \cdot 0.037 \cdot 253961 \cdot 3007.0 / (13500.0(14.0 + 1.0 \cdot 14.0)^2)$$

$$= -345.58 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	5300.0000	mm.
Baseplate Thickness	Bpthk	56.0000	mm.
Baseplate Width	Bpwid	740.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	28.0000	mm.
Web Thickness	Webtk	28.0000	mm.
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	N./mm2
Height of Web at Center	Hw,c	216.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(3000.0 \cdot 14.0)} + 900.0)14.0^3) / 12$$

$$= 27.891 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 27.891 + 170.759(161.712 - 7.0)^2$$

$$= 40900.508 \text{ cm}^4$$



	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ² (2) cm ⁴
Shell	1219.7	14.0	7.0	170.8	119531.1	0.409E+05
Wearplate	900.0	14.0	21.0	126.0	264600.0	0.250E+05
Web	28.0	216.0	136.0	60.5	822529.7	0.275E+04
BasePlate	740.0	56.0	272.0	414.4	11271692.0	0.515E+05
Totals	...	...	...	771.6	12478352.0	0.120E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 12478352 / 771.639$$

$$= 161.712 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 132.0 / 2$$

$$= 114.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(114.0) - 0.5 \cdot \sin(114.0)^2) / (\pi - 1.99 + \sin(114.0) \cos(114.0))$$

$$= 0.2255$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.226 \cdot 253960.906$$

$$= 57272.3711 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) =$$

$$9.3473 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} =$$

$$155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 3300.0 - 3000.0 \cdot \sin(132.0/2) / 1.1519$$

$$= 920.805 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d =$$

$$52737.7188 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) =$$

$$69.6323 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} =$$

$$173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3 \cdot (Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3 \cdot (253961 + 4953) \cdot 740.0 / (2 \cdot 5300.0 \cdot 173.33))}$$

$$= 55.390 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 132.0/2) (3000.0 + 14.0 + 14.0)$$

$$= 5532.431 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 5532.4307 / (6 - 1)$$

$$= 1106.486 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 1106.4861 \cdot 740.0 / 2$$

$$= 4093.999 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$



$$= 258913.688 / (5300.0 \cdot 740.0)$$

$$= 6.602 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 4094.0 \cdot 6.6$$

$$= 27026.830 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 90.16 + 154.908$$

$$= 245.068 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 27026.8 / 245.0681$$

$$= 10.815 \text{ N/mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	28.0	322.0	189.0	90.2	1704024.0	0.188E+05
Web	553.2	28.0	14.0	154.9	216871.3	0.652E+04
Totals	...	...	...	245.1	1920895.2	0.253E+05

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 350.0 - 28.0$$

$$= 322.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 1920895 / 245.068$$

$$= 78.382 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(78.382, 350.0 - 78.382)$$

$$= 271.618 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(25344.6 / 245.068)}$$

$$= 101.695 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 3300.0 - \cos(132.0/2) (3000.0 + 14.0 + 14.0) - 56.0$$

$$= 2012.402 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(132.0/2) (3000.0 + 14.0 + 14.0)$$

$$= 5532.431 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 5300.0 - 50.8$$

$$= 5249.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (L_w - B_d) / 2$$

$$= (5532.431 - 5249.2) / 2$$

$$= 141.615 \text{ mm}$$



Length of Outer Rib [L]:

$$= \sqrt{L_v^2 + D_d^2}$$

$$= \sqrt{2012.402^2 + 141.615^2}$$

$$= 2017.378 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 260.0}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 2017.378 / 101.695$$

$$= 19.838$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 2024.7 / (2 \cdot 5300.0) \cdot 1106.486 \cdot 2017.38 / 2$$

$$= 213.193 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($19.8375 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (19.84)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (19.84) / (8 \cdot 123.21) - (19.84)^3 / (8 \cdot 123.21^3))$$

$$= 148.6 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 10.82 / 148.64 + (213.19 \cdot 271.618 / 25345\text{E}+09) / 173.33$$

$$= 0.086$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	28.0	322.0	189.0	90.2	1704024.0	0.244E+05
Web	1106.5	28.0	14.0	309.8	433742.6	0.502E+04
Totals	...	...	...	400.0	2137766.5	0.294E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 2137767 / 399.976$$

$$= 53.447 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(53.447, \text{abs}(350.0 - 53.447))$$

$$= 296.553 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(Ro + W_{pdthk})^2 - (Pitch/2)^2} - B_{pthk}$$

$$= 3300.0 - \sqrt{(3028.0 + 14.0)^2 - (1106.486/2)^2} - 56.0$$

$$= 266.971 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{29380.0 / 399.976}$$

$$= 85.706 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 266.971 / 85.706$$

$$= 3.115$$

Unit Force [Force,u]:



$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 2024.731 / (2 \cdot 5300.0)$$

$$= 0.191 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.191 \cdot 1106.486 \cdot 266.971$$

$$= 56.426 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 56.426 / 990719$$

$$= 0.559 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < Cc$ ($3.115 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (3.11)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (3.11) / (8 \cdot 123.21) - (3.11^3) / (8 \cdot 123.21^3))$$

$$= 155.1 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$$

$$= 10.56 / 155.07 + (56.43 \cdot 296.553 / 29380.008) / 173.33$$

$$= 0.071$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	114.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	71.0 mm.
Nominal Bolt Diameter	Bnd	64.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	22.1008 cm ²
Saddle Load Q0 (Weight)	Q0	256364.1 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	824.9 Kg
Maximum Transverse Force	Ft	1364.9 Kg
Maximum Longitudinal Force	Fl	2024.7 Kg
Saddle Bolted to Steel Foundation	No	

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 2024.7 / (22.101 \cdot 4)$$

$$= 2.2 \text{ N./mm}^2. \text{ Must be less than } 68.4 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1364.9 / (22.101 \cdot 4)$$

$$= 1.5 \text{ N./mm}^2. \text{ Must be less than } 68.4 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (Q0 > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= FI / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 2024.73 / (68.4 \cdot 4.0)$$

$$= 0.7258 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$



$$= 3300.0 \cdot 1364.92 + 0.0$$

$$= 4504.31 \text{ Kg-m.}$$

Eccentricity (e):

$$= R_{mom} / QO$$

$$= 4504.31/256364.11$$

$$= 17.57 \text{ mm.} < B_{plen}/6 \rightarrow \text{No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 0.7258, 0.0]$$

$$= 0.7258 \text{ cm}^2$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	3007.00	mm.
Shell Thickness used in this Case	t	14.000	mm.
Stiffened Vessel Length per 4.15.6	L	13500.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	2700.00	mm.
Saddle Width	b1 or b	350.00	mm.
Saddle Bearing Angle	delta or theta	132.00	degrees
Wear Plate Width	b2 or b1	900.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	144.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm.
Wear Plate Allowable Stress	fw or Sr	137.20	N./mm ²
Inside or Mean Depth of Head	Hi or hm	1506.00	mm.
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	3300.00	mm.
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		254958.94	Kgf
Pressure used in Analysis	P	5.335	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	54.43	234.00
Long. Stress at Bottom of Midspan	60.16	234.00
Long. Stress at Top of Saddles	87.15	234.00
Long. Stress at Bottom of Saddles	40.51	234.00
Tangential Shear in Shell	30.90	138.58
Circ. Stress at Horn of Saddle	346.94	350.99
Circ. Compressive Stress in Shell	9.59	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0(2729.8/2 + 0) \cdot 3300.0/5300.0$$



= 2549.6 Kgf

Longitudinal Saddle Reaction Force [Fwl]:

= max(Fl,wind; Sum of X Forces) • B / Ls

= max(2024.73, 0) • 3300.0/8100.0

= 824.9 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)

= 252409 + max(825, 2550, 0, 0)

= 254958.9 Kgf

Longitudinal Wind Force [FI]:

= WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))

= 1.1 • 766.08(6.028 + $\pi/4$ (6.028 • 1.2)²)

= 39711.066 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	259911.72	Kgf
Transverse Shear Load	1364.92	Kgf
Longitudinal Shear Load	2024.73	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= Baseplate Length

= 5300.000 mm.

The Computed K values from Table 4.15.1:

K1 = 0.1266	K2 = 0.9959	K3 = 0.6937	K4 = 0.3546
K5 = 0.7196	K6 = 0.0434	K7 = 0.0368	K8 = 0.3272
K9 = 0.2516	K10 = 0.0493	K1* = 0.2252	K6p = 0.0353
K7p = 0.0299			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1- a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]
 = -254959•2700.0[1-(1-2700.0/13500.0+(3007.0²-1506.0²)/(2•2700.0•13500.0))/(1+(4•1506.0)/(3•13500.0))]
 = -153303.8 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm)/(3L))-4a/L)
 = 254959•13500/4((1+2(3007²-1506²)/(13500²))/(1+(4•1506)/(3•13500))-4•2700/13500)
 = 116366.6 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= P • Rm/(2t) - M2/(π •Rm²t)
 = 5.335 • 3007.0/(2•14.0) - 116366.6/(π •3007.0²•14.0)
 = 54.43 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= P • Rm/(2t) + M2/(π • Rm² • t)
 = 5.335 • 3007.0/(2 • 14.0) + 116366.6/(π • 3007.0² • 14.0)
 = 60.16 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

= P • Rm/(2t) - M1/(K1• π •Rm²t)
 = 5.335•3007.0/(2•14.0)--153303.8/(0.1266• π •3007.0²•14.0)
 = 87.15 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)



$$= 5.335 \cdot 3007.0 / (2 \cdot 14.0) + (-153303.8 / (0.2252 \cdot \pi \cdot 3007.0^2 \cdot 14.0))$$

$$= 40.51 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a)/(L+(4 \cdot h_m/3))$$

$$= 254959(13500.0 - 2 \cdot 2700.0)/(13500.0 + (4 \cdot 1506.0/3))$$

$$= 133167.9 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.9959 \cdot 133167.89 / (3007.0 \cdot 14.0)$$

$$= 30.90 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(3007.0 \cdot 14.0)}$$

$$= 160.039 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(350.0 + 1.56 \cdot \sqrt{(3007.0 \cdot 14.0)}, 2 \cdot 2700.0)$$

$$= 670.08 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7196 \cdot 254959 \cdot 0.1 / (670.078(14.0 + 1.0 \cdot 14.0))$$

$$= -9.59 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q/(4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -254959/(4(14.0 + 1.0 \cdot 14.0)670.078) -$$

$$12 \cdot 0.037 \cdot 254959 \cdot 3007.0 / (13500.0(14.0 + 1.0 \cdot 14.0)^2)$$

$$= -346.94 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	5300.0000	mm.
Baseplate Thickness	Bpthk	56.0000	mm.
Baseplate Width	Bpwid	740.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	28.0000	mm.
Web Thickness	Webtk	28.0000	mm.
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	216.0	mm.
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(3000.0 \cdot 14.0)} + 900.0)14.0^3)/12$$

$$= 27.891 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 27.891 + 170.759(161.712 - 7.0)^2$$

$$= 40900.508 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ² (2) cm ⁴
Shell	1219.7	14.0	7.0	170.8	119531.1	0.409E+05
Wearplate	900.0	14.0	21.0	126.0	264600.0	0.250E+05

V6501 ABC

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Saddle Calcs: Test Case:

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Web	28.0	216.0	136.0	60.5	822529.7	0.275E+04
BasePlate	740.0	56.0	272.0	414.4	11271692.0	0.515E+05
Totals	...	...	...	771.6	12478352.0	0.120E+06

Distance to Centroid [C1]:

= AY / A

= 12478352/771.639

= 161.712 mm.

Angle [β]:

= 180 - Saddle Angle/2

= 180 - 132.0/2

= 114.0

Saddle Splitting Coefficient [K1]:

= $(1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$ = $(1 + \cos(114.0) - 0.5 \cdot \sin(114.0)^2) / (\pi - 1.99 + \sin(114.0) \cos(114.0))$

= 0.2255

Saddle Splitting Force [Fh]:

= $K1 \cdot Q$

= 0.226 · 254958.938

= 57497.4453 Kgf

Tension Stress, $St = (Fh / As)$

=

9.3840 N./mm²Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str}$

=

155.9972 N./mm²

Saddle Splitting Dimension [d]:

= $B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$ = 3300.0 - 3000.0 · $\sin(132.0/2) / 1.1519$

= 920.805 mm.

Bending Moment, $M = Fh \cdot d$

=

52944.9727 Kg-m.

Bending Stress, $Sb = (M \cdot C1 / I)$

=

69.9059 N./mm²Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str}$

=

173.3302 N./mm²**Minimum Thickness of Baseplate per Moss:**= $\sqrt{(3 \cdot (Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$ = $\sqrt{(3 \cdot (254959 + 4953) \cdot 740.0) / (2 \cdot 5300.0 \cdot 173.33)}$

= 55.496 mm.

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

= $2 \cdot \cos(90 - \text{Saddle Angle}/2) \cdot (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$ = $2 \cdot \cos(90 - 132.0/2) \cdot (3000.0 + 14.0 + 14.0)$

= 5532.431 mm.

Distance between Ribs [e]:

= $\text{Web Length} / (\text{Nr ribs} - 1)$

= 5532.4307 / (6 - 1)

= 1106.486 mm.

Baseplate Pressure Area [Ap]:

= $e \cdot \text{Bpwid} / 2$

= 1106.4861 · 740.0/2

= 4093.999 cm²

Bearing Pressure [Bp]:

= $(Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$

= 259911.688 / (5300.0 · 740.0)

= 6.627 Kgf/cm²

Axial Load [P]:

= $Ap \cdot Bp$

= 4094.0 · 6.63



= 27131.008 Kgf

Area of the Rib and Web [Ar]:

= Rib Area + Web Area

= 90.16 + 154.908

= 245.068 cm²

Compressive Stress [Sc]:

= P/Ar

= 27131.0/245.0681

= 10.857 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	28.0	322.0	189.0	90.2	1704024.0	0.188E+05
Web	553.2	28.0	14.0	154.9	216871.3	0.652E+04
Totals	...	...	...	245.1	1920895.2	0.253E+05

Rib dimension [D]:

= Saddle Width - Web Thickness

= 350.0 - 28.0

= 322.000 mm.

Distance to Centroid from Datum [ytot]:

= AY / A

= 1920895/245.068

= 78.382 mm.

Distance to Centroid [C1]:

= max(ytot, Saddle Width - ytot)

= max(78.382, 350.0 - 78.382)

= 271.618 mm.

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{25344.6/245.068}$

= 101.695 mm.

Length of Outer Rib, step 1, [Lv]:

= Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk

= 3300.0 - cos(132.0/2)(3000.0 + 14.0 + 14.0) - 56.0

= 2012.402 mm.

Length of Outer Rib, step 2, [Lw]:

= 2 • sin(saddle bearing angle / 2)(radius + shlthk + wpdthk)

= 2 • sin(132.0/2)(3000.0 + 14.0 + 14.0)

= 5532.431 mm.

Length of Outer Rib, step 3, [Bd]:

= Baseplate length - clearance

= 5300.0 - 50.8

= 5249.200 mm.

Length of Outer Rib, step 4, [Dd]:

= (Lw - Bd)/2

= (5532.431 - 5249.2)/2

= 141.615 mm.

Length of Outer Rib [L]:

= $\sqrt{Lv^2 + Dd^2}$

= $\sqrt{2012.402^2 + 141.615^2}$

= 2017.378 mm.

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Saddle Calcs: Test Case:

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Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 260.0)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 2017.378 / 101.695$$

$$= 19.838$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 2024.7 / (2 \cdot 5300.0) \cdot 1106.486 \cdot 2017.38 / 2$$

$$= 213.193 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($19.8375 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (19.84)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (19.84) / (8 \cdot 123.21) - (19.84)^3 / (8 \cdot 123.21^3))$$

$$= 148.6 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 10.86 / 148.64 + (213.19 \cdot 271.618 / 25345\text{E}+09) / 173.33$$

$$= 0.086$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	28.0	322.0	189.0	90.2	1704024.0	0.244E+05
Web	1106.5	28.0	14.0	309.8	433742.6	0.502E+04
Totals	...	...	...	400.0	2137766.5	0.294E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 2137767 / 399.976$$

$$= 53.447 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(53.447, \text{abs}(350.0 - 53.447))$$

$$= 296.553 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{((Ro + W_{pdthk})^2 - (Pitch/2)^2)} - B_{pthk}$$

$$= 3300.0 - \sqrt{(3028.0 + 14.0)^2 - (1106.486/2)^2} - 56.0$$

$$= 266.971 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(29380.0 / 399.976)}$$

$$= 85.706 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 266.971 / 85.706$$

$$= 3.115$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 2024.731 / (2 \cdot 5300.0)$$

$$= 0.191 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$



$$= 0.191 \cdot 1106.486 \cdot 266.971$$

$$= 56.426 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 56.426/990719$$

$$= 0.559 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (3.115 < 123.2105) per AISC E2-1 [Sca]:

$$= (1 - (K/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (K/r) / (8 \cdot C_c) - (K/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (3.11)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (3.11) / (8 \cdot 123.21) - (3.11)^3 / (8 \cdot 123.21^3))$$

$$= 155.1 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 10.6/155.07 + (56.43 \cdot 296.553/29380.008) / 173.33$$

$$= 0.072$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	114.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	71.0 mm.
Nominal Bolt Diameter	Bnd	64.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	22.1008 cm ²
Saddle Load QO (Weight)	QO	257362.1 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	824.9 Kg
Maximum Transverse Force	Ft	1364.9 Kg
Maximum Longitudinal Force	Fl	2024.7 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 2024.7 / (22.101 \cdot 4)$$

$$= 2.2 \text{ N./mm}^2. \text{ Must be less than } 68.4 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1364.9 / (22.101 \cdot 4)$$

$$= 1.5 \text{ N./mm}^2. \text{ Must be less than } 68.4 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 2024.73 / (68.4 \cdot 4.0)$$

$$= 0.7258 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 3300.0 \cdot 1364.92 + 0.0$$

$$= 4504.31 \text{ Kg-m.}$$

Eccentricity (e):

$$= R_{mom} / QO$$

$$= 4504.31/257362.14$$

V6501 ABC

Ammonia Process Flare KOD

QUEP

PV Elite 28 U1

Licensee: SL User

FileName : Ammonia Flare KOD - V6501 ABC -7%

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Saddle Calcs: Test Case:

Step: 22 2:48pm Sep 16,2026

PV Elite® v28



HEXAGON

= 17.50 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0.0, 0.7258, 0.0]

= 0.7258 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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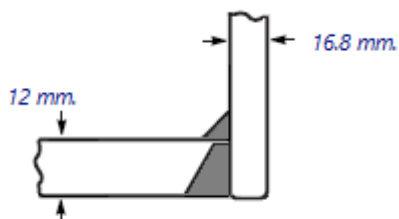
**Input, Nozzle Desc: K1A-2in****From: 10**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	270	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	137.20	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	6000.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	12.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1000.0000	mm.
User Entered Minimum Design Metal Temperature		-45.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: K1A-2in**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P)$ per Appendix 1-4(c)

= $3.5 \cdot 6006.0 \cdot 0.8991 / (2 \cdot 137.2 \cdot 1.0 \cdot 0.2 \cdot 3.5)$

= 6.8903 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

= $P \cdot R / (Sn \cdot E \cdot 0.6 \cdot P)$ per UG-27 (c)(1)

= $3.5 \cdot 28.4 / (133 \cdot 1.0 \cdot 0.6 \cdot 3.5)$

= 0.0747 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	115.3954 mm.
--	----	--------------

Parallel to Vessel Wall, opening length	d	57.6977 mm.
---	---	-------------

Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000 mm.
---	------	-------------

Taking a UG-36(c)(3)(a) exemption for nozzle: K1A-2in.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :****Note:**

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
--	--------

Gov. MDMT of the nozzle to shell joint welded assembly :	-46 °C
--	--------

Weld Size Calculations, Description: K1A-2in

Intermediate Calc. for nozzle/shell Welds	Tmin	13.0000 mm.
---	------	-------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K1A-2in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Nozzle input data check completed without errors.



Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 6006.0 \cdot 0.8991 / (2 \cdot 137.2 \cdot 1.0 - 0.2 \cdot 3.5)$
 $= 6.8903 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.5 \cdot 28.4 / (133 \cdot 1.0 - 0.6 \cdot 3.5)$
 $= 0.0747 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm.
Parallel to Vessel Wall, opening length	d	56.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K1A-2in.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.0747	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 11.0862	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 11.0862	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(11.0862, 4.5)]$
 $= 7.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.0747, 7.8)$
 $= 7.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Weld Size Calculations, Description: K1A-2in

Intermediate Calc. for nozzle/shell Welds	Tmin	13.0000	mm.
---	------	---------	-----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 8.1111 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.6962 mm.

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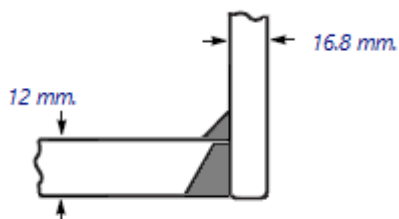
**Input, Nozzle Desc: K1B-2in****From: 10**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	270	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	137.20	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	6000.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	12.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1000.0000	mm.
User Entered Minimum Design Metal Temperature		-45.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		15.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: K1B-2in**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P)$ per Appendix 1-4(c)

= $3.5 \cdot 6006.0 \cdot 0.8991 / (2 \cdot 137.2 \cdot 1.0 \cdot 0.2 \cdot 3.5)$

= 6.8903 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

= $P \cdot R / (Sn \cdot E \cdot 0.6 \cdot P)$ per UG-27 (c)(1)

= $3.5 \cdot 28.4 / (133 \cdot 1.0 \cdot 0.6 \cdot 3.5)$

= 0.0747 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	115.3954 mm.
--	----	--------------

Parallel to Vessel Wall, opening length	d	57.6977 mm.
---	---	-------------

Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000 mm.
---	------	-------------

Taking a UG-36(c)(3)(a) exemption for nozzle: K1B-2in.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :****Note:**

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
--	--------

Gov. MDMT of the nozzle to shell joint welded assembly :	-46 °C
--	--------

Weld Size Calculations, Description: K1B-2in

Intermediate Calc. for nozzle/shell Welds	Tmin	13.0000 mm.
---	------	-------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K1B-2in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Nozzle input data check completed without errors.



Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 6006.0 \cdot 0.8991 / (2 \cdot 137.2 \cdot 1.0 - 0.2 \cdot 3.5)$
 $= 6.8903 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.5 \cdot 28.4 / (133 \cdot 1.0 - 0.6 \cdot 3.5)$
 $= 0.0747 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm.
Parallel to Vessel Wall, opening length	d	56.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K1B-2in.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.0747	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 11.0862	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 11.0862	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(11.0862, 4.5)]$
 $= 7.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.0747, 7.8)$
 $= 7.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Weld Size Calculations, Description: K1B-2in

Intermediate Calc. for nozzle/shell Welds	Tmin	13.0000	mm.
---	------	---------	-----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 8.1111 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.6962 mm.

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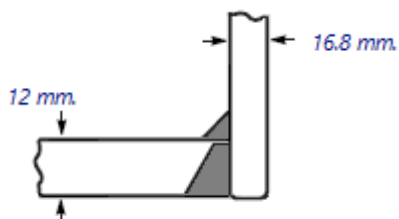
**Input, Nozzle Desc: K2A-2in****From: 10**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	270	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	137.20	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	6000.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	12.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1000.0000	mm.
User Entered Minimum Design Metal Temperature		-45.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		30.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: K2A-2in**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P)$ per Appendix 1-4(c)

= $3.5 \cdot 6006.0 \cdot 0.8991 / (2 \cdot 137.2 \cdot 1.0 \cdot 0.2 \cdot 3.5)$

= 6.8903 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

= $P \cdot R / (Sn \cdot E \cdot 0.6 \cdot P)$ per UG-27 (c)(1)

= $3.5 \cdot 28.4 / (133 \cdot 1.0 \cdot 0.6 \cdot 3.5)$

= 0.0747 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	115.3954 mm.
--	----	--------------

Parallel to Vessel Wall, opening length	d	57.6977 mm.
---	---	-------------

Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000 mm.
---	------	-------------

Taking a UG-36(c)(3)(a) exemption for nozzle: K2A-2in.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :****Note:**

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
--	--------

Gov. MDMT of the nozzle to shell joint welded assembly :	-46 °C
--	--------

Weld Size Calculations, Description: K2A-2in

Intermediate Calc. for nozzle/shell Welds	Tmin	13.0000 mm.
---	------	-------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K2A-2in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Nozzle input data check completed without errors.



Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 6006.0 \cdot 0.8991 / (2 \cdot 137.2 \cdot 1.0 - 0.2 \cdot 3.5)$
 $= 6.8903 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.5 \cdot 28.4 / (133 \cdot 1.0 - 0.6 \cdot 3.5)$
 $= 0.0747 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm.
Parallel to Vessel Wall, opening length	d	56.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K2A-2in.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.0747	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 11.0862	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 11.0862	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(11.0862, 4.5)]$
 $= 7.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.0747, 7.8)$
 $= 7.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Weld Size Calculations, Description: K2A-2in

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 8.1111 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.6962 mm.

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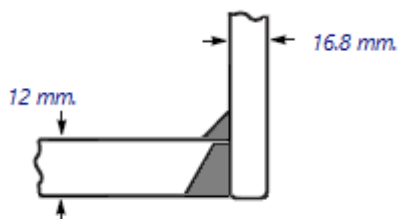
**Input, Nozzle Desc: K2B-2in****From: 10**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	270	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	137.20	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	6000.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	12.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1000.0000	mm.
User Entered Minimum Design Metal Temperature		-45.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		45.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: K2B-2in**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P)$ per Appendix 1-4(c)

= $3.5 \cdot 6006.0 \cdot 0.8991 / (2 \cdot 137.2 \cdot 1.0 \cdot 0.2 \cdot 3.5)$

= 6.8903 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

= $P \cdot R / (Sn \cdot E \cdot 0.6 \cdot P)$ per UG-27 (c)(1)

= $3.5 \cdot 28.4 / (133 \cdot 1.0 \cdot 0.6 \cdot 3.5)$

= 0.0747 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	115.3954 mm.
--	----	--------------

Parallel to Vessel Wall, opening length	d	57.6977 mm.
---	---	-------------

Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000 mm.
---	------	-------------

Taking a UG-36(c)(3)(a) exemption for nozzle: K2B-2in.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :****Note:**

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
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Gov. MDMT of the nozzle to shell joint welded assembly :	-46 °C
--	--------

Weld Size Calculations, Description: K2B-2in

Intermediate Calc. for nozzle/shell Welds	Tmin	13.0000 mm.
---	------	-------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K2B-2in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Nozzle input data check completed without errors.



Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 6006.0 \cdot 0.8991 / (2 \cdot 137.2 \cdot 1.0 - 0.2 \cdot 3.5)$
 $= 6.8903 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.5 \cdot 28.4 / (133 \cdot 1.0 - 0.6 \cdot 3.5)$
 $= 0.0747 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm.
Parallel to Vessel Wall, opening length	d	56.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K2B-2in.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.0747	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 11.0862	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 11.0862	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(11.0862, 4.5)]$
 $= 7.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.0747, 7.8)$
 $= 7.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Weld Size Calculations, Description: K2B-2in

Intermediate Calc. for nozzle/shell Welds	Tmin	13.0000	mm.
---	------	---------	-----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 8.1111 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.6962 mm.

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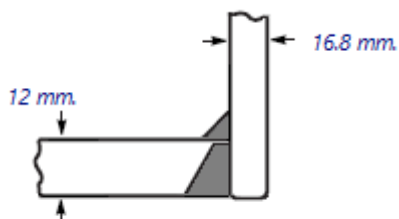
**Input, Nozzle Desc: K3A-2in****From: 10**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	270	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	137.20	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	6000.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	12.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1000.0000	mm.
User Entered Minimum Design Metal Temperature		-45.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		60.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: K3A-2in**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P)$ per Appendix 1-4(c)

= $3.5 \cdot 6006.0 \cdot 0.8991 / (2 \cdot 137.2 \cdot 1.0 \cdot 0.2 \cdot 3.5)$

= 6.8903 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

= $P \cdot R / (Sn \cdot E \cdot 0.6 \cdot P)$ per UG-27 (c)(1)

= $3.5 \cdot 28.4 / (133 \cdot 1.0 \cdot 0.6 \cdot 3.5)$

= 0.0747 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	115.3954 mm.
--	----	--------------

Parallel to Vessel Wall, opening length	d	57.6977 mm.
---	---	-------------

Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000 mm.
---	------	-------------

Taking a UG-36(c)(3)(a) exemption for nozzle: K3A-2in.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :****Note:**

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
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Gov. MDMT of the nozzle to shell joint welded assembly :	-46 °C
--	--------

Weld Size Calculations, Description: K3A-2in

Intermediate Calc. for nozzle/shell Welds	Tmin	13.0000 mm.
---	------	-------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K3A-2in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Nozzle input data check completed without errors.



Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 6006.0 \cdot 0.8991 / (2 \cdot 137.2 \cdot 1.0 - 0.2 \cdot 3.5)$
 $= 6.8903 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.5 \cdot 28.4 / (133 \cdot 1.0 - 0.6 \cdot 3.5)$
 $= 0.0747 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm.
Parallel to Vessel Wall, opening length	d	56.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K3A-2in.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.0747	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 11.0862	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 11.0862	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(11.0862, 4.5)]$
 $= 7.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.0747, 7.8)$
 $= 7.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Weld Size Calculations, Description: K3A-2in

Intermediate Calc. for nozzle/shell Welds	Tmin	13.0000	mm.
---	------	---------	-----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 8.1111 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.6962 mm.

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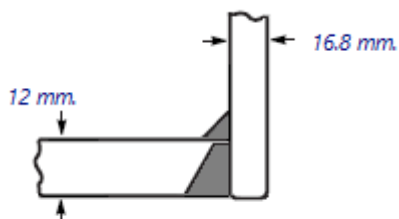
**Input, Nozzle Desc: K3B-2in****From: 10**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	270	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	137.20	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	6000.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	12.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1000.0000	mm.
User Entered Minimum Design Metal Temperature		-45.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		75.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: K3B-2in**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P)$ per Appendix 1-4(c)

= $3.5 \cdot 6006.0 \cdot 0.8991 / (2 \cdot 137.2 \cdot 1.0 \cdot 0.2 \cdot 3.5)$

= 6.8903 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

= $P \cdot R / (Sn \cdot E \cdot 0.6 \cdot P)$ per UG-27 (c)(1)

= $3.5 \cdot 28.4 / (133 \cdot 1.0 \cdot 0.6 \cdot 3.5)$

= 0.0747 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	115.3954 mm.
--	----	--------------

Parallel to Vessel Wall, opening length	d	57.6977 mm.
---	---	-------------

Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000 mm.
---	------	-------------

Taking a UG-36(c)(3)(a) exemption for nozzle: K3B-2in.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :****Note:**

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
--	--------

Gov. MDMT of the nozzle to shell joint welded assembly :	-46 °C
--	--------

Weld Size Calculations, Description: K3B-2in

Intermediate Calc. for nozzle/shell Welds	Tmin	13.0000 mm.
---	------	-------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K3B-2in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Nozzle input data check completed without errors.



Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 6006.0 \cdot 0.8991 / (2 \cdot 137.2 \cdot 1.0 - 0.2 \cdot 3.5)$
 $= 6.8903 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.5 \cdot 28.4 / (133 \cdot 1.0 - 0.6 \cdot 3.5)$
 $= 0.0747 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	113.6000	mm.
Parallel to Vessel Wall, opening length	d	56.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K3B-2in.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.0747	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 11.0862	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 11.0862	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(11.0862, 4.5)]$
 $= 7.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.0747, 7.8)$
 $= 7.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Weld Size Calculations, Description: K3B-2in

Intermediate Calc. for nozzle/shell Welds	Tmin	13.0000	mm.
---	------	---------	-----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 8.1111 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.6962 mm.

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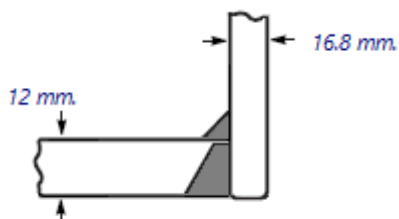
**Input, Nozzle Desc: K4A-2in****From: 10**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	270	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	137.20	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	6000.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	12.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1000.0000	mm.
User Entered Minimum Design Metal Temperature		-45.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		85.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: K4A-2in**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P)$ per Appendix 1-4(c)

= $3.5 \cdot 6006.0 \cdot 0.8991 / (2 \cdot 137.2 \cdot 1.0 \cdot 0.2 \cdot 3.5)$

= 6.8903 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

= $P \cdot R / (Sn \cdot E \cdot 0.6 \cdot P)$ per UG-27 (c)(1)

= $3.5 \cdot 28.4 / (133 \cdot 1.0 \cdot 0.6 \cdot 3.5)$

= 0.0747 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	115.3954 mm.
--	----	--------------

Parallel to Vessel Wall, opening length	d	57.6977 mm.
---	---	-------------

Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000 mm.
---	------	-------------

Taking a UG-36(c)(3)(a) exemption for nozzle: K4A-2in.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :****Note:**

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
--	--------

Gov. MDMT of the nozzle to shell joint welded assembly :	-46 °C
--	--------

Weld Size Calculations, Description: K4A-2in

Intermediate Calc. for nozzle/shell Welds	Tmin	13.0000 mm.
---	------	-------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K4A-2in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Nozzle input data check completed without errors.



Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 6006.0 \cdot 0.8991 / (2 \cdot 137.2 \cdot 1.0 - 0.2 \cdot 3.5)$
 $= 6.8903 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.5 \cdot 28.4 / (133 \cdot 1.0 - 0.6 \cdot 3.5)$
 $= 0.0747 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm.
Parallel to Vessel Wall, opening length	d	56.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K4A-2in.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.0747	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 11.0862	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 11.0862	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(11.0862, 4.5)]$
 $= 7.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.0747, 7.8)$
 $= 7.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Weld Size Calculations, Description: K4A-2in

Intermediate Calc. for nozzle/shell Welds	Tmin	13.0000	mm.
---	------	---------	-----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 8.1111 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.6962 mm.

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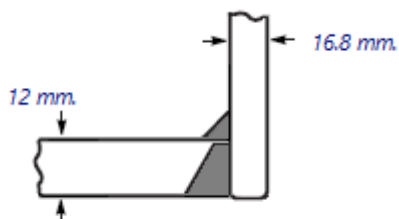
**Input, Nozzle Desc: K4B-2in****From: 10**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	270	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	137.20	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	6000.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	12.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1000.0000	mm.
User Entered Minimum Design Metal Temperature		-45.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		95.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: K4B-2in**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P)$ per Appendix 1-4(c)

= $3.5 \cdot 6006.0 \cdot 0.8991 / (2 \cdot 137.2 \cdot 1.0 \cdot 0.2 \cdot 3.5)$

= 6.8903 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

= $P \cdot R / (Sn \cdot E \cdot 0.6 \cdot P)$ per UG-27 (c)(1)

= $3.5 \cdot 28.4 / (133 \cdot 1.0 \cdot 0.6 \cdot 3.5)$

= 0.0747 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	115.3954 mm.
--	----	--------------

Parallel to Vessel Wall, opening length	d	57.6977 mm.
---	---	-------------

Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000 mm.
---	------	-------------

Taking a UG-36(c)(3)(a) exemption for nozzle: K4B-2in.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :****Note:**

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
--	--------

Gov. MDMT of the nozzle to shell joint welded assembly :	-46 °C
--	--------

Weld Size Calculations, Description: K4B-2in

Intermediate Calc. for nozzle/shell Welds	Tmin	13.0000 mm.
---	------	-------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K4B-2in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Nozzle input data check completed without errors.



Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 6006.0 \cdot 0.8991 / (2 \cdot 137.2 \cdot 1.0 - 0.2 \cdot 3.5)$
 $= 6.8903 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.5 \cdot 28.4 / (133 \cdot 1.0 - 0.6 \cdot 3.5)$
 $= 0.0747 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm.
Parallel to Vessel Wall, opening length	d	56.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K4B-2in.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.0747	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 11.0862	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 11.0862	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(11.0862, 4.5)]$
 $= 7.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.0747, 7.8)$
 $= 7.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Weld Size Calculations, Description: K4B-2in

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 8.1111 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.6962 mm.

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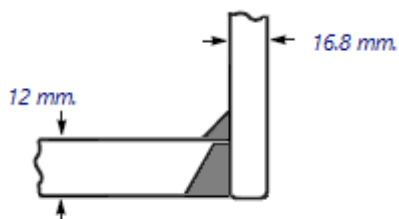
**Input, Nozzle Desc: K5A-2in****From: 10**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	270	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	137.20	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	6000.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	12.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1000.0000	mm.
User Entered Minimum Design Metal Temperature		-45.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		110.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: K5A-2in**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P)$ per Appendix 1-4(c)

= $3.5 \cdot 6006.0 \cdot 0.8991 / (2 \cdot 137.2 \cdot 1.0 \cdot 0.2 \cdot 3.5)$

= 6.8903 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

= $P \cdot R / (Sn \cdot E \cdot 0.6 \cdot P)$ per UG-27 (c)(1)

= $3.5 \cdot 28.4 / (133 \cdot 1.0 \cdot 0.6 \cdot 3.5)$

= 0.0747 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	115.3954 mm.
--	----	--------------

Parallel to Vessel Wall, opening length	d	57.6977 mm.
---	---	-------------

Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000 mm.
---	------	-------------

Taking a UG-36(c)(3)(a) exemption for nozzle: K5A-2in.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :****Note:**

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
--	--------

Gov. MDMT of the nozzle to shell joint welded assembly :	-46 °C
--	--------

Weld Size Calculations, Description: K5A-2in

Intermediate Calc. for nozzle/shell Welds	Tmin	13.0000 mm.
---	------	-------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K5A-2in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Nozzle input data check completed without errors.



Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 6006.0 \cdot 0.8991 / (2 \cdot 137.2 \cdot 1.0 - 0.2 \cdot 3.5)$
 $= 6.8903 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.5 \cdot 28.4 / (133 \cdot 1.0 - 0.6 \cdot 3.5)$
 $= 0.0747 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm.
Parallel to Vessel Wall, opening length	d	56.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K5A-2in.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.0747	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 11.0862	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 11.0862	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(11.0862, 4.5)]$
 $= 7.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.0747, 7.8)$
 $= 7.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Weld Size Calculations, Description: K5A-2in

Intermediate Calc. for nozzle/shell Welds	Tmin	13.0000	mm.
---	------	---------	-----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 8.1111 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.6962 mm.

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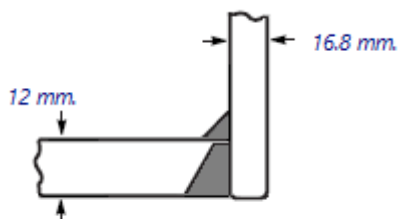
**Input, Nozzle Desc: K5B-2in****From: 10**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	270	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	137.20	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	6000.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	12.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1000.0000	mm.
User Entered Minimum Design Metal Temperature		-45.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		130.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: K5B-2in**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P)$ per Appendix 1-4(c)

= $3.5 \cdot 6006.0 \cdot 0.8991 / (2 \cdot 137.2 \cdot 1.0 \cdot 0.2 \cdot 3.5)$

= 6.8903 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

= $P \cdot R / (Sn \cdot E \cdot 0.6 \cdot P)$ per UG-27 (c)(1)

= $3.5 \cdot 28.4 / (133 \cdot 1.0 \cdot 0.6 \cdot 3.5)$

= 0.0747 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	115.3954 mm.
--	----	--------------

Parallel to Vessel Wall, opening length	d	57.6977 mm.
---	---	-------------

Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000 mm.
---	------	-------------

Taking a UG-36(c)(3)(a) exemption for nozzle: K5B-2in.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :****Note:**

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
--	--------

Gov. MDMT of the nozzle to shell joint welded assembly :	-46 °C
--	--------

Weld Size Calculations, Description: K5B-2in

Intermediate Calc. for nozzle/shell Welds	Tmin	13.0000 mm.
---	------	-------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K5B-2in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Nozzle input data check completed without errors.



Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 6006.0 \cdot 0.8991 / (2 \cdot 137.2 \cdot 1.0 - 0.2 \cdot 3.5)$
 $= 6.8903 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.5 \cdot 28.4 / (133 \cdot 1.0 - 0.6 \cdot 3.5)$
 $= 0.0747 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm.
Parallel to Vessel Wall, opening length	d	56.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K5B-2in.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.0747	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 11.0862	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 11.0862	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(11.0862, 4.5)]$
 $= 7.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.0747, 7.8)$
 $= 7.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Weld Size Calculations, Description: K5B-2in

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 8.1111 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.6962 mm.

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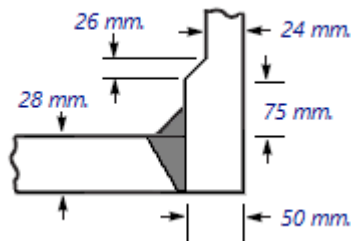
**Input, Nozzle Desc: N1A-34in****From: 20**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	270	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	137.20	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	6000.00	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1300.00	mm.
User Entered Minimum Design Metal Temperature		-45.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	+IT
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	137.20	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		34.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	24.0000	mm.
Flange Material		SA-350 LF2	+IT
Hub Height of Integral Nozzle	h	75.0000	mm.
Height of Beveled Transition	L'	26.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	353.6960	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		B	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N1A-34in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	34.000 in.
Actual Thickness Used in Calculation	0.945 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.5 \cdot 3003.0 / (137 \cdot 1.0 - 0.6 \cdot 3.5)$
 $= 7.6730 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 3.5 \cdot 431.8 / (137 \cdot 1.0 + 0.4 \cdot 3.5)$
 $= 1.1005 \text{ mm.}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 26.0 / 0.5773$
 $= 45.0334 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

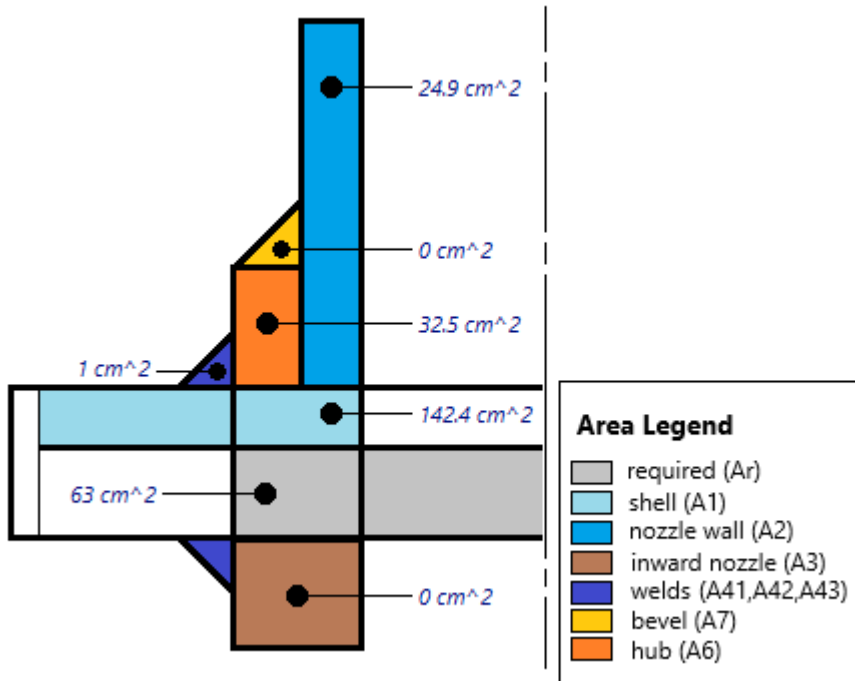
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1643.2001 mm.
Parallel to Vessel Wall, opening length	d	821.6000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	62.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 137.2 / 137.2)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 137.2 / 137.2)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	63.042	NA	NA
Area in Shell	A1	142.359	NA	NA
Area in Nozzle Wall	A2	24.874	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	32.500	NA	NA
TOTAL AREA AVAILABLE	Atot	200.733	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (821.6 \cdot 7.673 \cdot 1.0 + 2 \cdot 47.0 \cdot 7.673 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 63.042 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 821.6(1.0 \cdot 25.0 - 1.0 \cdot 7.673) - 2 \cdot 47.0$$

$$(1.0 \cdot 25.0 - 1.0 \cdot 7.673) \cdot (1 - 1.0)$$

$$= 142.359 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 62.5)(21.0 - 1.1)1.0$$

$$= 24.874 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$



$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lp}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(62.5, 353.7, 75.0)) \cdot (50.0 - 24.0) \cdot 1.0$$

$$= 32.500 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.1005 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 10.6730 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 10.6730 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 11.3312 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.331, \max(10.673, 4.5)]$$

$$= 10.6730 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(4.1005, 10.673)$$

$$= 10.6730 \text{ mm.}$$

Available Nozzle Neck Thickness = 24.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 17.0,	Allowable	: 137.2 N./mm2	Passed
Expansion	: 0.0,	Allowable	: 327.0 N./mm2	Passed
Occasional	: 3.3,	Allowable	: 182.5 N./mm2	Passed
Shear	: 8.7,	Allowable	: 96.0 N./mm2	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-104 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

$$\text{Design Pressure/Ambient Rating} = 3.50/19.60 = 0.179$$

Weld Size Calculations, Description: N1A-34in

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------



Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (63.0415 - 142.3585 + 2 \cdot 47.0 \cdot 1.0 \cdot (1.0 \cdot 25.0 - 7.673)) \cdot 137)$$

$$= \max(0, -88178.75) \text{ Kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (57.3744 + 1. - 0.0 \cdot 1.0) \cdot 137$$

$$= 81666.05 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (57.3744 + 0.0 + 1. + (23.5)) \cdot 137$$

$$= 114542.66 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (57.3744 + 0.0 + 1. + 0.0 + (23.5)) \cdot 137$$

$$= 114542.66 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 863.6 \cdot 10.0 \cdot 0.49 \cdot 137$$

$$= 92992. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 421.3) \cdot (50.0 - 3.0) \cdot 0.7 \cdot 137$$

$$= 609204. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnv} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 863.6 \cdot (12.0 - 0.0) \cdot 0.74 \cdot 137$$

$$= 168528. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (92992 + 609204) = 702196 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (92992 + 0 + 168528 + 0) = 261520 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (92992 + 168528 + 0) = 261520 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 702196 Kgf, must exceed W = 0 Kgf	or W1 = 81666 Kgf
Path 2-2 = 261520 Kgf, must exceed W = 0 Kgf	or W2 = 114542 Kgf
Path 3-3 = 261520 Kgf, must exceed W = 0 Kgf	or W3 = 114542 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

V6501 ABC

Ammonia Process Flare KOD

QUEP

PV Elite 28 U1

Licensee: SL User

FileName : Ammonia Flare KOD - V6501 ABC -7%

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Nozzle Calcs.: N1A-34in

Noz1: 43 2:48pm Sep 16,2026



HEXAGON

Converged Maximum Allowable Pressure in the Operating case: 5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 31.2378 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 412.9339 mm.

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 821.60 mm.

Input Echo, WRC297 Item 1, Description: N1A-34in :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	6000.000	mm.
Shell Thickness	28.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 137.20	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	915.600	mm.
Nozzle Thickness	50.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 137.20	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	3.50	bars
Radial Load	P	6498.64	Kgf
Circumferential Shear	Vc	4332.42	Kgf
Longitudinal Shear	Vl	4332.42	Kgf
Circumferential Moment	Mc	10311.69	Kg-m.
Longitudinal Moment	Ml	10311.69	Kg-m.
Torsional Moment	Mt	14731.05	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(\text{t} - \text{ca}))}$$

$$= 915.6 + 2 \cdot 1.65 \cdot \sqrt{(3015.5(28.0 - 3.0))}$$

$$= 1821.674 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6031.000 mm.
Nozzle Outside Diameter	(d) =	915.600 mm.
Vessel Thickness used	(T) =	25.000 mm.

V6501 ABC

Ammonia Process Flare KOD

QUEP

PV Elite 28 U1

Licensee: SL User

FileName : Ammonia Flare KOD - V6501 ABC -7%

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Nozzle Calcs.: N1A-34in

Nozl: 43

2:48pm

Sep 16, 2026



HEXAGON

Nozzle Thickness used (t) = 47.000 mm.
 T / t = 0.532
 d / t = 19.481

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.358$$

Nr/P = 0.049
 Mr/P = 0.125
 M0/P = 0.037
 N0/P = 0.099
 MrD/Mc = 0.268
 NrDL/Mc = 0.074
 M0d/Mc = 0.086
 N0DL/Mc = 0.096
 MrD/M1 = 0.109
 NrDL/M1 = 0.050
 M0D/M1 = 0.033
 N0DL/M1 = 0.140

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-10	-10	-10	-10
Outplane Bending (P)	-22	22	-22	22
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-24	-24	24	24
Outplane Bending (ML)	-34	34	34	-34
Normal Pressure Stress	41	42	41	42
Outplane Stress Summary	-50	64	68	45

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-5	-5	-5	-5
Inplane Bending (P)	-76	76	-76	76
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-8	-8	8	8
Inplane Bending (ML)	-115	115	115	-115
Inplane Pressure Stress	20	20	20	20
Inplane Stress Summary	-185	199	63	-13

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	185	199	69	59

**Vessel Stresses (N./mm²):**

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	-10	-10	-10	-10
Outplane Bending (P)	-22	22	-22	22
Outplane Membrane (Mc)	-17	-17	17	17
Outplane Bending (Mc)	-91	91	91	-91
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	20	20	20	20
Outplane Stress Summary	-119	107	96	-40

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	-5	-5	-5	-5
Inplane Bending (P)	-76	76	-76	76
Inplane Membrane (Mc)	-13	-13	13	13
Inplane Bending (Mc)	-284	284	284	-284
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	41	42	41	42
Inplane Stress Summary	-337	384	257	-157

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	3	3	5	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	337	384	257	157

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	-10	-10	-10	-10
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-24	-24	24	24
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	3	3	3
Hoop Stress Summary	-31	-30	18	18

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside

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Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-17	17	-17	17
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-25	25	25	-25
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-45	39	12	-2
----------------------	-----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2

Shear Stress Summary	2	2	1	1
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	45	71	18	21
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	-10	-10	-10	-10
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-17	-17	17	17
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	3	3	3

Hoop Stress Summary	-23	-23	10	10
---------------------	-----	-----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-17	17	-17	17
Axial Membrane (Mc)	-3	-3	3	3
Axial Bending (Mc)	-69	69	69	-69
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-89	84	56	-46
----------------------	-----	----	----	-----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2

Shear Stress Summary	1	1	2	2
----------------------	---	---	---	---

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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	89	107	57	57

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	41	42	41	42	41	42	41	42
Circ. Pl (SUS)	-34	-34	14	14	-18	-18	8	8
Circ. Q (SUS)	-56	56	12	-12	-360	360	208	-208
Long. Pm (SUS)	20	20	20	20	20	20	20	20
Long. Pl (SUS)	-13	-13	3	3	-27	-27	7	7
Long. Q (SUS)	-191	191	39	-39	-113	113	69	-69
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	41.0	42.0	41.0	42.0	41.0	42.0	41.0	42.0
Pm+Pl (SUS)	8.0	8.6	55.0	56.0	30.1	31.1	49.0	50.0
Pm+Pl+Q (Total)	184.2	198.2	68.4	60.3	337.0	384.0	257.2	158.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	42.00	137.20	Passed
Pm+Pl (SUS)	56.03	205.80	Passed
Pm+Pl+Q (TOTAL)	384.03	411.59	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	3	3	3	3	3	3	3
Circ. Pl (SUS)	-34	-34	14	14	-27	-27	7	7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-3	-3	3	3	-3	-3	3	3
Long. Q (SUS)	-42	42	8	-8	-86	86	52	-52
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
Pm+Pl (SUS)	31.0	31.0	17.0	17.0	24.0	24.0	10.0	10.0

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Pm+Pl+Q (Total)	44.3	71.1	17.7	21.4	88.1	108.1	56.1	58.1
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Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	3.00	137.20	Passed
Pm+Pl (SUS)	31.00	205.80	Passed
Pm+Pl+Q (TOTAL)	108.07	411.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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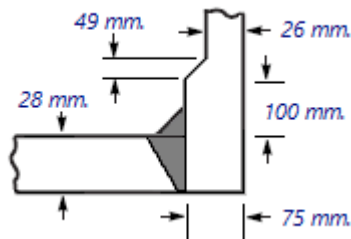
**Input, Nozzle Desc: N2-42in****From: 40**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	270	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	137.20	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	6000.00	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		6750.00	mm.
User Entered Minimum Design Metal Temperature		-45.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	+IT
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	137.20	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		42.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.0000	mm.
Flange Material		SA-350 LF2	+IT
Hub Height of Integral Nozzle	h	100.0000	mm.
Height of Beveled Transition	L'	49.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		75.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	331.5980	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		B	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N2-42in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	42.000 in.
Actual Thickness Used in Calculation	1.024 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.5 \cdot 3003.0 / (137 \cdot 1.0 - 0.6 \cdot 3.5)$
 $= 7.6730 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 3.5 \cdot 533.4 / (137 \cdot 1.0 + 0.4 \cdot 3.5)$
 $= 1.3594 \text{ mm.}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 49.0 / 0.5773$
 $= 84.8705 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

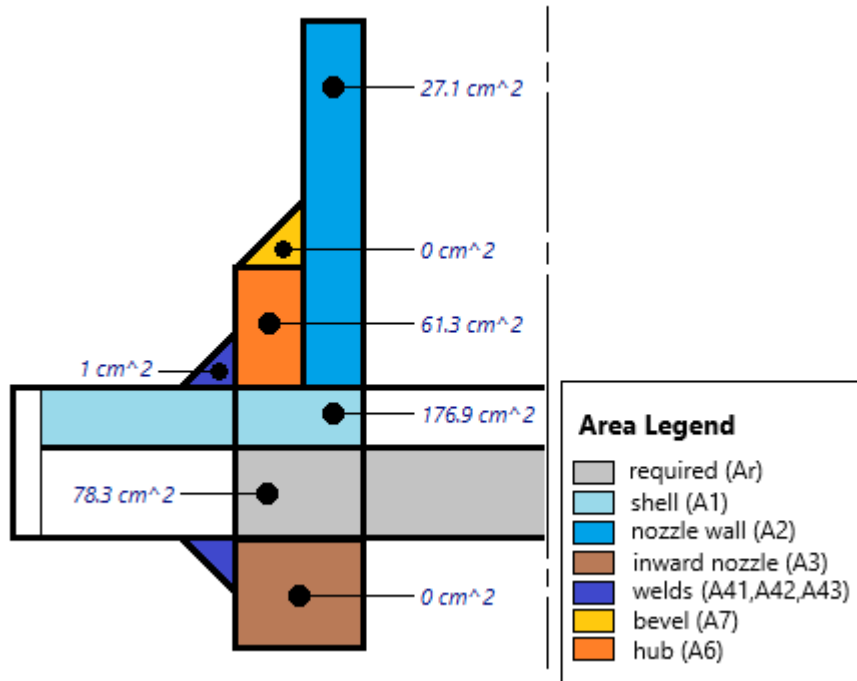
UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	2041.6000 mm.
Parallel to Vessel Wall, opening length	d	1020.8000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	62.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 137.2 / 137.2)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 137.2 / 137.2)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	78.326	NA	NA
Area in Shell	A1	176.874	NA	NA
Area in Nozzle Wall	A2	27.051	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	61.250	NA	NA
TOTAL AREA AVAILABLE	Atot	266.175	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (1020.8 \cdot 7.673 \cdot 1.0 + 2 \cdot 72.0 \cdot 7.673 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 78.326 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 1020.8(1.0 \cdot 25.0 - 1.0 \cdot 7.673) - 2 \cdot 72.0$$

$$(1.0 \cdot 25.0 - 1.0 \cdot 7.673) \cdot (1 - 1.0)$$

$$= 176.874 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 62.5)(23.0 - 1.36)1.0$$

$$= 27.051 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

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$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lp}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(62.5, 331.6, 100.0)) \cdot (75.0 - 26.0) \cdot 1.0$$

$$= 61.250 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.3594 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 10.6730 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 10.6730 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 11.3312 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.331, \max(10.673, 4.5)]$$

$$= 10.6730 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(4.3594, 10.673)$$

$$= 10.6730 \text{ mm.}$$

Available Nozzle Neck Thickness = 26.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 16.2,	Allowable	: 137.2 N./mm2	Passed
Expansion	: 0.0,	Allowable	: 327.8 N./mm2	Passed
Occasional	: 3.8,	Allowable	: 182.5 N./mm2	Passed
Shear	: 7.9,	Allowable	: 96.0 N./mm2	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-104 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

$$\text{Design Pressure/Ambient Rating} = 3.50/19.60 = 0.179$$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

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= 49.0/0.5773
= 84.8705 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1531.2000 mm.

Parallel to Vessel Wall, opening length d 765.6000 mm.

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 62.5000 mm.

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure]	AR	52.220	52.220
Area Available in Shell	A1	88.437	88.437
Area Available in Nozzle Wall	A2	27.051	27.051
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	1.000	1.000
Area Available in Pad	A5	0.000	0.000
Area Available in Hub	A6	61.250	61.250
TOTAL AREA AVAILABLE	Atot	177.738	177.738

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: N2-42in

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

Nozzle Weld Required Thickness Actual Thickness
6.0000 = Min per Code 7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (78.3262 - 176.8739 + 2 • 72.0 • 1.0 •
(1.0 • 25.0 - 7.673))) 137
= max(0, -102962.33) Kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (88.3008 + 1. - 0.0 • 1.0) • 137
= 124932.14 Kgf

Weld Load [W2]:

= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (88.3008 + 0.0 + 1. + (36.0)) • 137
= 175296.33 Kgf

Weld Load [W3]:

= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (88.3008 + 0.0 + 1. + 0.0 + (36.0)) • 137
= 175296.33 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

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$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.1416/2.0) \cdot 1066.7999 \cdot 10.0 \cdot 0.49 \cdot 137 \\
 &= 114873. \text{ Kg}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\
 &= (3.1416 \cdot 521.9) \cdot (75.0 - 3.0) \cdot 0.7 \cdot 137 \\
 &= 1156094. \text{ Kg}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_{gvi} \cdot 0.74 \cdot S_{ng} \\
 &= (3.1416/2.0) \cdot 1066.7999 \cdot (12.0 - 0.0) \cdot 0.74 \cdot 137 \\
 &= 208181. \text{ Kg}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (114873 + 1156094) = 1270967 \text{ Kg}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (114873 + 0 + 208181 + 0) = 323054 \text{ Kg}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Sonw + Tngw + Sinw) \\
 &= (114873 + 208181 + 0) = 323054 \text{ Kg}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 1270967 Kg, must exceed W = 0 Kg	or W1 = 124932 Kg
Path 2-2 = 323054 Kg, must exceed W = 0 Kg	or W2 = 175296 Kg
Path 3-3 = 323054 Kg, must exceed W = 0 Kg	or W3 = 175296 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 47.8000 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 407.3980 mm.

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 1020.80 mm.

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: N2-42in :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	6000.000	mm.
Shell Thickness	28.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 137.20	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm.



Nozzle Diameter	1164.800	mm.
Nozzle Thickness	75.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	137.20 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	3.50	bars
Radial Load	P	8027.72	Kgf
Circumferential Shear	Vc	5351.80	Kgf
Longitudinal Shear	Vl	5351.80	Kgf
Circumferential Moment	Mc	15735.09	Kg-m.
Longitudinal Moment	Ml	15735.09	Kg-m.
Torsional Moment	Mt	22478.73	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 1164.8 + 2 \cdot 1.65 \cdot \sqrt{3015.5(28.0 - 3.0)}$$

$$= 2070.874 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6031.000	mm.
Nozzle Outside Diameter	(d) =	1164.800	mm.
Vessel Thickness used	(T) =	25.000	mm.
Nozzle Thickness used	(t) =	72.000	mm.
T / t	=	0.347	
d / t	=	16.178	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.000$$

Nr/P	=	0.052
Mr/P	=	0.106
M0/P	=	0.031
N0/P	=	0.082
MrD/Mc	=	0.260
NrDL/Mc	=	0.080
M0d/Mc	=	0.080
N0DL/Mc	=	0.093
MrD/Ml	=	0.089
NrDL/Ml	=	0.051
M0D/Ml	=	0.025
N0DL/Ml	=	0.097

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-10	-10	-10	-10
Outplane Bending (P)	-23	23	-23	23
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0

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Outplane Membrane (ML)	-20	-20	20	20
Outplane Bending (ML)	-31	31	31	-31
Normal Pressure Stress	41	42	41	42

Outplane Stress Summary	-44	66	60	44
-------------------------	-----	----	----	----

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Inplane Membrane (P)	-6	-6	-6	-6
Inplane Bending (P)	-79	79	-79	79
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-10	-10	10	10
Inplane Bending (ML)	-112	112	112	-112
Inplane Pressure Stress	20	20	20	20

Inplane Stress Summary	-188	196	57	-7
------------------------	------	-----	----	----

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4

Shear Stress Summary	5	5	2	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	189	196	62	52
------------------------	-----	-----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	-10	-10	-10	-10
Outplane Bending (P)	-23	23	-23	23
Outplane Membrane (Mc)	-19	-19	19	19
Outplane Bending (Mc)	-101	101	101	-101
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	20	20	20	20

Outplane Stress Summary	-134	115	108	-47
-------------------------	------	-----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	-6	-6	-6	-6
Inplane Bending (P)	-79	79	-79	79
Inplane Membrane (Mc)	-16	-16	16	16
Inplane Bending (Mc)	-331	331	331	-331
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	41	42	41	42

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Inplane Stress Summary	-392	429	303	-198
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	2	2	5	5
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	392	429	303	199
------------------------	-----	-----	-----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	-10	-10	-10	-10
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-20	-20	20	20
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	3	2	3

Hoop Stress Summary	-28	-27	13	13
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-7	7	-7	7
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-9	9	9	-9
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-18	15	5	1
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	1	1	1	1
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	28	43	13	13
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):

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CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-10	-10	-10	-10
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-19	-19	19	19
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	3	2	3
Hoop Stress Summary	-27	-26	12	12

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-7	7	-7	7
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-33	33	33	-33
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-42	39	30	-23

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	1	1	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	42	66	30	35

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	41	42	41	42	41	42	41	42
Circ. Pl (SUS)	-30	-30	10	10	-22	-22	10	10
Circ. Q (SUS)	-54	54	8	-8	-410	410	252	-252
Long. Pm (SUS)	20	20	20	20	20	20	20	20
Long. Pl (SUS)	-16	-16	4	4	-29	-29	9	9
Long. Q (SUS)	-191	191	33	-33	-124	124	78	-78
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	41.0	42.0	41.0	42.0	41.0	42.0	41.0	42.0
Pm+Pl (SUS)	11.1	12.1	51.0	52.0	28.1	29.1	51.0	52.0

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Pm+Pl+Q (Total)	187.2	195.2	61.2	53.3	391.0	430.0	303.1	200.2
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Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	42.00	137.20	Passed
Pm+Pl (SUS)	52.04	205.80	Passed
Pm+Pl+Q (TOTAL)	430.03	411.59	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	3	2	3	2	3	2	3
Circ. Pl (SUS)	-30	-30	10	10	-29	-29	9	9
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	-16	16	2	-2	-40	40	26	-26
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	2.0	3.0	2.0	3.0	2.0	3.0	2.0	3.0
Pm+Pl (SUS)	28.0	27.0	12.0	13.0	27.0	26.0	11.0	12.0
Pm+Pl+Q (Total)	28.1	42.0	12.1	13.1	41.1	65.0	29.1	35.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	3.00	137.20	Passed
Pm+Pl (SUS)	28.00	205.80	Passed
Pm+Pl+Q (TOTAL)	65.03	411.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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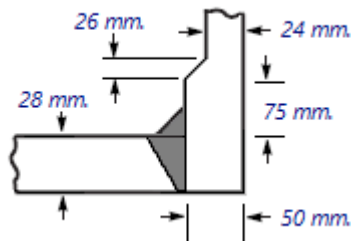
**Input, Nozzle Desc: N1B-34in****From: 60**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	270	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	137.20	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	6000.00	mm.
Shell Finished (Minimum) Thickness	t	28.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		12200.00	mm.
User Entered Minimum Design Metal Temperature		-45.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	+IT
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	137.20	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		34.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	24.0000	mm.
Flange Material		SA-350 LF2	+IT
Hub Height of Integral Nozzle	h	75.0000	mm.
Height of Beveled Transition	L'	26.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	353.6960	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		B	

The Pressure Design option was Design Pressure + static head.

**Reinforcement CALCULATION, Description: N1B-34in**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	34.000 in.
Actual Thickness Used in Calculation	0.945 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.5 \cdot 3003.0 / (137 \cdot 1.0 - 0.6 \cdot 3.5)$
 $= 7.6730 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 3.5 \cdot 431.8 / (137 \cdot 1.0 + 0.4 \cdot 3.5)$
 $= 1.1005 \text{ mm.}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 26.0 / 0.5773$
 $= 45.0334 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

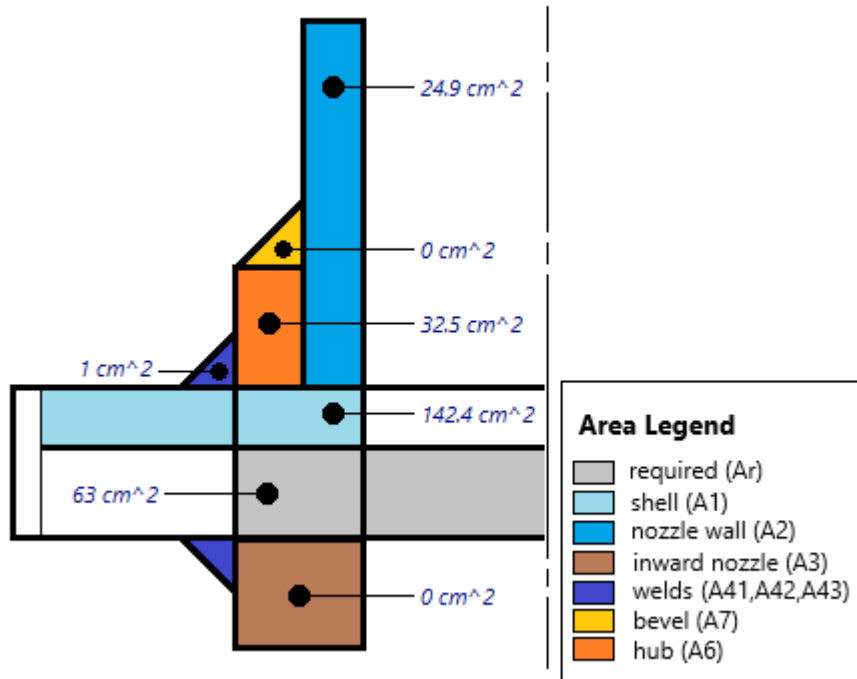
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1643.2001 mm.
Parallel to Vessel Wall, opening length	d	821.6000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	62.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 137.2 / 137.2)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 137.2 / 137.2)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	63.042	NA	NA
Area in Shell	A1	142.359	NA	NA
Area in Nozzle Wall	A2	24.874	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	32.500	NA	NA
TOTAL AREA AVAILABLE	Atot	200.733	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (821.6 \cdot 7.673 \cdot 1.0 + 2 \cdot 47.0 \cdot 7.673 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 63.042 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 821.6(1.0 \cdot 25.0 - 1.0 \cdot 7.673) - 2 \cdot 47.0$$

$$(1.0 \cdot 25.0 - 1.0 \cdot 7.673) \cdot (1 - 1.0)$$

$$= 142.359 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 62.5)(21.0 - 1.1)1.0$$

$$= 24.874 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$



$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lp}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(62.5, 353.7, 75.0)) \cdot (50.0 - 24.0) \cdot 1.0$$

$$= 32.500 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.1005 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 10.6730 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 10.6730 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 11.3312 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.331, \max(10.673, 4.5)]$$

$$= 10.6730 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(4.1005, 10.673)$$

$$= 10.6730 \text{ mm.}$$

Available Nozzle Neck Thickness = 24.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 17.0,	Allowable	: 137.2 N./mm2	Passed
Expansion	: 0.0,	Allowable	: 327.0 N./mm2	Passed
Occasional	: 3.3,	Allowable	: 182.5 N./mm2	Passed
Shear	: 8.7,	Allowable	: 96.0 N./mm2	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-104 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

$$\text{Design Pressure/Ambient Rating} = 3.50/19.60 = 0.179$$

Weld Size Calculations, Description: N1B-34in

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------



Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (63.0415 - 142.3585 + 2 \cdot 47.0 \cdot 1.0 \cdot (1.0 \cdot 25.0 - 7.673)) \cdot 137)$$

$$= \max(0, -88178.75) \text{ Kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (57.3744 + 1. - 0.0 \cdot 1.0) \cdot 137$$

$$= 81666.05 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (57.3744 + 0.0 + 1. + (23.5)) \cdot 137$$

$$= 114542.66 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (57.3744 + 0.0 + 1. + 0.0 + (23.5)) \cdot 137$$

$$= 114542.66 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 863.6 \cdot 10.0 \cdot 0.49 \cdot 137$$

$$= 92992. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 421.3) \cdot (50.0 - 3.0) \cdot 0.7 \cdot 137$$

$$= 609204. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 863.6 \cdot (12.0 - 0.0) \cdot 0.74 \cdot 137$$

$$= 168528. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (92992 + 609204) = 702196 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (92992 + 0 + 168528 + 0) = 261520 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (92992 + 168528 + 0) = 261520 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 702196 Kgf, must exceed W = 0 Kgf	or W1 = 81666 Kgf
Path 2-2 = 261520 Kgf, must exceed W = 0 Kgf	or W2 = 114542 Kgf
Path 3-3 = 261520 Kgf, must exceed W = 0 Kgf	or W3 = 114542 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

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Nozzle Calcs.: N1B-34in

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Converged Maximum Allowable Pressure in the Operating case: 5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 31.2378 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 412.9339 mm.

Input Echo, WRC297 Item 1, Description: N1B-34in :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	6000.000	mm.
Shell Thickness	28.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 137.20	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	915.600	mm.
Nozzle Thickness	50.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 137.20	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	3.50	bars
Radial Load	P	6498.64	Kgf
Circumferential Shear	Vc	4332.42	Kgf
Longitudinal Shear	Vl	4332.42	Kgf
Circumferential Moment	Mc	10311.69	Kg-m.
Longitudinal Moment	Ml	10311.69	Kg-m.
Torsional Moment	Mt	14731.05	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 915.6 + 2 \cdot 1.65 \cdot \sqrt{(3015.5(28.0 - 3.0))}$$

$$= 1821.674 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6031.000	mm.
Nozzle Outside Diameter	(d) =	915.600	mm.
Vessel Thickness used	(T) =	25.000	mm.
Nozzle Thickness used	(t) =	47.000	mm.
T / t	=	0.532	
d / t	=	19.481	



$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.358$$

Nr/P	=	0.049
Mr/P	=	0.125
M0/P	=	0.037
N0/P	=	0.099
MrD/Mc	=	0.268
NrDL/Mc	=	0.074
M0d/Mc	=	0.086
N0DL/Mc	=	0.096
MrD/Ml	=	0.109
NrDL/Ml	=	0.050
M0D/Ml	=	0.033
N0DL/Ml	=	0.140

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-10	-10	-10	-10
Outplane Bending (P)	-22	22	-22	22
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-24	-24	24	24
Outplane Bending (ML)	-34	34	34	-34
Normal Pressure Stress	41	42	41	42
Outplane Stress Summary	-50	64	68	45

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-5	-5	-5	-5
Inplane Bending (P)	-76	76	-76	76
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-8	-8	8	8
Inplane Bending (ML)	-115	115	115	-115
Inplane Pressure Stress	20	20	20	20
Inplane Stress Summary	-185	199	63	-13

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	185	199	69	59

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------



Outplane Membrane (P)	-10	-10	-10	-10
Outplane Bending (P)	-22	22	-22	22
Outplane Membrane (Mc)	-17	-17	17	17
Outplane Bending (Mc)	-91	91	91	-91
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	20	20	20	20

Outplane Stress Summary	-119	107	96	-40
-------------------------	------	-----	----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	-5	-5	-5	-5
Inplane Bending (P)	-76	76	-76	76
Inplane Membrane (Mc)	-13	-13	13	13
Inplane Bending (Mc)	-284	284	284	-284
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	41	42	41	42

Inplane Stress Summary	-337	384	257	-157
------------------------	------	-----	-----	------

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	3	3	5	5
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	337	384	257	157
------------------------	-----	-----	-----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	-10	-10	-10	-10
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-24	-24	24	24
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	3	3	3

Hoop Stress Summary	-31	-30	18	18
---------------------	-----	-----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-17	17	-17	17
Axial Membrane (Mc)	0	0	0	0

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Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-25	25	25	-25
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-45	39	12	-2
----------------------	-----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	2	2	1	1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	45	71	18	21

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	-10	-10	-10	-10
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-17	-17	17	17
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	3	3	3
Hoop Stress Summary	-23	-23	10	10

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-17	17	-17	17
Axial Membrane (Mc)	-3	-3	3	3
Axial Bending (Mc)	-69	69	69	-69
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-89	84	56	-46

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	1	1	2	2

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside

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Two * Max Shear Stress 89 107 57 57

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	41	42	41	42	41	42	41	42
Circ. Pl (SUS)	-34	-34	14	14	-18	-18	8	8
Circ. Q (SUS)	-56	56	12	-12	-360	360	208	-208
Long. Pm (SUS)	20	20	20	20	20	20	20	20
Long. Pl (SUS)	-13	-13	3	3	-27	-27	7	7
Long. Q (SUS)	-191	191	39	-39	-113	113	69	-69
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	41.0	42.0	41.0	42.0	41.0	42.0	41.0	42.0
Pm+Pl (SUS)	8.0	8.6	55.0	56.0	30.1	31.1	49.0	50.0
Pm+Pl+Q (Total)	184.2	198.2	68.4	60.3	337.0	384.0	257.2	158.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	42.00	137.20	Passed
Pm+Pl (SUS)	56.03	205.80	Passed
Pm+Pl+Q (TOTAL)	384.03	411.59	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	3	3	3	3	3	3	3
Circ. Pl (SUS)	-34	-34	14	14	-27	-27	7	7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-3	-3	3	3	-3	-3	3	3
Long. Q (SUS)	-42	42	8	-8	-86	86	52	-52
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
Pm+Pl (SUS)	31.0	31.0	17.0	17.0	24.0	24.0	10.0	10.0
Pm+Pl+Q (Total)	44.3	71.1	17.7	21.4	88.1	108.1	56.1	58.1

Nozzle Stress Summation Comparison (N./mm²):

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Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	3.00	137.20	Passed
Pm+Pl (SUS)	31.00	205.80	Passed
Pm+Pl+Q (TOTAL)	108.07	411.60	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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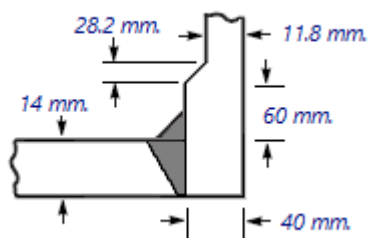
**Input, Nozzle Desc: N3-4in****From: 60**

Pressure for Reinforcement Calculations	P	3.530	bars
Temperature for Internal Pressure	Temp	270	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	137.20	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	6000.00	mm.
Shell Finished (Minimum) Thickness	t	14.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		13000.00	mm.
User Entered Minimum Design Metal Temperature		-45.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 3	+IT
Material UNS Number		K31918	
Material Specification/Type		Pipe	
Allowable Stress at Temperature	Sn	126.40	N./mm2
Allowable Stress At Ambient	Sna	128.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-350 LF2	+IT
Hub Height of Integral Nozzle	h	60.0000	mm.
Height of Beveled Transition	L'	28.1985	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		40.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	164.1480	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



**Reinforcement CALCULATION, Description: N3-4in**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.500 in.
Actual Thickness Used in Calculation	0.465 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.53 \cdot 3003.0 / (137 \cdot 1.0 - 0.6 \cdot 3.53)$
 $= 7.7391 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 3.53 \cdot 57.15 / (126 \cdot 1.0 + 0.4 \cdot 3.53)$
 $= 0.1594 \text{ mm.}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 28.199 / 0.5773$
 $= 48.8413 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

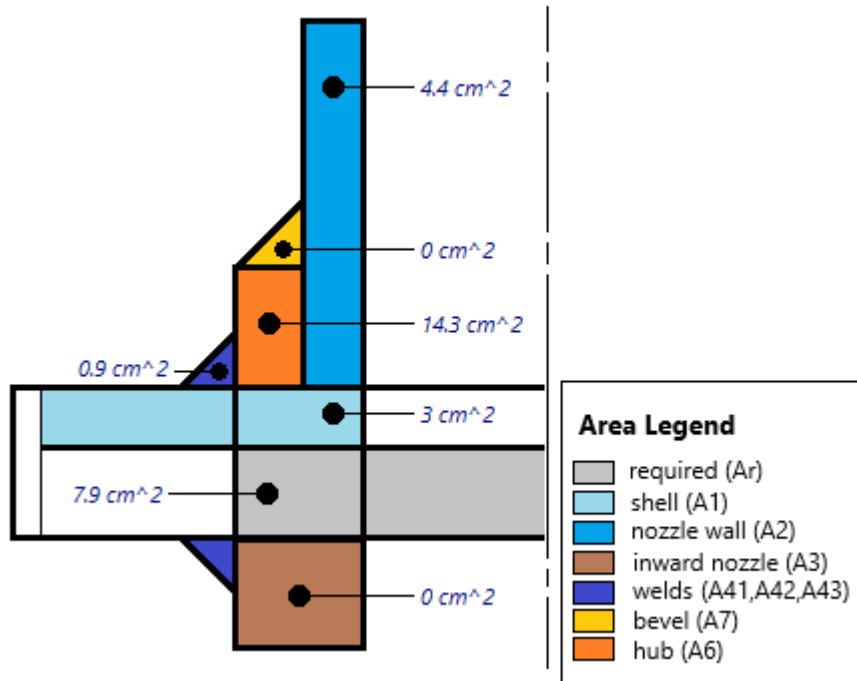
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	193.3941 mm.
Parallel to Vessel Wall, opening length	d	96.6970 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 126.4 / 137.2)$
 $= 0.921$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 126.4 / 137.2)$
 $= 0.921$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.921, 1.0)$
 $= 0.921$



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	7.934	NA	NA
Area in Shell	A1	2.963	NA	NA
Area in Nozzle Wall	A2	4.379	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.921	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	14.289	NA	NA
TOTAL AREA AVAILABLE	Atot	22.552	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (96.697 \cdot 7.7391 \cdot 1.0 + 2 \cdot 37.0 \cdot 7.7391 \cdot 1.0 \cdot (1 - 0.92))$$

$$= 7.934 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 96.697 (1.0 \cdot 11.0 - 1.0 \cdot 7.739) - 2 \cdot 37.0$$

$$(1.0 \cdot 11.0 - 1.0 \cdot 7.7391) \cdot (1 - 0.921)$$

$$= 2.963 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 27.5) (8.8 - 0.16) 0.9213$$

$$= 4.379 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$$

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$$= 10.0^2 \cdot 0.9213 + (0.0)^2 \cdot 0.9213$$

$$= 0.921 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(27.5, 164.1, 60.0)) \cdot (40.0 - 11.8) \cdot 0.921$$

$$= 14.289 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.1594 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 10.7391 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 10.7391 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 8.2578 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.258, \max(10.7391, 4.5)]$$

$$= 8.2578 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.1594, 8.2578)$$

$$= 8.2578 \text{ mm.}$$

Available Nozzle Neck Thickness = 11.8015 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 31.1,	Allowable	: 126.4 N./mm2	Passed
Expansion	: 0.0,	Allowable	: 286.9 N./mm2	Passed
Occasional	: 0.9,	Allowable	: 168.1 N./mm2	Passed
Shear	: 20.8,	Allowable	: 88.5 N./mm2	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld (Impact tested) :***Note:**This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-101 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :*Note:**This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-101 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-101 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-104 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

$$\text{Design Pressure/Ambient Rating} = 3.53/51.00 = 0.069$$



Weld Size Calculations, Description: N3-4in

Intermediate Calc. for nozzle/shell welds T_{min} 11.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (7.9342 - 2.9633 + 2 \cdot 37.0 \cdot 0.9213 \cdot (1.0 \cdot 11.0 - 7.7391)) \cdot 137) \\
 &= 10064.50 \text{ Kgf}
 \end{aligned}$$

For hub type nozzles, A₂ includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A_2 + A_4 - (W_i - \text{Can} / .707)^2 \cdot f_{r2}) \cdot S_v \\
 &= (18.6676 + 0.9213 - 0.0 \cdot 0.92) \cdot 137 \\
 &= 27404.94 \text{ Kgf}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (18.6676 + 0.0 + 0.9213 + (7.4994)) \cdot 137 \\
 &= 37896.57 \text{ Kgf}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\
 &= (18.6676 + 0.0 + 0.9213 + 0.0 + (7.4994)) \cdot 137 \\
 &= 37896.57 \text{ Kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.1416/2.0) \cdot 114.3 \cdot 10.0 \cdot 0.49 \cdot 126 \\
 &= 11339. \text{ Kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.1416 \cdot 52.7493) \cdot (40.0 - 3.0) \cdot 0.7 \cdot 126 \\
 &= 55320. \text{ Kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\
 &= (3.1416/2.0) \cdot 114.3 \cdot (14.0 - 3.0) \cdot 0.74 \cdot 137 \\
 &= 20446. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (11339 + 55320) = 66659 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\
 &= (11339 + 0 + 20446 + 0) = 31785 \text{ Kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\
 &= (11339 + 20446 + 0) = 31785 \text{ Kgf}
 \end{aligned}$$

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Nozzle Calcs.: N3-4in

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Summary of Failure Path Calculations:

Path 1-1 = 66659 Kgf, must exceed W = 10064 Kgf or W1 = 27404 Kgf
 Path 2-2 = 31785 Kgf, must exceed W = 10064 Kgf or W2 = 37896 Kgf
 Path 3-3 = 31785 Kgf, must exceed W = 10064 Kgf or W3 = 37896 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.5443 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 178.6923 mm.

Input Echo, WRC297 Item 1, Description: N3-4in :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	6000.000	mm.
Shell Thickness	14.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 137.20	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	170.697	mm.
Nozzle Thickness	40.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 3	
Nozzle Cold S.I. Allowable	SNmc 128.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 126.40	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUS tained loads.

Design Internal Pressure	Dp	3.53	bars
Radial Load	P	764.54	Kgf
Circumferential Shear	Vc	509.70	Kgf
Longitudinal Shear	Vl	509.70	Kgf
Circumferential Moment	Mc	142.72	Kg-m.
Longitudinal Moment	Ml	142.72	Kg-m.
Torsional Moment	Mt	203.89	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (4.267) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(\text{t} - \text{ca}))}$$

$$= 170.697 + 2 \cdot 1.65 \cdot \sqrt{(3008.5(14.0 - 3.0))}$$

$$= 771.020 \text{ mm.}$$

WRC 297 Curve Access Parameters:



Vessel Mean Diameter (D) = 6017.000 mm.
 Nozzle Outside Diameter (d) = 170.697 mm.
 Vessel Thickness used (T) = 11.000 mm.
 Nozzle Thickness used (t) = 37.000 mm.
 T / t = 0.297
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 0.663$$

Nr/P = 0.128
 Mr/P = 0.286
 M0/P = 0.084
 N0/P = 0.145
 MrD/Mc = 0.315
 NrDL/Mc = 0.063
 M0d/Mc = 0.098
 N0DL/Mc = 0.011
 MrD/Ml = 0.247
 NrDL/Ml = 0.084
 M0D/Ml = 0.077
 N0DL/Ml = 0.082

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-8	-8	-8	-8
Outplane Bending (P)	-31	31	-31	31
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5
Outplane Bending (ML)	-31	31	31	-31
Normal Pressure Stress	96	96	96	96
Outplane Stress Summary	19	144	93	92

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-7	-7	-7	-7
Inplane Bending (P)	-106	106	-106	106
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-100	100	100	-100
Inplane Pressure Stress	48	48	48	48
Inplane Stress Summary	-172	241	40	51

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	5	5	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top	Al Top	Bu Bottom	Bl Bottom
--	-----------	-----------	--------------	--------------

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	Outside	Inside	Outside	Inside
Two * Max Shear Stress	191	241	93	92

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	-8	-8	-8	-8
Outplane Bending (P)	-31	31	-31	31
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-39	39	39	-39
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	48	48	48	48
Outplane Stress Summary	-32	109	48	31

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	-7	-7	-7	-7
Inplane Bending (P)	-106	106	-106	106
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-127	127	127	-127
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	96	96	96	96
Inplane Stress Summary	-150	318	114	71

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	5	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	150	318	114	72

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	-8	-8	-8	-8
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-5	-5	5	5
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	0	0	0	0
Hoop Stress Summary	-14	-13	-3	-2

**Nozzle Stresses (N./mm²):**

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-7	7	-7	7
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-7	7	7	-7
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	-18	11	3	2

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	1	1	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	18	25	6	5

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-8	-8	-8	-8
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	0	0	0	0
Hoop Stress Summary	-9	-8	-7	-7

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-7	7	-7	7
Axial Membrane (Mc)	-3	-3	3	3
Axial Bending (Mc)	-10	10	10	-10
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	-20	13	5	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0

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Shear due to Torsion	1	1	1	1
Shear Stress Summary	0	0	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	20	22	14	8

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	96	96	96	96	96	96	96	96
Circ. Pl (SUS)	-13	-13	-3	-3	-11	-11	-3	-3
Circ. Q (SUS)	-62	62	0	0	-233	233	21	-21
Long. Pm (SUS)	48	48	48	48	48	48	48	48
Long. Pl (SUS)	-12	-12	-2	-2	-8	-8	-8	-8
Long. Q (SUS)	-206	206	-6	6	-70	70	8	-8
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	96.0	96.0	96.0	96.0	96.0	96.0	96.0	96.0
Pm+Pl (SUS)	83.0	83.0	93.0	93.0	85.0	85.0	93.0	93.0
Pm+Pl+Q (Total)	191.2	242.2	93.1	93.1	148.0	318.0	114.2	72.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	96.00	137.20	Passed
Pm+Pl (SUS)	93.02	205.80	Passed
Pm+Pl+Q (TOTAL)	318.02	411.59	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	0	0	0	0	0	0	0	0
Circ. Pl (SUS)	-13	-13	-3	-3	-8	-8	-8	-8
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-3	-3	3	3	-3	-3	3	3
Long. Q (SUS)	-14	14	0	0	-17	17	3	-3
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1

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Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)	13.0	13.0	6.0	6.0	8.0	8.0	11.0	11.0
Pm+Pl+Q (Total)	17.2	24.1	6.3	6.3	20.1	22.1	14.1	8.2

Nozzle Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	126.40	Passed
Pm+Pl (SUS)	13.00	189.60	Passed
Pm+Pl+Q (TOTAL)	24.08	379.20	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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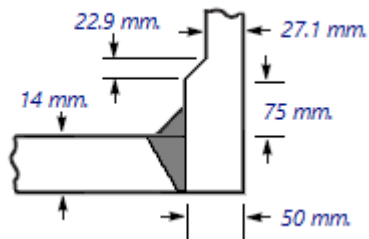
**Input, Nozzle Desc: M124in****From: 70**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	270	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	137.20	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	6000.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	14.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-45.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 3	+IT
Material UNS Number		K31918	
Material Specification/Type		Pipe	
Allowable Stress at Temperature	Sn	126.40	N./mm2
Allowable Stress At Ambient	Sna	128.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-350 LF2	+IT
Hub Height of Integral Nozzle	h	75.0000	mm.
Height of Beveled Transition	L'	22.9300	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	353.6960	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: M124in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	1.066 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 6006.0 \cdot 0.8991 / (2 \cdot 137.2 \cdot 1.0 - 0.2 \cdot 3.5)$
 $= 6.8903 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 3.5 \cdot 304.8 / (126 \cdot 1.0 + 0.4 \cdot 3.5)$
 $= 0.8431 \text{ mm.}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 22.93 / 0.5773$
 $= 39.7159 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

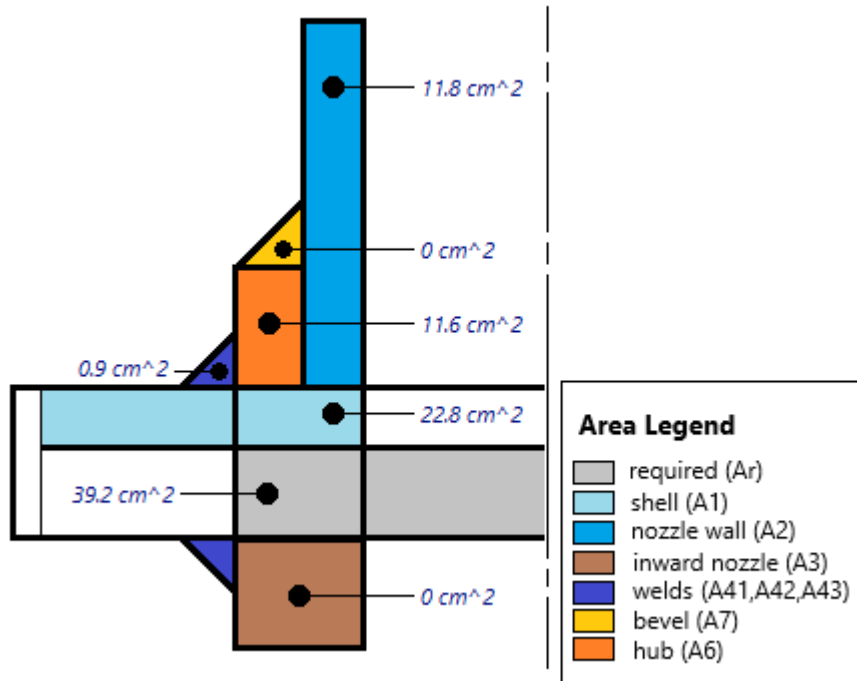
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1122.9198 mm.
Parallel to Vessel Wall, opening length	d	561.4599 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 126.4/137.2)$
 $= 0.921$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 126.4/137.2)$
 $= 0.921$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.921, 1.0)$
 $= 0.921$



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	39.196	NA	NA
Area in Shell	A1	22.770	NA	NA
Area in Nozzle Wall	A2	11.769	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.921	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	11.619	NA	NA
TOTAL AREA AVAILABLE	Atot	47.080	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (561.4599 \cdot 6.8903 \cdot 1.0 + 2 \cdot 47.0 \cdot 6.8903 \cdot 1.0 \cdot (1 - 0.92))$$

$$= 39.196 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 561.46 (1.0 \cdot 11.0 - 1.0 \cdot 6.89) - 2 \cdot 47.0$$

$$(1.0 \cdot 11.0 - 1.0 \cdot 6.8903) \cdot (1 - 0.921)$$

$$= 22.770 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 27.5) (24.07 - 0.84) \cdot 0.9213$$

$$= 11.769 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$$



$$= 10.0^2 \cdot 0.9213 + (0.0)^2 \cdot 0.9213$$

$$= 0.921 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot \text{fr}_2$$

$$= (2 \cdot \min(27.5, 353.7, 75.0)) \cdot (50.0 - 27.1) \cdot 0.921$$

$$= 11.619 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 3.8431 \text{ mm}$.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 3.8431 \text{ mm}.$$

Available Nozzle Neck Thickness = 27.0700 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification $-101 \text{ }^\circ\text{C}$

Calculated Minimum Design Metal Temperature $-104 \text{ }^\circ\text{C}$

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification $-101 \text{ }^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : $-101 \text{ }^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification $-46 \text{ }^\circ\text{C}$

Flange MDMT with Temp reduction per UCS-66(i)(2) $-104 \text{ }^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

$$\text{Design Pressure/Ambient Rating} = 3.50/19.60 = 0.179$$

Weld Size Calculations, Description: M124in

Intermediate Calc. for nozzle/shell Welds $T_{min} = 13.0000 \text{ mm}$.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot \text{fr}_1 \cdot (E_1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (39.1962 - 22.7701 + 2 \cdot 47.0 \cdot 0.9213 \cdot (1.0 \cdot 11.0 - 6.8903)) \cdot 137)$$

$$= 27959.26 \text{ Kg}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_4 - (W_i - \text{Can}/.707)^2 \cdot \text{fr}_2) \cdot S_v$$



$$= (23.3883 + 0.9213 - 0.0 \cdot 0.92) \cdot 137$$

$$= 34009.23 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (23.3883 + 0.0 + 0.9213 + (9.5262)) \cdot 137$$

$$= 47336.44 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (23.3883 + 0.0 + 0.9213 + 0.0 + (9.5262)) \cdot 137$$

$$= 47336.44 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 126$$

$$= 60475. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 292.765) \cdot (50.0 - 3.0) \cdot 0.7 \cdot 126$$

$$= 390017. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 609.6 \cdot (14.0 - 3.0) \cdot 0.74 \cdot 137$$

$$= 109046. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (60475 + 390017) = 450492 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (60475 + 0 + 109046 + 0) = 169521 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (60475 + 109046 + 0) = 169521 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 450492 Kgf, must exceed W = 27959 Kgf	or W1 = 34009 Kgf
Path 2-2 = 169521 Kgf, must exceed W = 27959 Kgf	or W2 = 47336 Kgf
Path 3-3 = 169521 Kgf, must exceed W = 27959 Kgf	or W3 = 47336 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4 bars

The Drop for this Nozzle is : 8.5909 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 376.2869 mm.

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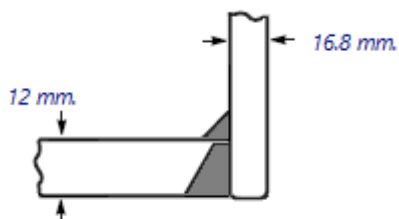
**Input, Nozzle Desc: K6-2in****From: 70**

Pressure for Reinforcement Calculations	P	3.500	bars
Temperature for Internal Pressure	Temp	270	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	137.20	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	6000.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	12.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1000.0000	mm.
User Entered Minimum Design Metal Temperature		-45.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

**Reinforcement CALCULATION, Description: K6-2in**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

$$= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P) \text{ per Appendix 1-4(c)}$$

$$= 3.5 \cdot 6006.0 \cdot 0.8991 / (2 \cdot 137.2 \cdot 1.0 \cdot 0.2 \cdot 3.5)$$

$$= 6.8903 \text{ mm.}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (Sn \cdot E \cdot 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 3.5 \cdot 28.4 / (133 \cdot 1.0 \cdot 0.6 \cdot 3.5)$$

$$= 0.0747 \text{ mm.}$$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	115.3954 mm.
Parallel to Vessel Wall, opening length	d	57.6977 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000 mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K6-2in.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :****Note:**

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
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Gov. MDMT of the nozzle to shell joint welded assembly :	-46 °C
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Weld Size Calculations, Description: K6-2in

Intermediate Calc. for nozzle/shell Welds	Tmin	13.0000 mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K6-2in

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.661 in.

Nozzle input data check completed without errors.



Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.5 \cdot 6006.0 \cdot 0.8991 / (2 \cdot 137.2 \cdot 1.0 - 0.2 \cdot 3.5)$
 $= 6.8903 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.5 \cdot 28.4 / (133 \cdot 1.0 - 0.6 \cdot 3.5)$
 $= 0.0747 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	113.6000	mm.
Parallel to Vessel Wall, opening length	d	56.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K6-2in.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.0747	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 11.0862	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 11.0862	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(11.0862, 4.5)]$
 $= 7.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.0747, 7.8)$
 $= 7.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Weld Size Calculations, Description: K6-2in

Intermediate Calc. for nozzle/shell Welds	Tmin	13.0000	mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 8.1111 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.6962 mm.

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: